

HG-DC ACB

DC Air Circuit Breakers

HD HYUNDAI ELECTRIC

Essential for Today, Potential for Tomorrow

HD Hyundai Electric solely pursues the growth of our customers' business. From power generation to power distribution, we focus on developing and commercializing products and solutions aimed at increasing the efficiency of energy equipment as well as at proactively monitoring and controlling assets in an integrated manner to improve our customers' productivity and management efficiency. We are well aware that our efforts add to the driving force behind our customers' growth and contribute to the creation and maintenance of a more dynamic world. We focus on achieving innovation and strive to evolve continuously to shape a better tomorrow based on today's technological advancement

Solution

Utility

INTEGRICT

Energy Solution

Energy solution business refers to the business of designing, procuring and establishing a system that enables the efficient use of power energy through integrated management of the production, consumption, sales and operation.

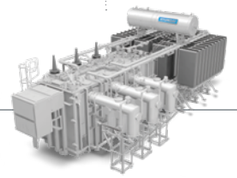
Asset Management Solution

Asset management solution is a business that maximizes the overall business efficiency by systematically managing the performance, risk, maintenance cost and others as well as by providing an asset management solution suitable to the customer's circumstance depending on the product lifecycle (PLC) of various products.

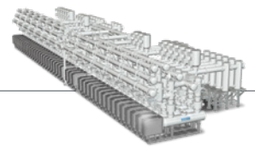
Generation(step-up)

Power Plants

- Supplied to more than 70 countries around the world for the past 45 over years since 1978
- Satisfies the various demands of customers through the acquisition of quality certifications from international accredited institute
- Participates in the world's key technical committee such as GIGRE and others, pioneering the establishment of technology standard related to power network



Power Transformer
· up to 800 kV, 1,500 MVA



Gas Insulated Switchgear
· up to 800 kV

- Can be installed in spaces smaller than the open type of substation by using SF_6 gas with outstanding insulation and arc extinguishing characteristics
- Manufacturing advanced products that are resistant to external environment and climate effects through the sealing at the charge part
- Extensive project experiences around the world
- Reduces installation period and cost due to simple installation and transportation, convenient maintenance
- Design considering the safety of the workers as priority

GREENTRIC



ECO Transformer
· up to 500 kV, 1,000 MVA

- Eco-friendly oil (Natural/Synthetic)
- Dynamic Pressure Resistant System(DPRS)
- Guarantees excellence in safety

GREENTRIC



SF_6 Free Gas Insulated Switchgear
· GIS for 170 kV 50 kA 60 Hz
· ECO GIS for 72.5 kV, 145 kV

- Korea's first solely developed technology
- Eco-friendly Novec mixture gas instead of SF_6 gas

Transmission

Primary Substation/Secondary Substation



Gas Insulated Switchgear
· up to 800 kV



Power Transformer
· up to 800 kV, 1,500 MVA



Shunt Reactor
· up to 765 kV, 300 Mvar

Distribution



Cubicle GIS
· up to 40.5 kV
· IEC, ANSI, GOST, UL

- Produces high quality products using angle-less type
- Multi-functional digital protection relay (HiMAP) applied
- High reliability secured, provides various operation information such as protection, measurement and control
- Firm external box, size and compact, making it safe
- Maintains high quality through stringent quality control system and continuous research and development

Motor & Generator



Synchronous Generator
· Rated Output: < 50 MVA
· Rated Voltage: < 15 kV
· Ploes: > 2P

Marine

Marine Solution

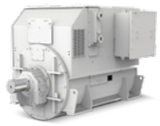
- Production of high quality marine devices satisfying the regulations and standards of key marine associations (LRS, ABS, DNV, GL, BV, NK etc.) and world's renowned institutes
- High quality safety secured through the latest equipment and stringent quality control system
- Realization of optimal high efficiency by converging SWGR, Generator, Motor, Telecom, Automation, Intelligent Motor Control Unit and others



Marine Switchgear



Marine Motor



Generator



Shaft Generator



Metal Clad Switchgear
· up to 38 kV
· IEC, ANSI, UL



Low Voltage Switchgear & Motor Control Center
· H8PU : 480 V, 3,200 A, 65 kA
· H5600 : 480 V, 1,200 A, 65 kA
· HiMCC : 480 V, 5,000 A, 100 kA



VCB
· IEC, ANSI, UL
· up to 36/38 kV, 50 kA, 4,000 A



ACB
· AC : up to 150 kA, 6,300 A
· DC : up to 60 kA, 4,000 A



MCCB
· AC : up to 150 kA, 1,600 A
· DC : up to 100 kA, 800 A



ELCB
· up to 85 kA, 800 A, 1,000 mA



MCCB / ELCB
· up to 10 kA, 100 A, 100 mA



MS / MC
· up to 800 A



MMS
· up to 100 kA, 80 A



Distribution Transformer
· up to 36 kV, 50 MVA



Cast Resin Transformer
· up to 36 kV, 20 MVA



VC
· up to 12 kV, 400 A



Switch-Disconnector
· DC : up to 100 kA/1 s, 4,000 A



Distribution Box
· Up to 36 ways
· Surface/Flush type



MCB
· up to 15 kA, 125 A



RCCB / RCBO
· up to 100 A, 500 mA



Isolator
· up to 125 A



Electronic Circuit Breaker
· up to 20 A



Fuse Links & Switches
· up to 1,250 A

- Can be applied to various places of demand due to various domestic and international certifications and marine certifications
- Full HG-Series line-up established
- High standard of breaking capacity and various product line offers production in any load environments
- Increased reliability with reinforced breaking performance of high-breaking MCCB
- Reinforced usage stability through internal reliability verification



Installation Contactor
· up to 63 A



Synchronous Motor
· Rated Output: < 40 MW
· Rated Voltage: < 15 kV
· Ploes: > 2P



MV&HV Induction Motor
· Rated Output: < 35 MW
· Rated Voltage: < 15 kV
· Poles: 2P-30P



HGMAP
· HGMAP Series



HGCAM
· HGCAM Series



SPD
· up to 200 kA
· AC, DC



HGDM



EOCR
· up to 60 A



LV Motor
· Rated Output: < 750 kW
· Rated Voltage: < 690 V
· Poles: 2P-12P

- Enhanced reliability and secured safety with production of products based on the world's best equipment and stringent quality system
- Realized high efficiency by selecting slot based on FEM
- Realized small and lightweight with optimal design based on FEM analysis method
- Satisfies the quality standards of international accredited institutes (IEC, IEEE, CSA, NEMA, API etc.)

HG-DC
Air Circuit Breakers

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DC ACB

Model Selection Table

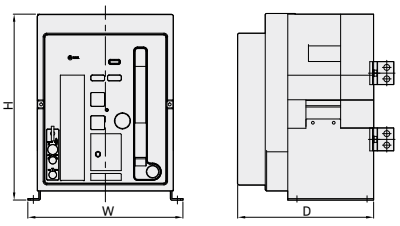
Air Circuit Breakers (HGN DC)

Model Name			HGN
Item			**I Frame**
Rated Current [In max]	Based on 40 ℃	A	16:1,600
			25:2,500
			32:3,200
			40:4,000
Rated Operational Voltage [Ue]		Vdc	1,000 (3P) / 1,500 (4P)
Rated Insulation Voltage [Ui]		Vdc	1,500
No. of Poles		P	3, 4
Current Setting Range (…× In max)		A	0.4 - 1.0
Service Short-Circuit Breaking Capacity (Ics)		%	100 %
Rated Breaking Capacity [Icu] [Sym]			
DC	1,000 V (3P)	kA	60
	1,500 V (4P)		60
Rated Service Short-Circuit Breaking Capacity [Ics] …%×Icu		kA	100 %
Rated Closing Current [Icm] [Peak]			
DC	1,000 V (3P)	kA	100
	1,500 V (4P)		100
Rated Short-Time withstand Voltage [Icw] (Without Inst)			
1 Second		kA	50 (3P) / 50 (4P)
Rated Impulse withstand Voltage [Uimp]		kV	12
Closing Operational Time			
Motor Charging Time (sec) max.			10
Closing Time (ms) max.			80
Lifecycle (Cycles)			
Mechanical			10,000
Electrical (at 1,000 Vdc)			3,000(500) [1)]
Weight			
3 Pole	Draw-Out Type	kg	95(115) [1)]
	Fixed Type		44(61) [1)]
4 Pole	Draw-Out Type		112(149) [1)]
	Fixed Type		55(80) [1)]
External Dimensions (W×H×D)			
3 Pole	Draw-Out Type	mm	399 × 480 × 368.4
	Fixed Type		408.4 × 404.4 × 295.8
4 Pole	Draw-Out Type		514 × 480 × 368.4
	Fixed Type		523.4 × 404.4 × 295.8

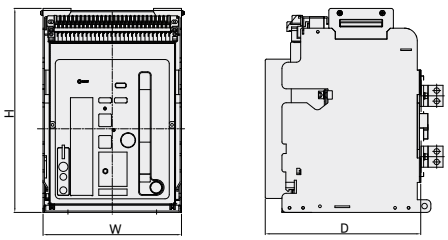
※ 1) 4,000 AF

Life time is the limit lifespan and is not the guaranteed lifespan. In case of maintenance, it is charged. In the event of abnormalities in accessories during use, it can be replaced. Quality Assurance : Based on IEC 60947-2's number of opening/closing within the warranty period.

Fixed Type



Draw-Out Type



Switch-Disconnector (HGN DC)

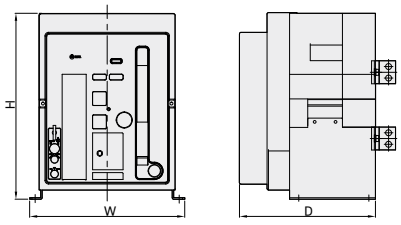
Model Name — HGN

Item			H Frame
Rated Current [In max]	Based on 40 ℃	A	08:800
			10:1,000
			12:1,250
			16:1,600
			20:2,000
			25:2,500
			32:3,200
			40:4,000
Rated Operational Voltage [Ue]		Vdc	1,500 (4P)
Rated Insulation Voltage [Ui]		Vdc	1,500
No. of Poles		P	4
Rated Closing Current [Icm] [Peak]			
DC	1,500 V (4P)	kA	100
Rated Short-Time withstand Voltage [Icw] (Without Inst)			
1 Second		kA	100
Rated Impulse withstand Voltage [Uimp]		kV	12
Closing Operational Time			
Motor Charging Time (sec) max.			10
Closing Time (ms) max.			80
Lifecycle (Cycles)			
Mechanical			10,000
Electrical (Time constant 2 ms)			3,000 (2,000) 1)
Weight			
4 Pole	Draw-Out Type	kg	112(149) 1)
	Fixed Type		55(80) 1)
External Dimensions (W×H×D)			
4 Pole	Draw-Out Type	mm	514 × 480 × 368.4
	Fixed Type		523.4 × 404.4 × 295.8

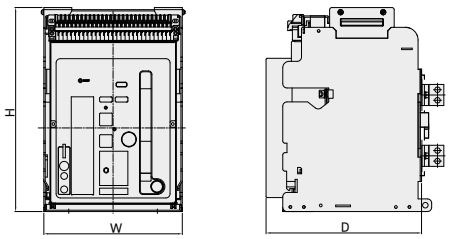
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Fixed Type



Draw-Out Type



Model Selection Table

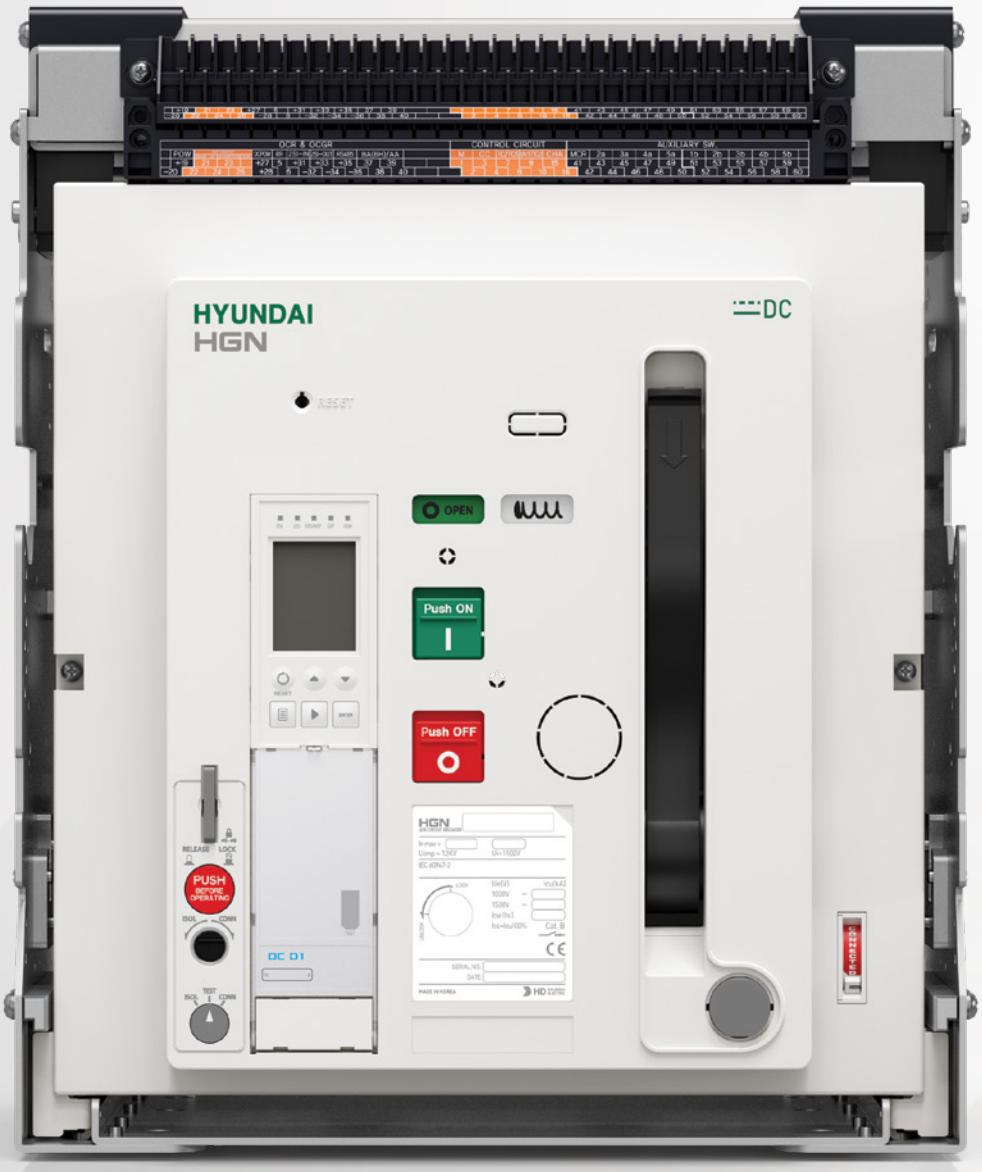
Accessories - Protective Relay (OCR)

Function	General Feeder
Model Name	**DC D1**
Control Power	
External Power	●
Self-Power	-
Protection Function	
LTD (Long Time)	●
STD (Short Time)	●
INST (Instantaneous)	●
Pre-Trip Alarm	●
Ground Fault Trip	●
ELT Function	-
Thermal Function	●
Field Test	●
Indication	
True RMS Detection Method	●
LED Indication per Trip Type	●
Fault LED	PTA, L, S/I, G
Real-Time LCD Indication of Load Rate per Phase	●
Measurement LCD	●
Output Contact	
Individual Continuous Contact (4a)	●
Option	
MCR	●
Communication	Modbus-RTU
Event/Fault Recording	●

HG Series

HGN DC Type Air Circuit Breakers

This product is suitable for protecting DC power/loads such as solar power, ESS, data centers, and wind power generators with a maximum operating voltage of DC 1,500 V (4P).



Maximized Insulation Performance

- Rated Insulation Voltage Ui : 1,000 V / 1,500 V
- Rated Impulse Withstand Voltage Uimp : 12 kV

Equipped with DC-specific Protection Relays (OCR) and Common Accessories

- Equipped with DC-specific protection relays for versatile applications in solar energy, ESS, and more.
- Temperature monitoring, accident recording, diagnostics, digital output (DO), communication capabilities, etc.
- Compatible with AC circuit breakers and various accessories like motors, coils, and more.

Specifications and Certifications

- IEC 60947-2 CB certification
- IEC 60947-3 CB certification



Air Circuit Breaker (I Frame)
Icu 60 kA @ 1,500 Vdc
Icw 50 kA/1 s



Switch-Disconnector (H Frame)
Icw 100 kA/1 s

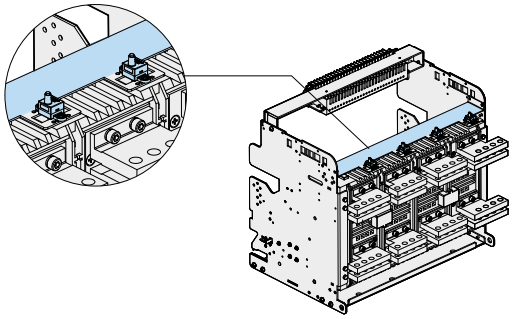
Overview and Characteristics

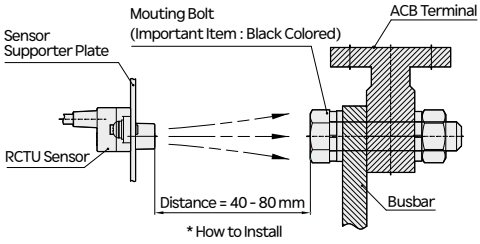
Air Circuit Breaker (HGN DC)

Equipped with Temperature Sensor

Reliable high temperature measurement is possible following the sensor measurement of the heat source (Range of measurement : -5 - 250 degrees)

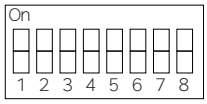
Example of IR Sensor Application



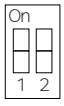


* How to Install

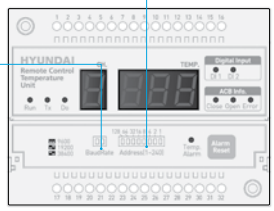
Address Setting : 1 - 240



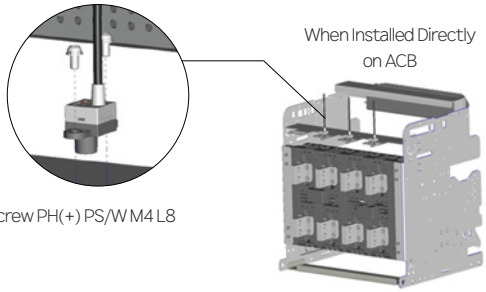
Baud Rate Setting

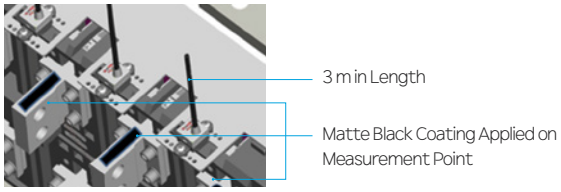


Set		Description
Off	Off	9,600
Off	On	19,200
On	Off	38,400



Installation of IR Sensor





- IR temperature sensor has to be installed with sufficient insulation distance from the point to be measured
- The recommended distance between the measurement point and temperature sensor is 50 - 80 mm.
- The measurement point has to be a surface without reflection due to the characteristics of the IR sensor and matte black painting is recommended for the measurement point.

⚠ Caution

1. The measurement value of the IR sensor differs depending on the reflection rate of the metal surface. Surfaces coated with matte black or surface with varnish excluding metallic varnish must be measured.
2. The size of measurement point differs depending on the D : S Ratio for the surface to be measured and the IR distance. This sensor has a ratio of 8 : 1.

Applied Standards and Certifications

HG-DC Series air circuit breaker has acquired testing/certifications from IEC 60947-2, 3 certified testing institute and can be installed and applied according to the usage environment and conditions permitted by the standards.

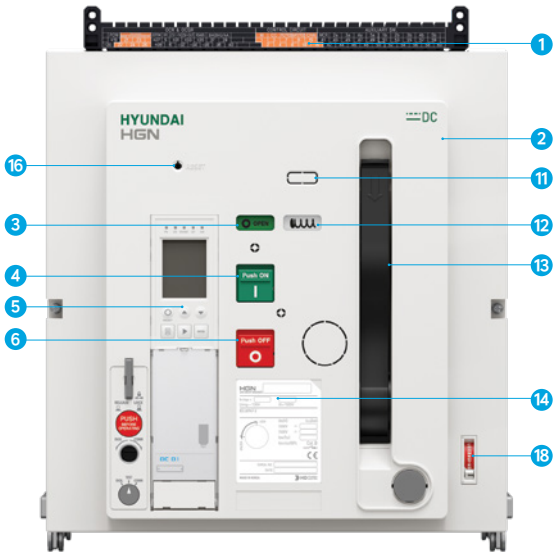
CE

Acquired Standards and Certifications

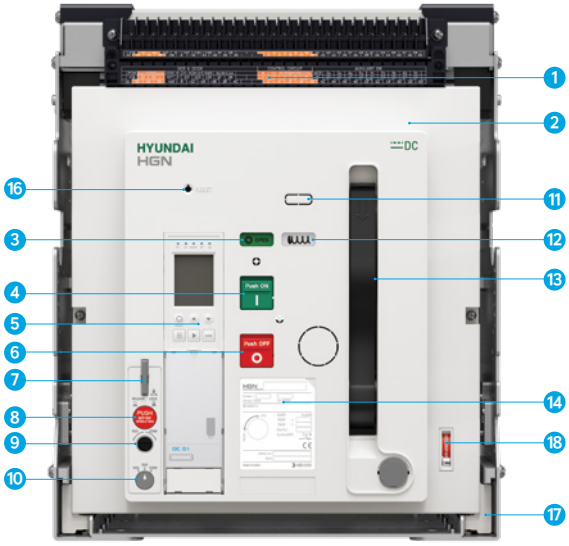
- CB Certification : IEC 60947-2, 3
- CE Mark

External·Internal Structure

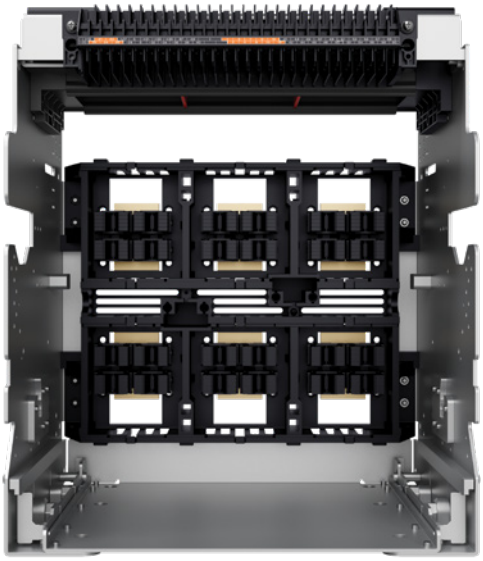
Air Circuit Breakers (HGN DC)



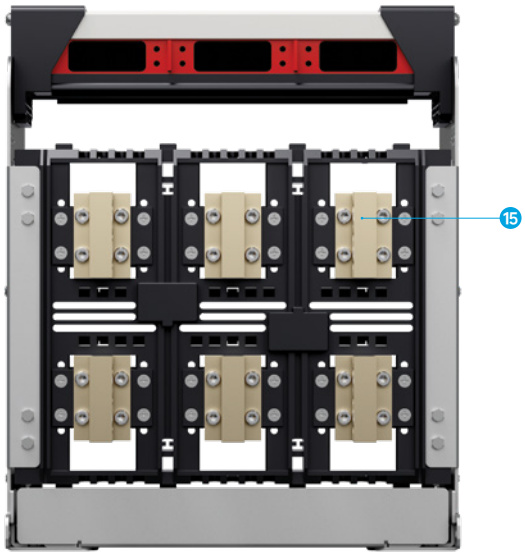
Draw Type



Draw Type (Include Cradle)

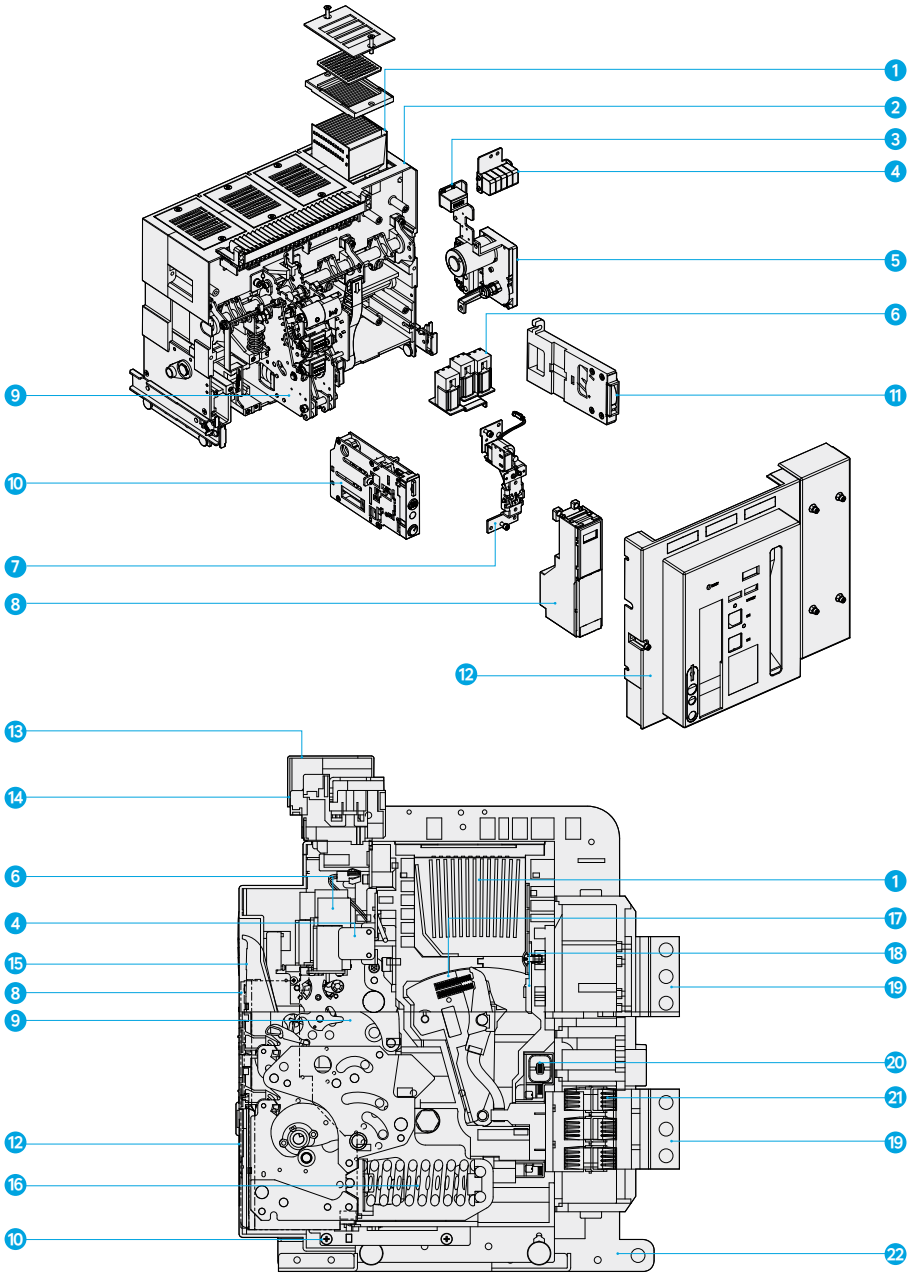


Cradle (Inside)



Cradle (Outside)

1. Control circuit terminal
2. Front cover
3. Close/Open Indicator
4. Close button
5. Protection relay (GPR)
6. Open button
7. Position padlock
8. Position lock release button
9. Draw-in/out handle insertion hole
10. Position indicator
11. Cycle counter
12. Charged/Discharged indicator
13. Charging handle
14. Name plate
15. Busbar terminal
16. OCR manual reset
17. Draw-out guide rail
18. Isol switch



1 DI Grid
2 CO Unit
3 Counter
4 AUX Switch
5 Motor
6 Closing/Trip/UVT coil
7 MHT device
8 OCR
9 Mechanism
10 DR Device
11 Isol Switch
12 Cover
13 Control Terminal Protection Cover
14 Control Terminal
15 Manual Charging Handle
16 Closing Spring
17 Moving Contact
18 Fixed Contact
19 Terminal
20 Current Transformer(CT)
21 Terminal Clip
22 Cradle

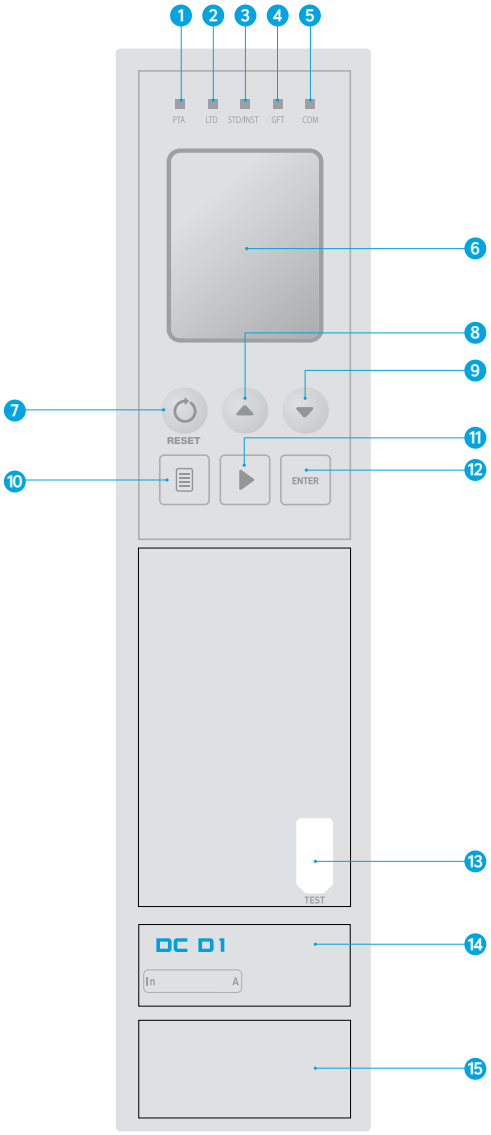
※ HG Series air circuit breaker has been designed so that upon closing, the N phase is closed earlier than R, S, T phase and upon opening, the N phase is disconnected last in order to reduce burden of main contact and to prevent ripple effect of accident of N phase.

Accessories

Over Current Relay (OCR)

External

Electronic trip unit - DC



1 PTA signal LED	6 LCD	11 Movement button
2 LTD signal LED	7 RESET button	12 Enter button
3 STD/INST signal LED	8 UP setting button	13 Temporary test
4 GFT signal LED	9 DOWN setting button	14 Model / Rated current
5 COMM. signal LED	10 LIST button	15 Battery / Rated current setting

※ The battery life is 10 years under normal conditions, and when it is time to replace it, please contact our customer support department for paid service. (Buy separately)

Appearance and Setting Value

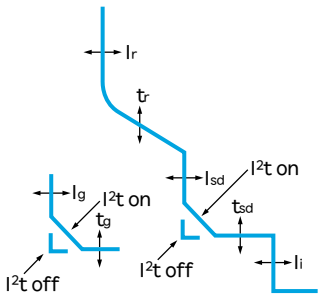


Rated Current Setting

800 AF	800 A
1,000 AF	800, 1,000 A
1,250 AF	800, 1,000, 1,250 A
1,600 AF	800, 1,000, 1,250, 1,600 A
2,000 AF	1,000, 1,250, 1,600, 2,000 A
2,500 AF	1,600, 2,000, 2,500 A
3,200 AF	1,600, 2,000, 2,500, 3,200 A
4,000 AF	2,000, 2,500, 3,200, 4,000 A

Key Functions

- Overload Protection
  - Long time delay
  - Thermal operation
- Short Circuit Protection
  - Short time delay, instantaneous trip
  - Selectable time-delay settings (definite/Inverse)
- Ground Fault Protection
  - Selectable time-delay settings (definite/Inverse)
- Over-Voltage, Under-Voltage, Imbalance, Reverse Power
- ZSI (Zone Selective Interlocking)
- Start-up Function
  - Selectable from short time delay, instantaneous, or ground fault protection
  - Adjustable start-up current and delay time settings
- Fault Recording
  - Records up to 128 fault information
  - Fault type, fault value, occurrence time of fault
- Event Recording
  - Records changes in device settings, operations, and status changes
  - Records up to 128 events of devices
- Load Profile
  - Records load usage every hour
- Diagnostic Function
  - Confirmation of trip device connection
  - Verification of rated current setting
  - Trip operation testing
- 4a DO (Digital Output)
  - Configuration of trip unit protection functions
- Communication Function
  - Modbus-RTU / RS-485
- Protection Features

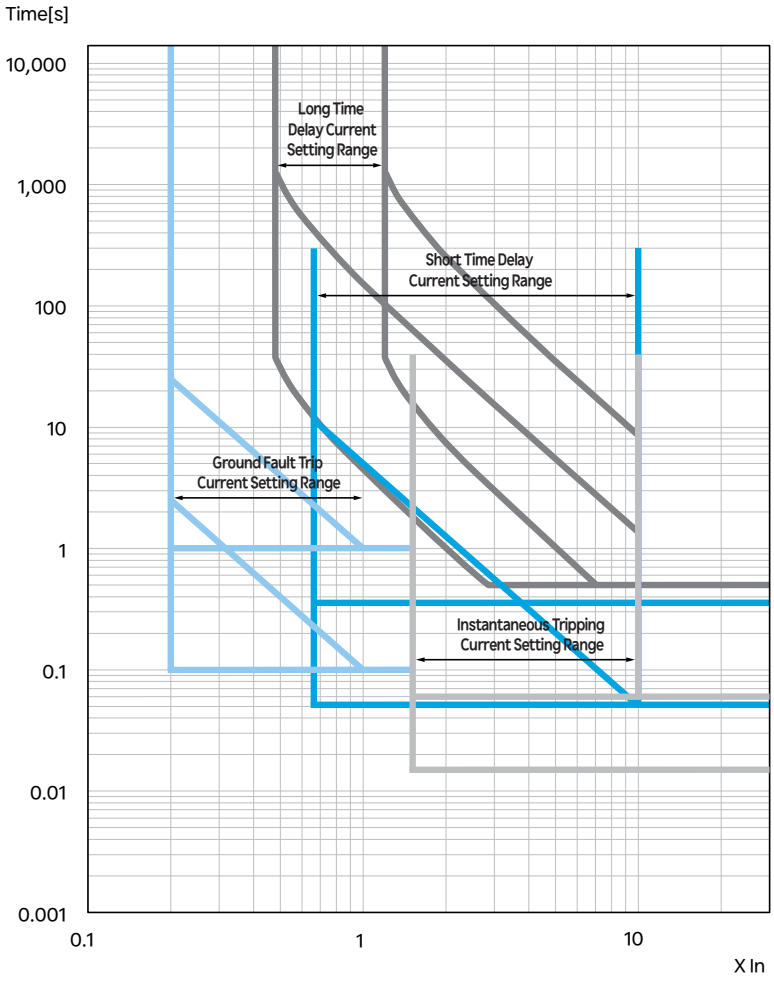


Accessories

Over Current Relay (OCR)

Characteristic Curve

General Feeder



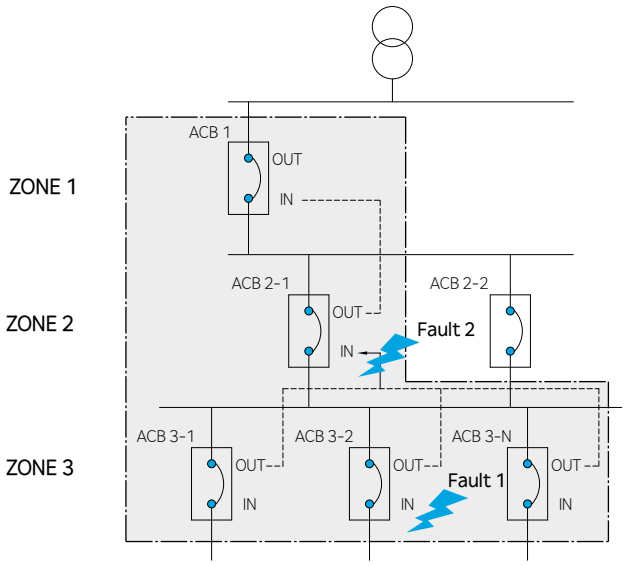
Zone Selective Interlocking System

Zone selective interlocking function involves the breaker in the fault sector detecting malfunction for selective protection coordination. The circuit breaker in the fault sector sends a lock signal to the back-up protection circuit breaker by force to delay the operation and the circuit breaker in the fault sector operates instantaneously to enable selective protection coordination of the system.

Examples

- Fault 1 Occurrence
  In case fault 1 occurs in selective zone (Zone 3), ACB 3-2 functions immediately and ACB 3-2 relay triggers a ZSI signal in ACB 2-1 relay to maintain a delay so that the circuit breaker does not function, preventing the expansion of fault region.

- Fault 2 Occurrence
  In case a fault occurs at the lower end of ACB 2-1, the circuit breaker in zone 3 region is unable to detect. Therefore, the relay of ACB 2-1 is unable to receive any signal from the lower end so, immediately, ACB 2-1 functions, triggering a ZSI signal to ACB 1 to prevent the expansion of fault range.



------- ZSI Connection

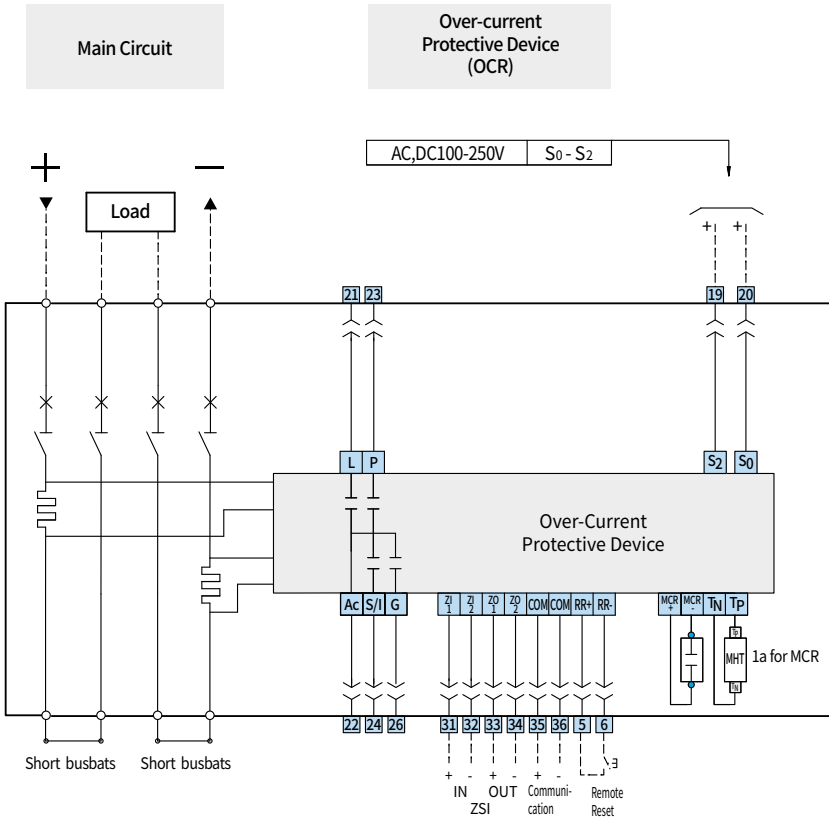
※ZSI connection wire's maximum distance has to be constituted within 3 m.
ZSI function cannot be operated during self-power.
Upon the release of the product, ZSI function is not activated and the 2 terminals (31, 32) at the control power input area are short so please cut the wire before connection when using ZSI.

Accessories

Over Current Relay (OCR)

Circuit Diagrams

DC



Time Chart for LTD Inverse Time (General Feeder)

Item	Tr / x Ir	LTD Time Setting (msec)									
		3,000	6,000	9,000	12,000	15,000	18,000	21,000	24,000	27,000	30,000
Overload Rate (%)	120	38,113	76,226	114,340	152,453	190,567	228,680	266,794	304,907	343,021	381,134
	125	30,938	61,876	92,814	123,753	154,691	185,629	216,567	247,506	278,444	309,382
	130	26,153	52,306	78,459	104,612	130,766	156,919	183,072	209,225	235,378	261,532
	135	22,656	45,313	67,969	90,626	113,283	135,939	158,596	181,253	203,909	226,566
	140	19,959	39,919	59,879	79,839	99,799	119,759	139,719	159,679	179,639	199,598
	145	17,803	35,607	53,411	71,215	89,018	106,822	124,626	142,430	160,233	178,037
	150	16,034	32,069	48,103	64,138	80,172	96,207	112,242	128,276	144,311	160,345
	155	14,554	29,108	43,662	58,216	72,770	87,324	101,878	116,432	130,987	145,541
	160	13,295	26,591	39,887	53,183	66,479	79,775	93,071	106,367	119,663	132,958
	165	12,213	24,426	36,639	48,852	61,065	73,278	85,491	97,704	109,917	122,130
	170	11,271	22,542	33,813	45,085	56,356	67,627	78,899	90,170	101,441	112,712
	175	10,445	20,890	31,335	41,780	52,225	62,670	73,115	83,560	94,005	104,450
	180	9,714	19,428	29,143	38,857	48,572	58,286	68,001	77,715	87,430	97,144
	185	9,064	18,128	27,193	36,257	45,321	54,386	63,450	72,514	81,579	90,643
	190	8,482	16,964	25,447	33,929	42,411	50,894	59,376	67,859	76,341	84,823
	195	7,958	15,917	23,876	31,834	39,793	47,752	55,711	63,669	71,628	79,587
	200	7,485	14,970	22,455	29,941	37,426	44,911	52,397	59,882	67,367	74,853
	205	7,055	14,111	21,166	28,222	35,277	42,333	49,389	56,444	63,500	70,555
	210	6,664	13,328	19,992	26,656	33,320	39,984	46,648	53,312	59,976	66,640
	215	6,305	12,611	18,917	25,223	31,529	37,835	44,141	50,447	56,753	63,059
	220	5,977	11,954	17,932	23,909	29,887	35,864	41,842	47,819	53,797	59,774
	225	5,675	11,350	17,025	22,701	28,376	34,051	39,726	45,402	51,077	56,752
	230	5,396	10,792	16,189	21,585	26,982	32,378	37,775	43,171	48,568	53,964
	235	5,138	10,277	15,416	20,554	25,693	30,832	35,970	41,109	46,248	51,386
	240	4,899	9,799	14,698	19,598	24,498	29,397	34,297	39,197	44,096	48,996
	245	4,677	9,355	14,032	18,710	23,388	28,065	32,743	37,420	42,098	46,776
	250	4,470	8,941	13,412	17,883	22,354	26,825	31,296	35,767	40,238	44,709
	255	4,278	8,556	12,834	17,112	21,390	25,668	29,946	34,224	38,502	42,781
	260	4,097	8,195	12,293	16,391	20,489	24,587	28,685	32,783	36,881	40,979
	265	3,929	7,858	11,787	15,717	19,646	23,575	27,505	31,434	35,363	39,292
	270	3,771	7,542	11,313	15,084	18,855	22,627	26,398	30,169	33,940	37,711
	275	3,622	7,245	10,868	14,490	18,113	21,736	25,359	28,981	32,604	36,227
	280	3,483	6,966	10,449	13,932	17,415	20,898	24,381	27,864	31,347	34,830
	285	3,351	6,703	10,054	13,406	16,758	20,109	23,461	26,813	30,164	33,516
	290	3,227	6,455	9,682	12,910	16,138	19,365	22,593	25,821	29,048	32,276
	295	3,110	6,221	9,331	12,442	15,553	18,663	21,774	24,885	27,995	31,106
	300	3,000	6,000	9,000	12,000	15,000	18,000	21,000	24,000	27,000	30,000
	305	2,895	5,790	8,685	11,581	14,476	17,371	20,267	23,162	26,057	28,953
	310	2,796	5,592	8,388	11,184	13,980	16,776	19,573	22,369	25,165	27,961
	315	2,702	5,404	8,106	10,808	13,510	16,212	18,914	21,616	24,318	27,020
	320	2,612	5,225	7,838	10,451	13,064	15,676	18,289	20,902	23,515	26,128
	325	2,527	5,055	7,583	10,111	12,639	15,167	17,695	20,223	22,751	25,279
	330	2,447	4,894	7,341	9,789	12,236	14,683	17,131	19,578	22,025	24,472
	335	2,370	4,740	7,111	9,481	11,852	14,222	16,593	18,963	21,334	23,704
	340	2,297	4,594	6,891	9,189	11,486	13,783	16,081	18,378	20,675	22,973
	345	2,227	4,455	6,682	8,910	11,137	13,365	15,592	17,820	20,047	22,275
	350	2,160	4,321	6,482	8,643	10,804	12,965	15,126	17,287	19,448	21,609

Accessories

Over Current Relay (OCR)

Time Chart for LTD Inverse Time (General Feeder)

Item	Tr / x Ir	LTD Time Setting (msec)									
		3,000	6,000	9,000	12,000	15,000	18,000	21,000	24,000	27,000	30,000
Overload Rate (%)	355	2,097	4,194	6,291	8,389	10,486	12,583	14,681	16,778	18,875	20,973
	360	2,036	4,073	6,109	8,146	10,182	12,219	14,255	16,292	18,328	20,365
	365	1,978	3,956	5,935	7,913	9,892	11,870	13,848	15,827	17,805	19,784
	370	1,922	3,845	5,768	7,691	9,613	11,536	13,459	15,382	17,304	19,227
	375	1,869	3,738	5,608	7,477	9,347	11,216	13,086	14,955	16,825	18,694
	380	1,818	3,636	5,455	7,273	9,091	10,910	12,728	14,547	16,365	18,183
	385	1,769	3,538	5,308	7,077	8,847	10,616	12,385	14,155	15,924	17,694
	390	1,722	3,444	5,167	6,889	8,612	10,334	12,056	13,779	15,501	17,224
	395	1,677	3,354	5,031	6,709	8,386	10,063	11,740	13,418	15,095	16,772
	400	1,633	3,267	4,901	6,535	8,169	9,803	11,437	13,071	14,705	16,339
	405	1,592	3,184	4,776	6,369	7,961	9,553	11,145	12,738	14,330	15,922
	410	1,552	3,104	4,656	6,208	7,760	9,313	10,865	12,417	13,969	15,521
	415	1,513	3,027	4,540	6,054	7,568	9,081	10,595	12,108	13,622	15,136
	420	1,476	2,952	4,429	5,905	7,382	8,858	10,335	11,811	13,288	14,764
	425	1,440	2,881	4,322	5,762	7,203	8,644	10,085	11,525	12,966	14,407
	430	1,406	2,812	4,218	5,625	7,031	8,437	9,843	11,250	12,656	14,062
	435	1,373	2,746	4,119	5,492	6,865	8,238	9,611	10,984	12,357	13,730
	440	1,340	2,681	4,022	5,363	6,704	8,045	9,386	10,727	12,068	13,409
	445	1,310	2,620	3,930	5,240	6,550	7,860	9,170	10,480	11,790	13,100
	450	1,280	2,560	3,840	5,120	6,400	7,681	8,961	10,241	11,521	12,801
	455	1,251	2,502	3,754	5,005	6,256	7,508	8,759	10,010	11,262	12,513
	460	1,223	2,446	3,670	4,893	6,117	7,340	8,564	9,787	11,011	12,234
	465	1,196	2,393	3,589	4,786	5,982	7,179	8,375	9,572	10,768	11,965
	470	1,170	2,340	3,511	4,681	5,852	7,022	8,193	9,363	10,534	11,704
	475	1,145	2,290	3,435	4,581	5,726	6,871	8,017	9,162	10,307	11,452
	480	1,120	2,241	3,362	4,483	5,604	6,725	7,846	8,967	10,088	11,209
	485	1,097	2,194	3,291	4,389	5,486	6,583	7,681	8,778	9,875	10,972
	490	1,074	2,148	3,223	4,297	5,372	6,446	7,520	8,595	9,669	10,744
	495	1,052	2,104	3,156	4,209	5,261	6,313	7,365	8,418	9,470	10,522
	500	1,030	2,061	3,092	4,123	5,154	6,184	7,215	8,246	9,277	10,308
	505	1,009	2,019	3,029	4,039	5,049	6,059	7,069	8,079	9,089	10,099
	510	989	1,979	2,969	3,959	4,949	5,938	6,928	7,918	8,908	9,898
	515	970	1,940	2,910	3,880	4,851	5,821	6,791	7,761	8,732	9,702
	520	951	1,902	2,853	3,804	4,756	5,707	6,658	7,609	8,561	9,512
	525	932	1,865	2,798	3,731	4,663	5,596	6,529	7,462	8,395	9,327
	530	914	1,829	2,744	3,659	4,574	5,489	6,404	7,319	8,233	9,148
	535	897	1,794	2,692	3,589	4,487	5,384	6,282	7,179	8,077	8,974
	540	880	1,761	2,641	3,522	4,402	5,283	6,164	7,044	7,925	8,805
	545	864	1,728	2,592	3,456	4,320	5,184	6,049	6,913	7,777	8,641
	550	848	1,696	2,544	3,392	4,240	5,089	5,937	6,785	7,633	8,481
	555	832	1,665	2,498	3,330	4,163	4,996	5,828	6,661	7,494	8,326
	560	817	1,635	2,452	3,270	4,087	4,905	5,723	6,540	7,358	8,175
	565	802	1,605	2,408	3,211	4,014	4,817	5,620	6,423	7,226	8,028
	570	788	1,577	2,365	3,154	3,942	4,731	5,520	6,308	7,097	7,885
	575	774	1,549	2,324	3,098	3,873	4,648	5,422	6,197	6,972	7,746
	580	761	1,522	2,283	3,044	3,805	4,566	5,327	6,089	6,850	7,611
	585	747	1,495	2,243	2,991	3,739	4,487	5,235	5,983	6,731	7,479

Time Chart for LTD Inverse Time (General Feeder)

Item	x Ir \ Tr	LTD Time Setting (msec)									
		3,000	6,000	9,000	12,000	15,000	18,000	21,000	24,000	27,000	30,000
Overload Rate (%)	590	735	1,470	2,205	2,940	3,675	4,410	5,145	5,880	6,615	7,350
	595	722	1,445	2,167	2,890	3,612	4,335	5,058	5,780	6,503	7,225
	600	710	1,420	2,131	2,841	3,551	4,262	4,972	5,683	6,393	7,103
	605	698	1,396	2,095	2,793	3,492	4,190	4,889	5,587	6,286	6,984
	610	686	1,373	2,060	2,747	3,434	4,121	4,808	5,495	6,182	6,868
	615	675	1,351	2,026	2,702	3,377	4,053	4,729	5,404	6,080	6,755
	620	664	1,329	1,993	2,658	3,322	3,987	4,651	5,316	5,981	6,645
	625	653	1,307	1,961	2,615	3,268	3,922	4,576	5,230	5,884	6,537
	630	643	1,286	1,929	2,573	3,216	3,859	4,503	5,146	5,789	6,433
	635	633	1,266	1,899	2,532	3,165	3,798	4,431	5,064	5,697	6,330
	640	623	1,246	1,869	2,492	3,115	3,738	4,361	4,984	5,607	6,230
	645	613	1,226	1,839	2,453	3,066	3,679	4,293	4,906	5,519	6,132
	650	603	1,207	1,811	2,414	3,018	3,622	4,226	4,829	5,433	6,037
	655	594	1,188	1,783	2,377	2,972	3,566	4,161	4,755	5,349	5,944
	660	585	1,170	1,756	2,341	2,926	3,512	4,097	4,682	5,268	5,853
	665	576	1,152	1,729	2,305	2,882	3,458	4,035	4,611	5,187	5,764
	670	567	1,135	1,703	2,271	2,838	3,406	3,974	4,542	5,109	5,677
	675	559	1,118	1,677	2,237	2,796	3,355	3,914	4,474	5,033	5,592
	680	550	1,101	1,652	2,203	2,754	3,305	3,856	4,407	4,958	5,509
	685	542	1,085	1,628	2,171	2,714	3,257	3,799	4,342	4,885	5,428
	690	534	1,069	1,604	2,139	2,674	3,209	3,744	4,279	4,814	5,348
	695	527	1,054	1,581	2,108	2,635	3,162	3,689	4,217	4,744	5,271
	700	519	1,039	1,558	2,078	2,597	3,117	3,636	4,156	4,675	5,195
	705	512	1,024	1,536	2,048	2,560	3,072	3,584	4,096	4,608	5,120
	710	504	1,009	1,514	2,019	2,524	3,028	3,533	4,038	4,543	5,048
	715	497	995	1,493	1,990	2,488	2,986	3,483	3,981	4,479	4,977
	720	490	981	1,472	1,962	2,453	2,944	3,435	3,925	4,416	4,907
	725	483	967	1,451	1,935	2,419	2,903	3,387	3,871	4,355	4,839
	730	477	954	1,431	1,908	2,386	2,863	3,340	3,817	4,295	4,772
	735	470	941	1,412	1,882	2,353	2,824	3,294	3,765	4,236	4,706
	740	464	928	1,392	1,857	2,321	2,785	3,249	3,714	4,178	4,642
	745	457	915	1,373	1,831	2,289	2,747	3,205	3,663	4,121	4,579
	750	451	903	1,355	1,807	2,259	2,711	3,162	3,614	4,066	4,518
	755	445	891	1,337	1,783	2,229	2,674	3,120	3,566	4,012	4,458
	760	439	879	1,319	1,759	2,199	2,639	3,079	3,519	3,959	4,398
	765	434	868	1,302	1,736	2,170	2,604	3,038	3,472	3,906	4,341
	770	428	856	1,285	1,713	2,142	2,570	2,998	3,427	3,855	4,284
	775	422	845	1,268	1,691	2,114	2,537	2,960	3,382	3,805	4,228
	780	417	834	1,252	1,669	2,087	2,504	2,921	3,339	3,756	4,174
	785	412	824	1,236	1,648	2,060	2,472	2,884	3,296	3,708	4,120
	790	406	813	1,220	1,627	2,033	2,440	2,847	3,254	3,661	4,067
	795	401	803	1,204	1,606	2,008	2,409	2,811	3,213	3,614	4,016
	800	396	793	1,189	1,586	1,982	2,379	2,776	3,172	3,569	3,965
	805	391	783	1,174	1,566	1,958	2,349	2,741	3,133	3,524	3,916
	810	386	773	1,160	1,547	1,933	2,320	2,707	3,094	3,480	3,867
	815	381	763	1,145	1,527	1,909	2,291	2,673	3,055	3,437	3,819
	820	377	754	1,131	1,509	1,886	2,263	2,641	3,018	3,395	3,773

Accessories

Over Current Relay (OCR)

Time Chart for LTD Inverse Time (General Feeder)

Item	x Ir \ Tr	LTD Time Setting (msec)									
		3,000	6,000	9,000	12,000	15,000	18,000	21,000	24,000	27,000	30,000
Overload Rate (%)	825	372	745	1,118	1,490	1,863	2,236	2,608	2,981	3,354	3,727
	830	368	736	1,104	1,472	1,840	2,209	2,577	2,945	3,313	3,681
	835	363	727	1,091	1,455	1,818	2,182	2,546	2,910	3,273	3,637
	840	359	718	1,078	1,437	1,797	2,156	2,515	2,875	3,234	3,594
	845	355	710	1,065	1,420	1,775	2,130	2,485	2,841	3,196	3,551
	850	350	701	1,052	1,403	1,754	2,105	2,456	2,807	3,158	3,509
	855	346	693	1,040	1,387	1,733	2,080	2,427	2,774	3,121	3,467
	860	342	685	1,028	1,370	1,713	2,056	2,399	2,741	3,084	3,427
	865	338	677	1,016	1,355	1,693	2,032	2,371	2,710	3,048	3,387
	870	334	669	1,004	1,339	1,674	2,009	2,343	2,678	3,013	3,348
	875	330	661	992	1,323	1,654	1,985	2,316	2,647	2,978	3,309
	880	327	654	981	1,308	1,636	1,963	2,290	2,617	2,944	3,272
	885	323	647	970	1,294	1,617	1,941	2,264	2,588	2,911	3,235
	890	319	639	959	1,279	1,599	1,919	2,238	2,558	2,878	3,198
	895	316	632	948	1,265	1,581	1,897	2,213	2,530	2,846	3,162
	900	312	625	938	1,250	1,563	1,876	2,189	2,501	2,814	3,127
	905	309	618	927	1,237	1,546	1,855	2,164	2,474	2,783	3,092
	910	305	611	917	1,223	1,529	1,835	2,140	2,446	2,752	3,058
	915	302	604	907	1,209	1,512	1,814	2,117	2,419	2,722	3,024
	920	299	598	897	1,196	1,495	1,795	2,094	2,393	2,692	2,991
	925	295	591	887	1,183	1,479	1,775	2,071	2,367	2,663	2,959
	930	292	585	878	1,170	1,463	1,756	2,049	2,341	2,634	2,927
	935	289	579	868	1,158	1,447	1,737	2,027	2,316	2,606	2,895
	940	286	572	859	1,145	1,432	1,718	2,005	2,291	2,578	2,864
	945	283	566	850	1,133	1,417	1,700	1,984	2,267	2,551	2,834
	950	280	560	841	1,121	1,402	1,682	1,963	2,243	2,524	2,804
	955	277	555	832	1,110	1,387	1,665	1,942	2,220	2,497	2,775
	960	274	549	823	1,098	1,373	1,647	1,922	2,196	2,471	2,746
	965	271	543	815	1,087	1,358	1,630	1,902	2,174	2,445	2,717
	970	268	537	806	1,075	1,344	1,613	1,882	2,151	2,420	2,689
	975	266	532	798	1,064	1,330	1,597	1,863	2,129	2,395	2,661
	980	263	526	790	1,053	1,317	1,580	1,844	2,107	2,370	2,634
	985	260	521	782	1,043	1,303	1,564	1,825	2,086	2,346	2,607
	990	258	516	774	1,032	1,290	1,548	1,806	2,064	2,323	2,581
	995	255	511	766	1,022	1,277	1,533	1,788	2,044	2,299	2,555
	1,000	252	505	758	1,011	1,264	1,517	1,770	2,023	2,276	2,529

Time Chart for STD Inverse Time (General Feeder)

Item	Tsd / x In	STD Time Setting (ms) Inland Type															
		50	70	90	110	130	150	170	190	210	230	250	270	290	310	330	350
Overload Rate (%)	0.66	11,478	16,069	20,661	25,252	29,843	34,435	39,026	43,617	48,209	52,800	57,392	61,983	66,574	71,166	75,757	80,348
	0.7	10,204	14,285	18,367	22,448	26,530	30,612	34,693	38,775	42,857	46,938	51,020	55,102	59,183	63,265	67,346	71,428
	0.75	8,888	12,444	16,000	19,555	23,111	26,666	30,222	33,777	37,333	40,888	44,444	48,000	51,555	55,111	58,666	62,222
	0.8	7,812	10,937	14,062	17,187	20,312	23,437	26,562	29,687	32,812	35,937	39,062	42,187	45,312	48,437	51,562	54,687
	0.85	6,920	9,688	12,456	15,224	17,993	20,761	23,529	26,297	29,065	31,833	34,602	37,370	40,138	42,906	45,674	48,442
	0.9	6,172	8,641	11,111	13,580	16,049	18,518	20,987	23,456	25,925	28,395	30,864	33,333	35,802	38,271	40,740	43,209
	0.95	5,540	7,756	9,972	12,188	14,404	16,620	18,836	21,052	23,268	25,484	27,700	29,916	32,132	34,349	36,565	38,781
	1	5,000	7,000	9,000	11,000	13,000	15,000	17,000	19,000	21,000	23,000	25,000	27,000	29,000	31,000	33,000	35,000
	1.05	4,535	6,349	8,163	9,977	11,791	13,605	15,419	17,233	19,047	20,861	22,675	24,489	26,303	28,117	29,931	31,746
	1.1	4,132	5,785	7,438	9,090	10,743	12,396	14,049	15,702	17,355	19,008	20,661	22,314	23,966	25,619	27,272	28,925
	1.15	3,780	5,293	6,805	8,317	9,829	11,342	12,854	14,366	15,879	17,391	18,903	20,415	21,928	23,440	24,952	26,465
	1.2	3,472	4,861	6,250	7,638	9,027	10,416	11,805	13,194	14,583	15,972	17,361	18,750	20,138	21,527	22,916	24,305
	1.25	3,200	4,480	5,760	7,040	8,320	9,600	10,880	12,160	13,440	14,720	16,000	17,280	18,560	19,840	21,120	22,400
	1.3	2,958	4,142	5,325	6,508	7,692	8,875	10,059	11,242	12,426	13,609	14,792	15,976	17,159	18,343	19,526	20,710
	1.35	2,743	3,840	4,938	6,035	7,133	8,230	9,327	10,425	11,522	12,620	13,717	14,814	15,912	17,009	18,106	19,204
	1.4	2,551	3,571	4,591	5,612	6,632	7,653	8,673	9,693	10,714	11,734	12,755	13,775	14,795	15,816	16,836	17,857
	1.45	2,378	3,329	4,280	5,231	6,183	7,134	8,085	9,036	9,988	10,939	11,890	12,841	13,793	14,744	15,695	16,646
	1.5	2,222	3,111	4,000	4,888	5,777	6,666	7,555	8,444	9,333	10,222	11,111	12,000	12,888	13,777	14,666	15,555
	1.55	2,081	2,913	3,746	4,578	5,411	6,243	7,075	7,908	8,740	9,573	10,405	11,238	12,070	12,903	13,735	14,568
	1.6	1,953	2,734	3,515	4,296	5,078	5,859	6,640	7,421	8,203	8,984	9,765	10,546	11,328	12,109	12,890	13,671
	1.65	1,836	2,571	3,305	4,040	4,775	5,509	6,244	6,978	7,713	8,448	9,182	9,917	10,651	11,386	12,121	12,855
	1.7	1,730	2,422	3,114	3,806	4,498	5,190	5,882	6,574	7,266	7,958	8,650	9,342	10,034	10,726	11,418	12,110
	1.75	1,632	2,285	2,938	3,591	4,244	4,897	5,551	6,204	6,857	7,510	8,163	8,816	9,469	10,122	10,775	11,428
	1.8	1,543	2,160	2,777	3,395	4,012	4,629	5,246	5,864	6,481	7,098	7,716	8,333	8,950	9,567	10,185	10,802
	1.85	1,460	2,045	2,629	3,214	3,798	4,382	4,967	5,551	6,135	6,720	7,304	7,888	8,473	9,057	9,642	10,226
	1.9	1,385	1,939	2,493	3,047	3,601	4,155	4,709	5,263	5,817	6,371	6,925	7,479	8,033	8,587	9,141	9,695
	1.95	1,314	1,840	2,366	2,892	3,418	3,944	4,470	4,996	5,522	6,048	6,574	7,100	7,626	8,152	8,678	9,204
	2	1,250	1,750	2,250	2,750	3,250	3,750	4,250	4,750	5,250	5,750	6,250	6,750	7,250	7,750	8,250	8,750
	2.05	1,189	1,665	2,141	2,617	3,093	3,569	4,045	4,521	4,997	5,472	5,948	6,424	6,900	7,376	7,852	8,328
	2.1	1,133	1,587	2,040	2,494	2,947	3,401	3,854	4,308	4,761	5,215	5,668	6,122	6,575	7,029	7,482	7,936
	2.15	1,081	1,514	1,946	2,379	2,812	3,244	3,677	4,110	4,542	4,975	5,408	5,840	6,273	6,706	7,138	7,571
	2.2	1,033	1,446	1,859	2,272	2,685	3,099	3,512	3,925	4,338	4,752	5,165	5,578	5,991	6,404	6,818	7,231
	2.25	987	1,382	1,777	2,172	2,567	2,962	3,358	3,753	4,148	4,543	4,938	5,333	5,728	6,123	6,518	6,913
	2.3	945	1,323	1,701	2,079	2,457	2,835	3,213	3,591	3,969	4,347	4,725	5,103	5,482	5,860	6,238	6,616
	2.35	905	1,267	1,629	1,991	2,354	2,716	3,078	3,440	3,802	4,164	4,526	4,889	5,251	5,613	5,975	6,337
	2.4	868	1,215	1,562	1,909	2,256	2,604	2,951	3,298	3,645	3,993	4,340	4,687	5,034	5,381	5,729	6,076
	2.45	832	1,166	1,499	1,832	2,165	2,498	2,832	3,165	3,498	3,831	4,164	4,498	4,831	5,164	5,497	5,830
	2.5	800	1,120	1,440	1,760	2,080	2,400	2,720	3,040	3,360	3,680	4,000	4,320	4,640	4,960	5,280	5,600
	2.55	768	1,076	1,384	1,691	1,999	2,306	2,614	2,921	3,229	3,537	3,844	4,152	4,459	4,767	5,074	5,382
	2.6	739	1,035	1,331	1,627	1,923	2,218	2,514	2,810	3,106	3,402	3,698	3,994	4,289	4,585	4,881	5,177
	2.65	711	996	1,281	1,566	1,851	2,135	2,420	2,705	2,990	3,275	3,559	3,844	4,129	4,414	4,699	4,983
	2.7	685	960	1,234	1,508	1,783	2,057	2,331	2,606	2,880	3,155	3,429	3,703	3,978	4,252	4,526	4,801
	2.75	661	925	1,190	1,454	1,719	1,983	2,247	2,512	2,776	3,041	3,305	3,570	3,834	4,099	4,363	4,628
	2.8	637	892	1,147	1,403	1,658	1,913	2,168	2,423	2,678	2,933	3,188	3,443	3,698	3,954	4,209	4,464
	2.85	615	861	1,108	1,354	1,600	1,846	2,092	2,339	2,585	2,831	3,077	3,324	3,570	3,816	4,062	4,309
	2.9	594	832	1,070	1,307	1,545	1,783	2,021	2,259	2,497	2,734	2,972	3,210	3,448	3,686	3,923	4,161
	2.95	574	804	1,034	1,264	1,493	1,723	1,953	2,183	2,413	2,642	2,872	3,102	3,332	3,562	3,792	4,021

Accessories

Over Current Relay (OCR)

Time Chart for STD Inverse Time (General Feeder)

Item	Tsd / x In	STD Time Setting (ms) Inland Type															
		50	70	90	110	130	150	170	190	210	230	250	270	290	310	330	350
Overload Rate (%)	3	555	777	1,000	1,222	1,444	1,666	1,888	2,111	2,333	2,555	2,777	3,000	3,222	3,444	3,666	3,888
	3.05	537	752	967	1,182	1,397	1,612	1,827	2,042	2,257	2,472	2,687	2,902	3,117	3,332	3,547	3,762
	3.1	520	728	936	1,144	1,352	1,560	1,768	1,977	2,185	2,393	2,601	2,809	3,017	3,225	3,433	3,642
	3.15	503	705	907	1,108	1,310	1,511	1,713	1,914	2,116	2,317	2,519	2,721	2,922	3,124	3,325	3,527
	3.2	488	683	878	1,074	1,269	1,464	1,660	1,855	2,050	2,246	2,441	2,636	2,832	3,027	3,222	3,417
	3.25	473	662	852	1,041	1,230	1,420	1,609	1,798	1,988	2,177	2,366	2,556	2,745	2,934	3,124	3,313
	3.3	459	642	826	1,010	1,193	1,377	1,561	1,744	1,928	2,112	2,295	2,479	2,662	2,846	3,030	3,213
	3.35	445	623	801	980	1,158	1,336	1,514	1,693	1,871	2,049	2,227	2,405	2,584	2,762	2,940	3,118
	3.4	432	605	778	951	1,124	1,297	1,470	1,643	1,816	1,989	2,162	2,335	2,508	2,681	2,854	3,027
	3.45	420	588	756	924	1,092	1,260	1,428	1,596	1,764	1,932	2,100	2,268	2,436	2,604	2,772	2,940
	3.5	408	571	734	897	1,061	1,224	1,387	1,551	1,714	1,877	2,040	2,204	2,367	2,530	2,693	2,857
	3.55	396	555	714	872	1,031	1,190	1,348	1,507	1,666	1,825	1,983	2,142	2,301	2,459	2,618	2,777
	3.6	385	540	694	848	1,003	1,157	1,311	1,466	1,620	1,774	1,929	2,083	2,237	2,391	2,546	2,700
	3.65	375	525	675	825	975	1,125	1,276	1,426	1,576	1,726	1,876	2,026	2,176	2,326	2,477	2,627
	3.7	365	511	657	803	949	1,095	1,241	1,387	1,533	1,680	1,826	1,972	2,118	2,264	2,410	2,556
	3.75	355	497	640	782	924	1,066	1,208	1,351	1,493	1,635	1,777	1,920	2,062	2,204	2,346	2,488
	3.8	346	484	623	761	900	1,038	1,177	1,315	1,454	1,592	1,731	1,869	2,008	2,146	2,285	2,423
	3.85	337	472	607	742	877	1,011	1,146	1,281	1,416	1,551	1,686	1,821	1,956	2,091	2,226	2,361
	3.9	328	460	591	723	854	986	1,117	1,249	1,380	1,512	1,643	1,775	1,906	2,038	2,169	2,301
	3.95	320	448	576	705	833	961	1,089	1,217	1,345	1,474	1,602	1,730	1,858	1,986	2,115	2,243
	4	312	437	562	687	812	937	1,062	1,187	1,312	1,437	1,562	1,687	1,812	1,937	2,062	2,187
	4.05	304	426	548	670	792	914	1,036	1,158	1,280	1,402	1,524	1,646	1,768	1,889	2,011	2,133
	4.1	297	416	535	654	773	892	1,011	1,130	1,249	1,368	1,487	1,606	1,725	1,844	1,963	2,082
	4.15	290	406	522	638	754	870	987	1,103	1,219	1,335	1,451	1,567	1,683	1,799	1,916	2,032
	4.2	283	396	510	623	736	850	963	1,077	1,190	1,303	1,417	1,530	1,643	1,757	1,870	1,984
	4.25	276	387	498	608	719	830	941	1,051	1,162	1,273	1,384	1,494	1,605	1,716	1,826	1,937
	4.3	270	378	486	594	703	811	919	1,027	1,135	1,243	1,352	1,460	1,568	1,676	1,784	1,892
	4.35	264	369	475	581	687	792	898	1,004	1,109	1,215	1,321	1,426	1,532	1,638	1,743	1,849
	4.4	258	361	464	568	671	774	878	981	1,084	1,188	1,291	1,394	1,497	1,601	1,704	1,807
	4.45	252	353	454	555	656	757	858	959	1,060	1,161	1,262	1,363	1,464	1,565	1,666	1,767
	4.5	246	345	444	543	641	740	839	938	1,037	1,135	1,234	1,333	1,432	1,530	1,629	1,728
	4.55	241	338	434	531	627	724	821	917	1,014	1,110	1,207	1,304	1,400	1,497	1,594	1,690
	4.6	236	330	425	519	614	708	803	897	992	1,086	1,181	1,275	1,370	1,465	1,559	1,654
	4.65	231	323	416	508	601	693	786	878	971	1,063	1,156	1,248	1,341	1,433	1,526	1,618
	4.7	226	316	407	497	588	679	769	860	950	1,041	1,131	1,222	1,312	1,403	1,493	1,584
	4.75	221	310	398	487	576	664	753	842	930	1,019	1,108	1,196	1,285	1,373	1,462	1,551
	4.8	217	303	390	477	564	651	737	824	911	998	1,085	1,171	1,258	1,345	1,432	1,519
	4.85	212	297	382	467	552	637	722	807	892	977	1,062	1,147	1,232	1,317	1,402	1,487
	4.9	208	291	374	458	541	624	708	791	874	957	1,041	1,124	1,207	1,291	1,374	1,457
	4.95	204	285	367	448	530	612	693	775	857	938	1,020	1,101	1,183	1,265	1,346	1,428
	5	200	280	360	440	520	600	680	760	840	920	1,000	1,080	1,160	1,240	1,320	1,400
	5.05	196	274	352	431	509	588	666	745	823	901	980	1,058	1,137	1,215	1,293	1,372
	5.1	192	269	346	422	499	576	653	730	807	884	961	1,038	1,114	1,191	1,268	1,345
	5.15	188	263	339	414	490	565	640	716	791	867	942	1,018	1,093	1,168	1,244	1,319
	5.2	184	258	332	406	480	554	628	702	776	850	924	998	1,072	1,146	1,220	1,294
	5.25	181	253	326	399	471	544	616	689	761	834	907	979	1,052	1,124	1,197	1,269
	5.3	177	249	320	391	462	533	605	676	747	818	889	961	1,032	1,103	1,174	1,245

Time Chart for STD Inverse Time (General Feeder)

Item	Tsd / x In	STD Time Setting (ms) Inland Type															
		50	70	90	110	130	150	170	190	210	230	250	270	290	310	330	350
Overload Rate (%)	5.35	174	244	314	384	454	524	593	663	733	803	873	943	1,013	1,083	1,152	1,222
	5.4	171	240	308	377	445	514	582	651	720	788	857	925	994	1,063	1,131	1,200
	5.45	168	235	303	370	437	505	572	639	707	774	841	909	976	1,043	1,111	1,178
	5.5	165	231	297	363	429	495	561	628	694	760	826	892	958	1,024	1,090	1,157
	5.55	162	227	292	357	422	486	551	616	681	746	811	876	941	1,006	1,071	1,136
	5.6	159	223	286	350	414	478	542	605	669	733	797	860	924	988	1,052	1,116
	5.65	156	219	281	344	407	469	532	595	657	720	783	845	908	971	1,033	1,096
	5.7	153	215	277	338	400	461	523	584	646	707	769	831	892	954	1,015	1,077
	5.75	151	211	272	332	393	453	514	574	635	695	756	816	877	937	998	1,058
	5.8	148	208	267	326	386	445	505	564	624	683	743	802	862	921	980	1,040
	5.85	146	204	262	321	379	438	496	555	613	672	730	788	847	905	964	1,022
	5.9	143	201	258	316	373	430	488	545	603	660	718	775	833	890	948	1,005
	5.95	141	197	254	310	367	423	480	536	593	649	706	762	819	875	932	988
	6	138	194	250	305	361	416	472	527	583	638	694	750	805	861	916	972
	6.05	136	191	245	300	355	409	464	519	573	628	683	737	792	846	901	956
	6.1	134	188	241	295	349	403	456	510	564	618	671	725	779	833	886	940
	6.15	132	185	237	290	343	396	449	502	555	608	660	713	766	819	872	925
	6.2	130	182	234	286	338	390	442	494	546	598	650	702	754	806	858	910
	6.25	128	179	230	281	332	383	435	486	537	588	639	691	742	793	844	895
	6.3	125	176	226	277	327	377	428	478	529	579	629	680	730	781	831	881
	6.35	124	173	223	272	322	372	421	471	520	570	620	669	719	768	818	868
	6.4	122	170	219	268	317	366	415	463	512	561	610	659	708	756	805	854
	6.45	120	168	216	264	312	360	408	456	504	552	600	648	697	745	793	841
	6.5	118	165	213	260	307	355	402	449	497	544	591	639	686	733	781	828
	6.55	116	163	209	256	303	349	396	442	489	536	582	629	675	722	769	815
	6.6	114	160	206	252	298	344	390	436	482	528	573	619	665	711	757	803
	6.65	113	158	203	248	293	339	384	429	474	520	565	610	655	701	746	791
	6.7	111	155	200	245	289	334	378	423	467	512	556	601	646	690	735	779
	6.75	109	153	197	241	285	329	373	417	460	504	548	592	636	680	724	768
	6.8	108	151	194	237	281	324	367	410	454	497	540	583	627	670	713	756
	6.85	106	149	191	234	277	319	362	404	447	490	532	575	618	660	703	745
	6.9	105	147	189	231	273	315	357	399	441	483	525	567	609	651	693	735
	6.95	103	144	186	227	269	310	351	393	434	476	517	558	600	641	683	724
	7	102	142	183	224	265	306	346	387	428	469	510	551	591	632	673	714
	7.05	100	140	181	221	261	301	342	382	422	462	502	543	583	623	663	704
	7.1	99	138	178	218	257	297	337	376	416	456	495	535	575	614	654	694
	7.15	97	136	176	215	254	293	332	371	410	449	489	528	567	606	645	684
	7.2	96	135	173	212	250	289	327	366	405	443	482	520	559	597	636	675
	7.25	95	133	171	209	247	285	323	361	399	437	475	513	551	589	627	665
	7.3	93	131	168	206	243	281	319	356	394	431	469	506	544	581	619	656
	7.35	92	129	166	203	240	277	314	351	388	425	462	499	536	573	610	647
	7.4	91	127	164	200	237	273	310	346	383	420	456	493	529	566	602	639
	7.45	90	126	162	198	234	270	306	342	378	414	450	486	522	558	594	630
	7.5	88	124	160	195	231	266	302	337	373	408	444	479	515	551	586	622
	7.55	87	122	157	192	228	263	298	333	368	403	438	473	508	543	578	614
	7.6	86	121	155	190	225	259	294	328	363	398	432	467	502	536	571	605
	7.65	85	119	153	187	222	256	290	324	358	393	427	461	495	529	563	598

Accessories

Over Current Relay (OCR)

Time Chart for STD Inverse Time (General Feeder)

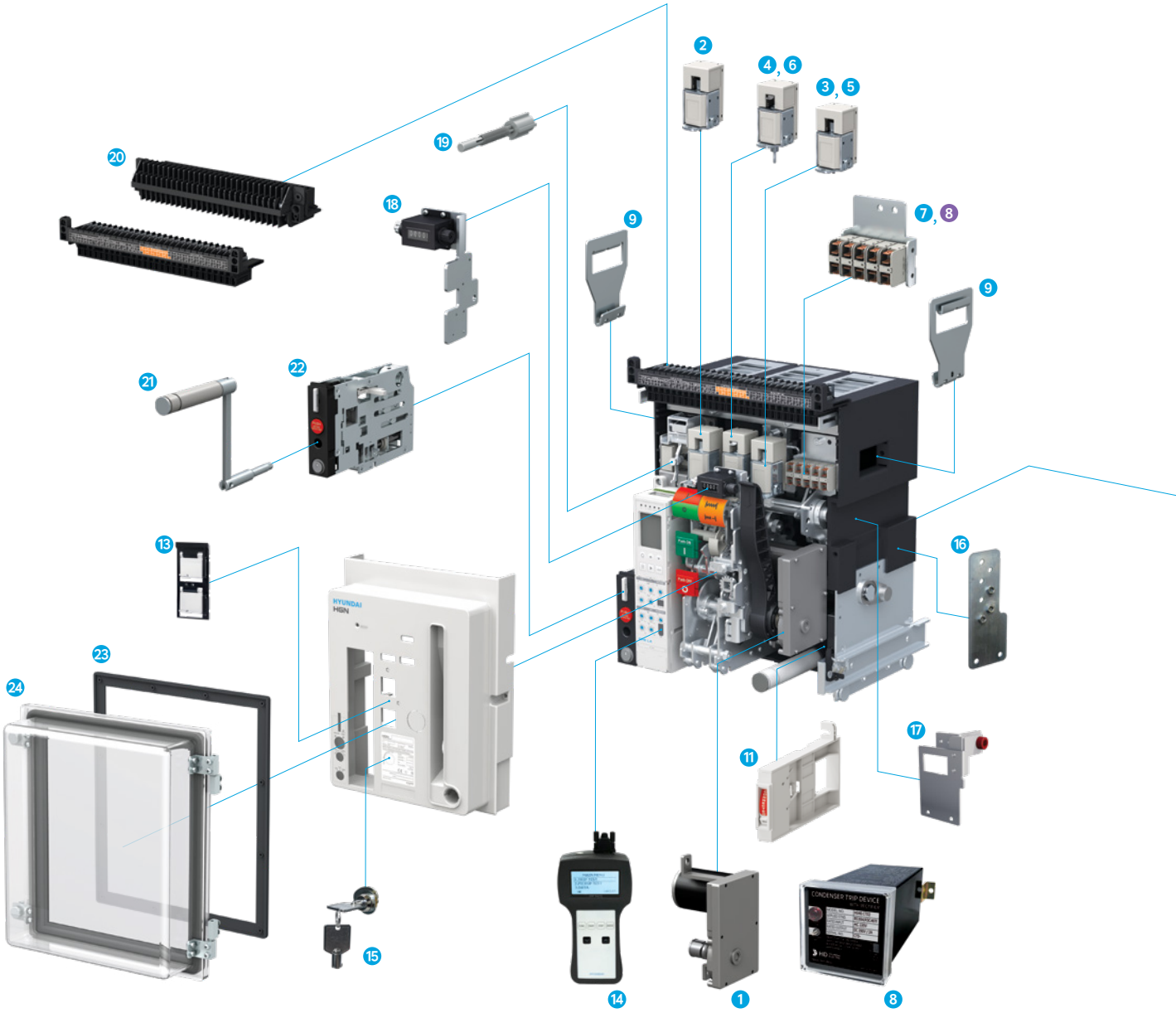
Item	Tsd / x In	STD Time Setting (ms) Inland Type															
		50	70	90	110	130	150	170	190	210	230	250	270	290	310	330	350
Overload Rate (%)	7.7	84	118	151	185	219	252	286	320	354	387	421	455	489	522	556	590
	7.75	83	116	149	183	216	249	283	316	349	382	416	449	482	516	549	582
	7.8	82	115	147	180	213	246	279	312	345	378	410	443	476	509	542	575
	7.85	81	113	146	178	210	243	275	308	340	373	405	438	470	503	535	567
	7.9	80	112	144	176	208	240	272	304	336	368	400	432	464	496	528	560
	7.95	79	110	142	174	205	237	268	300	332	363	395	427	458	490	522	553
	8	78	109	140	171	203	234	265	296	328	359	390	421	453	484	515	546
	8.05	77	108	138	169	200	231	262	293	324	354	385	416	447	478	509	540
	8.1	76	106	137	167	198	228	259	289	320	350	381	411	442	472	502	533
	8.15	75	105	135	165	195	225	255	286	316	346	376	406	436	466	496	526
	8.2	74	104	133	163	193	223	252	282	312	342	371	401	431	461	490	520
	8.25	73	102	132	161	191	220	249	279	308	337	367	396	426	455	484	514
	8.3	72	101	130	159	188	217	246	275	304	333	362	391	420	449	479	508
	8.35	71	100	129	157	186	215	243	272	301	329	358	387	415	444	473	501
	8.4	70	99	127	155	184	212	240	269	297	325	354	382	410	439	467	496
	8.45	70	98	126	154	182	210	238	266	294	322	350	378	406	434	462	490
	8.5	69	96	124	152	179	207	235	262	290	318	346	373	401	429	456	484
	8.55	68	95	123	150	177	205	232	259	287	314	341	369	396	424	451	478
	8.6	67	94	121	148	175	202	229	256	283	310	338	365	392	419	446	473
	8.65	66	93	120	147	173	200	227	253	280	307	334	360	387	414	441	467
	8.7	66	92	118	145	171	198	224	251	277	303	330	356	383	409	435	462
	8.75	65	91	117	143	169	195	222	248	274	300	326	352	378	404	431	457
	8.8	64	90	116	142	167	193	219	245	271	297	322	348	374	400	426	451
	8.85	63	89	114	140	165	191	217	242	268	293	319	344	370	395	421	446
	8.9	63	88	113	138	164	189	214	239	265	290	315	340	366	391	416	441
	8.95	62	87	112	137	162	187	212	237	262	287	312	337	362	387	411	436
	9	61	86	111	135	160	185	209	234	259	283	308	333	358	382	407	432
	9.05	61	85	109	134	158	183	207	231	256	280	305	329	354	378	402	427
	9.1	60	84	108	132	156	181	205	229	253	277	301	326	350	374	398	422
	9.15	59	83	107	131	155	179	203	226	250	274	298	322	346	370	394	418
	9.2	59	82	106	129	153	177	200	224	248	271	295	318	342	366	389	413
	9.25	58	81	105	128	151	175	198	222	245	268	292	315	338	362	385	409
	9.3	57	80	104	127	150	173	196	219	242	265	289	312	335	358	381	404
	9.35	57	80	102	125	148	171	194	217	240	263	285	308	331	354	377	400
	9.4	56	79	101	124	147	169	192	215	237	260	282	305	328	350	373	396
	9.45	55	78	100	123	145	167	190	212	235	257	279	302	324	347	369	391
	9.5	55	77	99	121	144	166	188	210	232	254	277	299	321	343	365	387
	9.55	54	76	98	120	142	164	186	208	230	252	274	296	317	339	361	383
	9.6	54	75	97	119	141	162	184	206	227	249	271	292	314	336	358	379
	9.65	53	75	96	118	139	161	182	204	225	246	268	289	311	332	354	375
	9.7	53	74	95	116	138	159	180	201	223	244	265	286	308	329	350	371
	9.75	52	73	94	115	136	157	178	199	220	241	262	284	305	326	347	368
	9.8	52	72	93	114	135	156	177	197	218	239	260	281	301	322	343	364
	9.85	51	72	92	113	133	154	175	195	216	237	257	278	298	319	340	360
	9.9	51	71	91	112	132	153	173	193	214	234	255	275	295	316	336	357
	9.95	50	70	90	111	131	151	171	191	212	232	252	272	292	313	333	353
	10	50	70	90	110	130	150	170	190	210	230	250	270	290	310	330	350

Time Chart for GFT Inverse Time (General Feeder)

Item	Tg / x In	GFT Time Setting (ms)																		
		100	150	200	250	300	350	400	450	500	550	600	650	700	750	800	850	900	950	1,000
Detection Setting Current (%)	0.2	2,500	3,750	5,000	6,250	7,500	8,750	10,000	11,250	12,500	13,750	15,000	16,250	17,500	18,750	20,000	21,250	22,500	23,750	25,000
	0.22	2,066	3,099	4,132	5,165	6,198	7,231	8,264	9,297	10,330	11,363	12,396	13,429	14,462	15,495	16,528	17,561	18,595	19,628	20,661
	0.24	1,736	2,604	3,472	4,340	5,208	6,076	6,944	7,812	8,680	9,548	10,416	11,284	12,152	13,020	13,888	14,756	15,625	16,493	17,361
	0.26	1,479	2,218	2,958	3,698	4,437	5,177	5,917	6,656	7,396	8,136	8,875	9,615	10,355	11,094	11,834	12,573	13,313	14,053	14,792
	0.28	1,275	1,913	2,551	3,188	3,826	4,464	5,102	5,739	6,377	7,015	7,653	8,290	8,928	9,566	10,204	10,841	11,479	12,117	12,755
	0.3	1,111	1,666	2,222	2,777	3,333	3,888	4,444	5,000	5,555	6,111	6,666	7,222	7,777	8,333	8,888	9,444	10,000	10,555	11,111
	0.32	976	1,464	1,953	2,441	2,929	3,417	3,906	4,394	4,882	5,371	5,859	6,347	6,835	7,324	7,812	8,300	8,789	9,277	9,765
	0.34	865	1,297	1,730	2,162	2,595	3,027	3,460	3,892	4,325	4,757	5,190	5,622	6,055	6,487	6,920	7,352	7,785	8,217	8,650
	0.36	771	1,157	1,543	1,929	2,314	2,700	3,086	3,472	3,858	4,243	4,629	5,015	5,401	5,787	6,172	6,558	6,944	7,330	7,716
	0.38	692	1,038	1,385	1,731	2,077	2,423	2,770	3,116	3,462	3,808	4,155	4,501	4,847	5,193	5,540	5,886	6,232	6,578	6,925
	0.4	625	937	1,250	1,562	1,875	2,187	2,500	2,812	3,125	3,437	3,750	4,062	4,375	4,687	5,000	5,312	5,625	5,937	6,250
	0.42	566	850	1,133	1,417	1,700	1,984	2,267	2,551	2,834	3,117	3,401	3,684	3,968	4,251	4,535	4,818	5,102	5,385	5,668
	0.44	516	774	1,033	1,291	1,549	1,807	2,066	2,324	2,582	2,840	3,099	3,357	3,615	3,873	4,132	4,390	4,648	4,907	5,165
	0.46	472	708	945	1,181	1,417	1,654	1,890	2,126	2,362	2,599	2,835	3,071	3,308	3,544	3,780	4,017	4,253	4,489	4,725
	0.48	434	651	868	1,085	1,302	1,519	1,736	1,953	2,170	2,387	2,604	2,821	3,038	3,255	3,472	3,689	3,906	4,123	4,340
	0.5	400	600	800	1,000	1,200	1,400	1,600	1,800	2,000	2,200	2,400	2,600	2,800	3,000	3,200	3,400	3,600	3,800	4,000
	0.52	369	554	739	924	1,109	1,294	1,479	1,664	1,849	2,034	2,218	2,403	2,588	2,773	2,958	3,143	3,328	3,513	3,698
	0.54	342	514	685	857	1,028	1,200	1,371	1,543	1,714	1,886	2,057	2,229	2,400	2,572	2,743	2,914	3,086	3,257	3,429
	0.56	318	478	637	797	956	1,116	1,275	1,434	1,594	1,753	1,913	2,072	2,232	2,391	2,551	2,710	2,869	3,029	3,188
	0.58	297	445	594	743	891	1,040	1,189	1,337	1,486	1,634	1,783	1,932	2,080	2,229	2,378	2,526	2,675	2,824	2,972
	0.6	277	416	555	694	833	972	1,111	1,250	1,388	1,527	1,666	1,805	1,944	2,083	2,222	2,361	2,500	2,638	2,777
	0.62	260	390	520	650	780	910	1,040	1,170	1,300	1,430	1,560	1,690	1,821	1,951	2,081	2,211	2,341	2,471	2,601
	0.64	244	366	488	610	732	854	976	1,098	1,220	1,342	1,464	1,586	1,708	1,831	1,953	2,075	2,197	2,319	2,441
	0.66	229	344	459	573	688	803	918	1,033	1,147	1,262	1,377	1,492	1,606	1,721	1,836	1,951	2,066	2,180	2,295
	0.68	216	324	432	540	648	756	865	973	1,081	1,189	1,297	1,405	1,513	1,621	1,730	1,838	1,946	2,054	2,162
	0.7	204	306	408	510	612	714	816	918	1,020	1,122	1,224	1,326	1,428	1,530	1,632	1,734	1,836	1,938	2,040
	0.72	192	289	385	482	578	675	771	868	964	1,060	1,157	1,253	1,350	1,446	1,543	1,639	1,736	1,832	1,929
	0.74	182	273	365	456	547	639	730	821	913	1,004	1,095	1,186	1,278	1,369	1,460	1,552	1,643	1,734	1,826
	0.76	173	259	346	432	519	605	692	779	865	952	1,038	1,125	1,211	1,298	1,385	1,471	1,558	1,644	1,731
	0.78	164	246	328	410	493	575	657	739	821	904	986	1,068	1,150	1,232	1,314	1,397	1,479	1,561	1,643
	0.8	156	234	312	390	468	546	625	703	781	859	937	1,015	1,093	1,171	1,250	1,328	1,406	1,484	1,562
	0.82	148	223	297	371	446	520	594	669	743	817	892	966	1,041	1,115	1,189	1,264	1,338	1,412	1,487
	0.84	141	212	283	354	425	496	566	637	708	779	850	921	992	1,062	1,133	1,204	1,275	1,346	1,417
	0.86	135	202	270	338	405	473	540	608	676	743	811	878	946	1,014	1,081	1,149	1,216	1,284	1,352
	0.88	129	193	258	322	387	451	516	581	645	710	774	839	903	968	1,033	1,097	1,162	1,226	1,291
	0.9	123	185	246	308	370	432	493	555	617	679	740	802	864	925	987	1,049	1,111	1,172	1,234
	0.92	118	177	236	295	354	413	472	531	590	649	708	767	827	886	945	1,004	1,063	1,122	1,181
	0.94	113	169	226	282	339	396	452	509	565	622	679	735	792	848	905	961	1,018	1,075	1,131
	0.96	108	162	217	271	325	379	434	488	542	596	651	705	759	813	868	922	976	1,030	1,085
	0.98	104	156	208	260	312	364	416	468	520	572	624	676	728	780	832	885	937	989	1,041
	1	100	150	200	250	300	350	400	450	500	550	600	650	700	750	800	850	900	950	1,000

Accessories

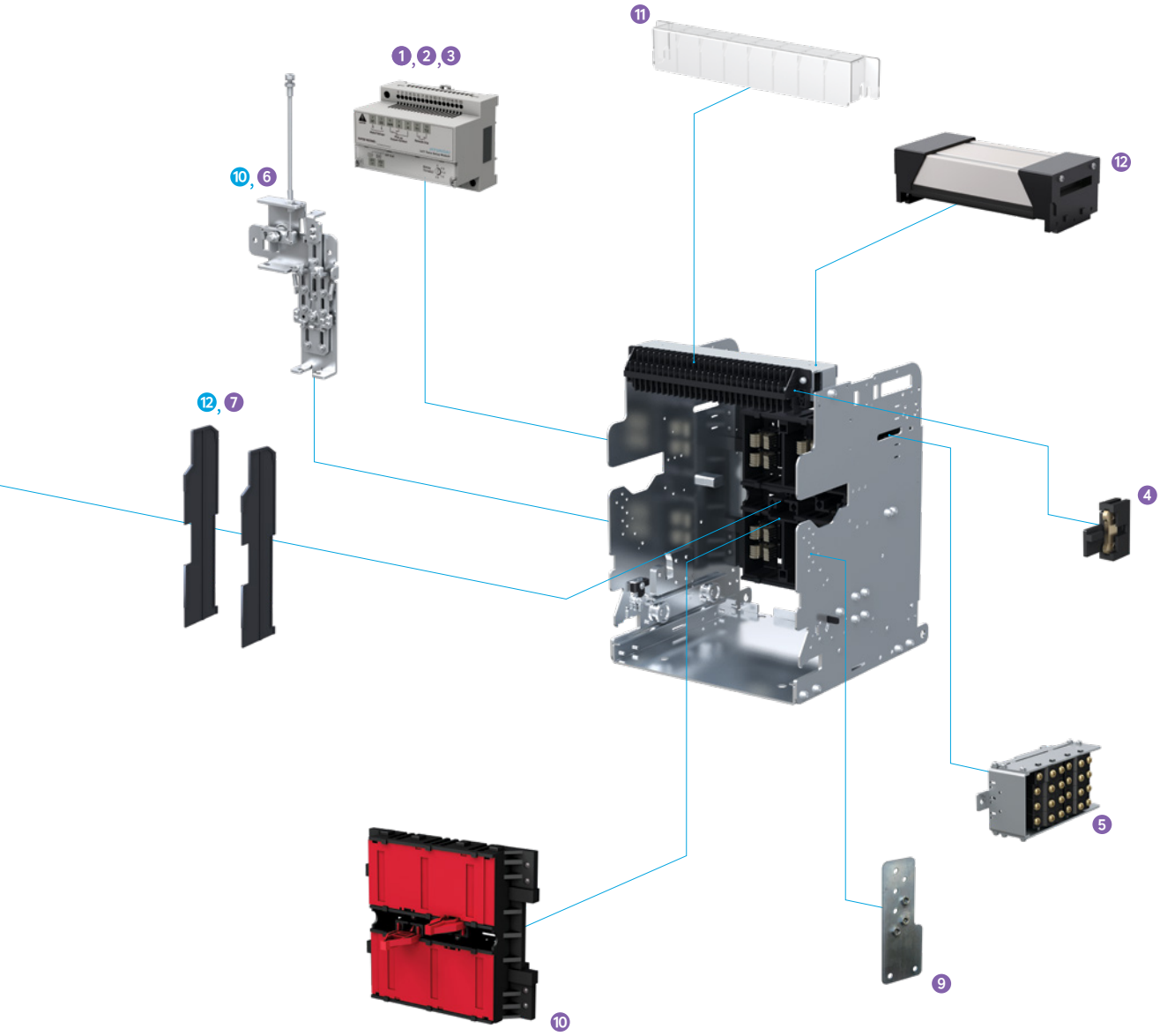
Main Unit



Accessories for Circuit Breaker

1. Spring Charge Geared Motor
2. Closing Coil
3. Trip Coil
4. Secondary Trip Coil
5. Trip Coil Supervision
6. UVT Coil
7. AUX Switch
8. Condenser Trip Device (CTD)
9. Lifting Lug
10. Mechanical Interlock
11. Isol Switch
12. Phase Insulation Barrier
13. On/Off Button Lock
14. OCR Portable Checker
15. Key Lock
16. Miss-Insertion Preventer
17. Fixing Block
18. Counter
19. OCR & Alarm S/W Reset Button
20. Test Jumper
21. Draw-In/Out Handle
22. Position Pad Lock
23. Door Flange
24. Dust Cover

Cradle



Accessories for Cradle

1. UVT Time Delay Controller
2. Remote Closing Prevention Module
3. Temperature Monitoring Device Module
4. Short "b" Contact
5. Position Switch
6. Mechanical Interlock
7. Phase Insulation Barrier
8. Mechanical Operated Cell Switch
9. Miss-Insertion Preventer
10. Safety Shutter
11. Control Block Protection Cover
12. Arc Shied

* Standard supply for HGN DC model

Accessories

Spring Charging Geared Motor

- Manual charging method and geared motor charging method using external power source for charging the spring in the circuit breaker.
- As for the geared motor charging method, if power is applied to the control power terminal number (1), (2), spring charge is possible at 85 - 110 % power of the operating voltage range.
- Add M1 - M9 in the order form.
- It can be purchased separately.

Ordering Method

Name of Accessory	Order Name Placed upon Outgoing	Detailed Specification
Spring Charging Geared Motor	M1	AC/DC 110 V
	M2	AC/DC 220 V
	M7	DC 24 V
	M8	DC 48 V
	M9	DC 125 V

Ratings

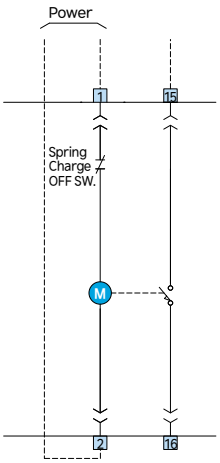
Item	DC 110	DC 220
Rated Current (A)	1.2	0.5
Starting Pick-Up (A)	5 - 6 times the rated current	
Final Rated rpm	16±3	
Charging Time (sec)	Within 5 seconds 1)	
Rated Torque (kgf)	300 kgf×cm	
Rated Watt (W)	100	100
Dielectric Strength (V/min)	2,000	
Insulation Resistance (MΩ)	100	
Workable Temperature (℃)	-15 - 60	

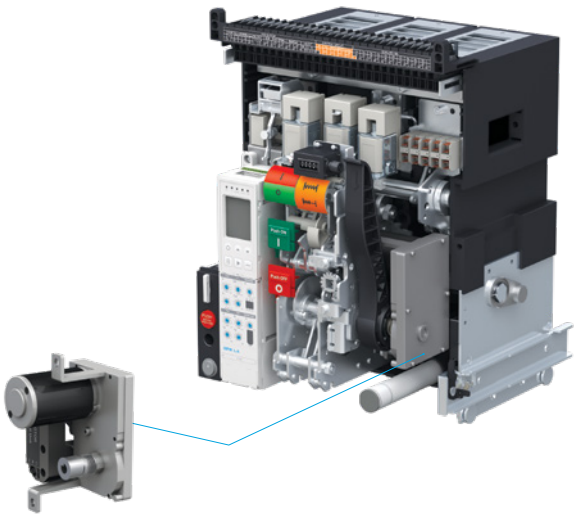
※ 1) Charging time is within 10 sec of rated, in case of DC 24/48 V.

Wire Ratings

Rated Voltage	DC 24, 48		DC 110, 220	
	AWG (mm^2)	Insulation Level (V)	AWG (mm^2)	Insulation Level (V)
Specification	18	600	18	600

Connection Diagram





Spring Charging Geared Motor

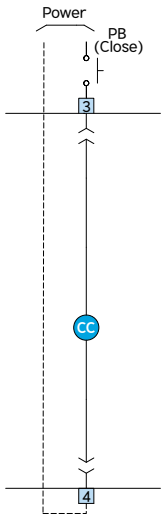
Spring Charging Switch or Ready to Close Switch

- Spring charging switch delivers (15, 16) the charged status when mechanism spring charge is complete.
- Read to close switch delivers (15, 16) only when the circuit breaker is open and simultaneously only when the mechanism spring charge is complete.
- Order name of spring charging switch : B6, Order name of ready to close switch : BT
- Two accessories cannot be ordered simultaneously.

Closing Coil (CC)

- A control device which closes a circuit breaker remotely from outside.
- The circuit breaker is closed by applying power of at least more than 150 ms within the range of 85 - 110 % of the rated control voltage to the control power terminal number (3) and (4).
- Add C1 - C9 in the order form when ordering the main unit.
- It can be purchased separately.
- A circuit to cut off the power using the internal 1a1b auxiliary switch cannot be implemented with DC 125 V coils, because the Pick Up current exceeds the auxiliary switch inductive load rating.
- Use a separate switch externally to apply power to the closing coil.

Connection Diagram



Ordering Method

Name of Accessory	Order Name Placed upon Outgoing	Detailed Specification
Closing Coil	C1	AC/DC 110 V
	C2	AC/DC 220 V
	C3	AC 380 V
	C4	AC 440 V
	C7	DC 24 V
	C8	DC 48 V
	C9	DC 125 V

Wire Ratings

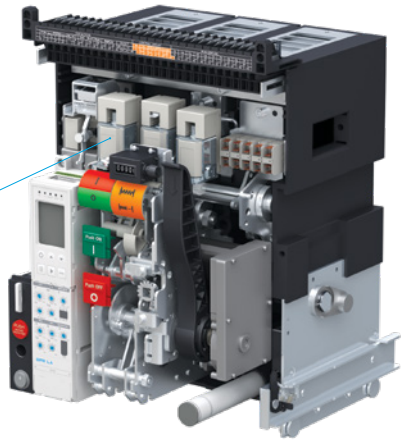
Rated Voltage	DC 24		AC/DC 110, 220	
	AWG (mm^2)	Insulation Level (V)	AWG (mm^2)	Insulation Level (V)
Specification	20	600	20	600

Ratings

Item	DC (V)	AC (V)
Rated Voltage (Vn)	24 - 48	-
	100 - 130	100 - 130
	200 - 250	200 - 250
Operating Limits (Vn)	0.85 - 1.1	
Power Consumption (VA)	Inrush (300), Steady-State (10)	
Trip Time (ms)	80	
Pick up Current	24 - 48 V : 3.0 A(±10 %)	-
	100 - 130 V : 2.0 A(±10 %)	100 - 130 V : 2.0 A(±10 %)
	200 - 250 V : 1.4 A(±10 %)	200 - 250 V : 1.4 A(±10 %)
Holing Current	24 - 48 V : 330 mA(±10 %)	-
	100 - 130 V : 150 mA(±10 %)	100 - 130 V : 150 mA(±10 %)
	200 - 250 V : 120 mA(±10 %)	200 - 250 V : 120 mA(±10 %)



Closing Coil



Accessories

Trip Coil (TC)

- A control device which trips a circuit breaker remotely.
- The circuit breaker is tripped by applying power of at least more than 150 ms within the range of 70 – 110 % of the rated control voltage to the control power terminal number (7) and (8).
- Add S1 – S9 in the order form when ordering the main unit.
- It can be purchased separately.
- A circuit to cut off the power using the internal 1a1b auxiliary switch cannot be implemented with DC 125 V coils, because the Pick Up current exceeds the auxiliary switch inductive load rating.
- Use a separate switch externally to apply power to the closing coil.

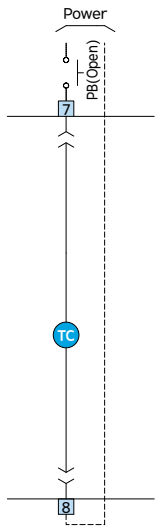
Ordering Method

Name of Accessory	Order Name Placed upon Outgoing	Detailed Specification
Trip Coil	S1	AC/DC 110V
	S2	AC/DC 220V
	S3	AC 380V
	S4	AC 440V
	S7	DC 24V
	S8	DC 48V
	S9	DC 125V

Ratings

Item	DC (V)	AC (V)
Rated Voltage (Vn)	24 - 48	-
	100 - 130	100 - 130
	200 - 250	200 - 250
Operating Limits (Vn)	0.7 - 1.1	
Power Consumption (VA)	Inrush (300), Steady-State (10)	
Trip Time (ms)	40	
Pick up Current	24 - 48 V : 2.0 A(±10 %)	-
	100 - 130 V : 2.0 A(±10 %)	100 - 130 V : 2.0 A(±10 %)
	200 - 250 V : 1.4 A(±10 %)	200 - 250 V : 1.4 A(±10 %)
Holing Current	24 - 48 V : 330 mA(±10 %)	-
	100 - 130 V : 120 mA(±10 %)	100 - 130 V : 120 mA(±10 %)
	200 - 250 V : 100 mA(±10 %)	200 - 250 V : 100 mA(±10 %)

Connection Diagram

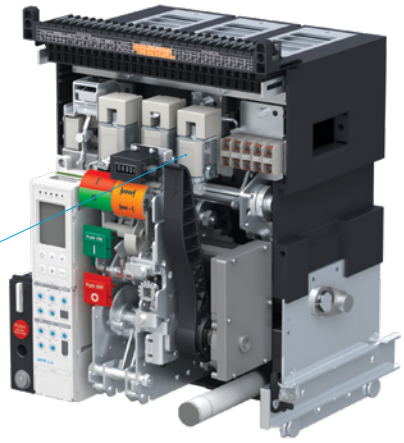


Wire Ratings

Rated Voltage	DC 24		AC/DC 110, 220	
	AWG (mm^2)	Insulation Level (V)	AWG (mm^2)	Insulation Level (V)
Specification	20	600	20	600



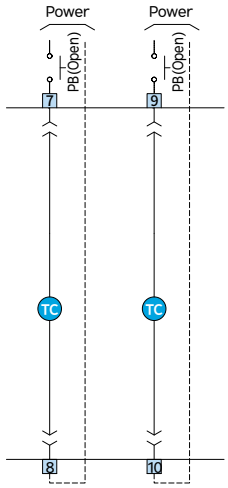
Trip Coil



Secondary Trip Coil

- It is a control device which trips a circuit breaker remotely from outside and dual remote trip is possible with an additional installation of trip coil.
- The circuit breaker is tripped by applying power of at least more than 150 ms within the range of 70 - 110 % of the rated control voltage by applying power to the control power terminal number (9) and (10).
- When dual trip coil is used, it cannot be used together with UVT coil.
- Add S1 - S9 in the order form when ordering the main unit and under the option, add S1 - S9 once again.
- It can be purchased separately.
- A circuit to cut off the power using the internal 1a1b auxiliary switch cannot be implemented with DC 125 V coils, because the Pick Up current exceeds the auxiliary switch inductive load rating.
- Use a separate switch externally to apply power to the dual trip coil.

Connection Diagram



Ordering Method

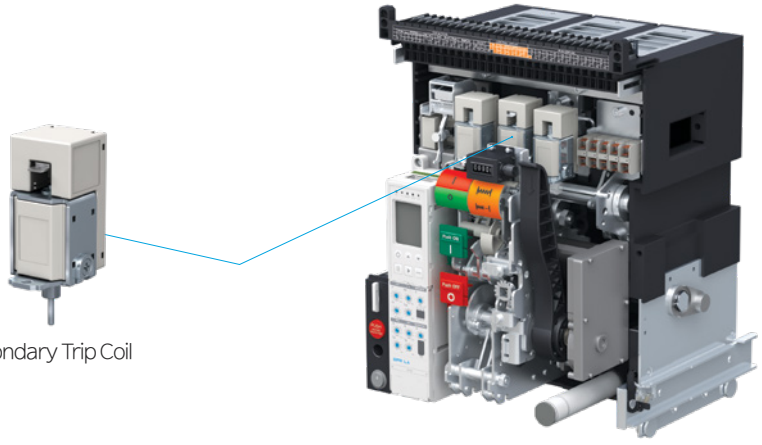
Name of Accessory	Order Name Placed upon Outgoing	Detailed Specification
Secondary Trip Coil	S1	AC/DC 110 V
	S2	AC/DC 220 V
	S3	AC 380 V
	S4	AC 440 V
	S7	DC 24 V
	S8	DC 48 V
	S9	DC 125 V

Wire Ratings

Rated Voltage	DC 24		AC/DC 110, 220	
	AWG (mm^2)	Insulation Level (V)	AWG (mm^2)	Insulation Level (V)
Specification	20	600	20	600

Ratings

Item	DC (V)	AC (V)
Rated Voltage (Vn)	24 - 48	-
	100 - 130	100 - 130
	200 - 250	200 - 250
Operating Limits (Vn)	0.7 - 1.1	
Power Consumption (VA)	Inrush (300), Steady-State (10)	
Trip Time (ms)	40	
Pick up Current	24 - 48 V : 2.0 A(±10 %)	-
	100 - 130 V : 2.0 A(±10 %)	100 - 130 V : 2.0 A(±10 %)
	200 - 250 V : 1.4 A(±10 %)	200 - 250 V : 1.4 A(±10 %)
Holing Current	24 - 48 V : 330 mA(±10 %)	-
	100 - 130 V : 120 mA(±10 %)	100 - 130 V : 120 mA(±10 %)
	200 - 250 V : 100 mA(±10 %)	200 - 250 V : 100 mA(±10 %)



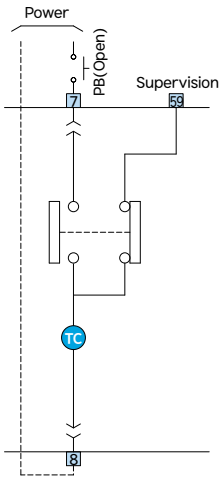
Secondary Trip Coil

Accessories

Trip Coil Supervision

- A control device which trips a circuit breaker remotely from outside.
- The circuit breaker is tripped by applying power of at least more than 150 ms within the range of 70 - 110 % of the rated control voltage by applying power to the control power terminal number (7) and (8).
- When using trip coil supervision, auxiliary contact is 4a3b and when 1a1b is added, it becomes 5a4b. (AUX S/W 43/44, 53/54 cannot be used.)
- In order to monitor the permanent normal status when TCS is closed and tripped, the circuit sequence cannot be created by using the additional 1a1b (Order code AA). (Possibility of contact damage) However, if it is compulsory, select 1a contact within 4a3b for use.
- The trip coil's monitoring output contact No. 59.
- Coil supervision checks not only the circuit's mechanical contact but also the condition of the coil.
- Add T1 - T9 in the order form when ordering the main unit.
- It can be purchased separately.

Connection Diagram



Ordering Method

Name of Accessory	Order Name Placed upon Outgoing	Detailed Specification
Trip Coil Supervision	T1	AC/DC 110 V
	T2	AC/DC 220 V
	T3	AC 380 V
	T4	AC 440 V
	T7	DC 24 V
	T8	DC 48 V
	T9	DC 125 V

Ratings

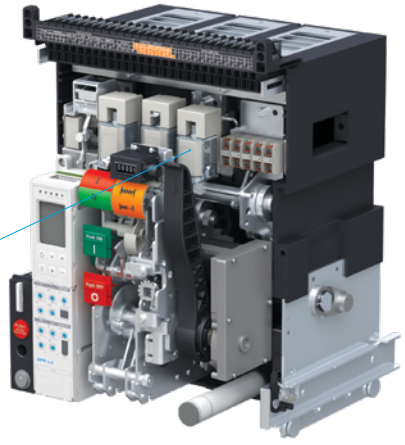
Item	DC (V)	AC (V)
Rated Voltage (Vn)	24	-
	100 - 130	100 - 130
	200 - 250	200 - 250
Operating Limits (Vn)	0.7 - 1.1	
Power Consumption (VA)	Inrush (300), Steady-State (10)	
Trip Time (ms)	40	

Wire Ratings

Rated Voltage	DC 24		AC/DC 110, 220	
	AWG (mm^2)	Insulation Level (V)	AWG (mm^2)	Insulation Level (V)
Specification	20	600	20	600



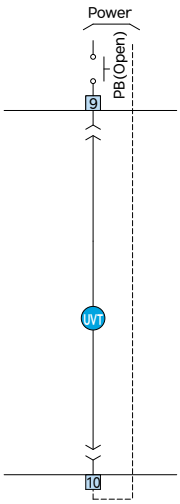
Trip Coil Supervision



UVT Coil

- Under-voltage trip device is a device that automatically trips the circuit breaker if the load voltage drops to below the standard or to prevent accidents at the load part during a black out.
- Under-voltage trip device is classified into instantaneous and time delay type for use. As for instantaneous type, directly connect to control power terminal number (9), (10) for use and as for time delay type, the Time Delay Controller can be used.
- When instantaneous type of UVT is used, dual trip coil cannot be used.
- When instantaneous type of UVT is used, add U1 - U9 in the order form when ordering the main unit.
- It can be purchased separately.

Connection Diagram



Ordering Method

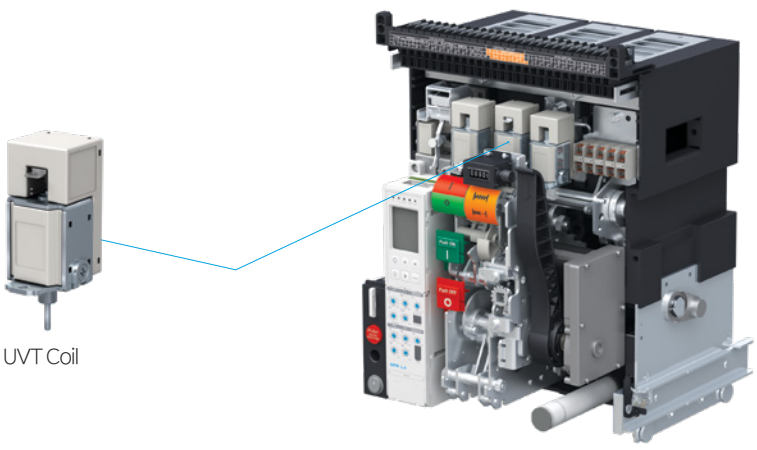
Name of Accessory	Order Name Placed upon Outgoing	Detailed Specification
UVT Coil	U1	AC/DC 110 V
	U2	AC/DC 220 V
	U3	AC 380 V
	U4	AC 440 V
	U7	DC 24 V
	U8	DC 48 V
	U9	DC 125 V

Ratings

Item	DC (V)	AC (V)
Rated Voltage (Vn)	24	-
	110 (100 - 130)	110 (100 - 130)
	220 (200 - 250)	220 (200 - 250)
	-	380
	-	440
Pickup	Above 0.85 Vn	
Drop	Below 0.35 Vn	
Power Consumption (VA)	Inrush (300), Steady-State (10)	
Trip Time (ms)	50	

Wire Ratings

Rated Voltage	DC 24		AC/DC 110, 220		AC 380, 440	
	AWG (mm^2)	Insulation Level (V)	AWG (mm^2)	Insulation Level (V)	AWG (mm^2)	Insulation Level (V)
Specification	20	600	20	600	20	600

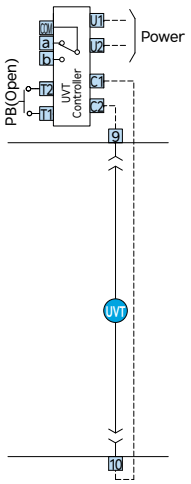


Accessories

UVT Time Delay Controller

- Under-voltage trip device is a device that automatically trips the circuit breaker if the load voltage drops to below operating voltage or to prevent accidents at the load part during a black out.
- This device is capable of delaying (0.5, 1.0, 1.5, 3 sec).
- UVT Time Delay Controller can be mounted at the left side of the cradle and the external rail and it can also be mounted at the position desired by the customer.
- Add V1 - V9 in the order form when ordering the main unit. Time Delay Controller will be included in the package.
- When purchasing the UVT time delay controller only, place the order for the separate product, HGNS UT1 - UT9.
- DC 24 V and DC 48 V are only available in instantaneous type.
- In the event the UVT Time Delay Controller and RCTU are purchased simultaneously, the front mounting can be installed in the front for 1 option designated by the customer. The remaining 1 shall be installed by designating a position in the DIN RAIL inside the panel.

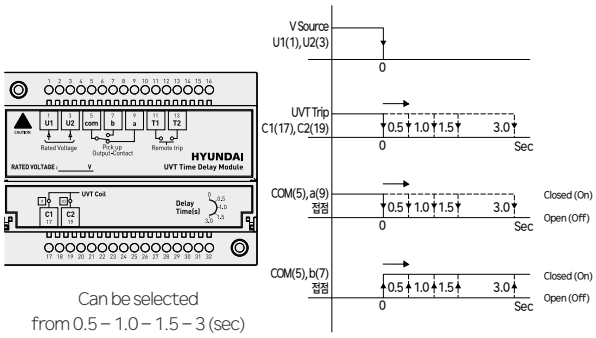
Connection Diagram



Ordering Method

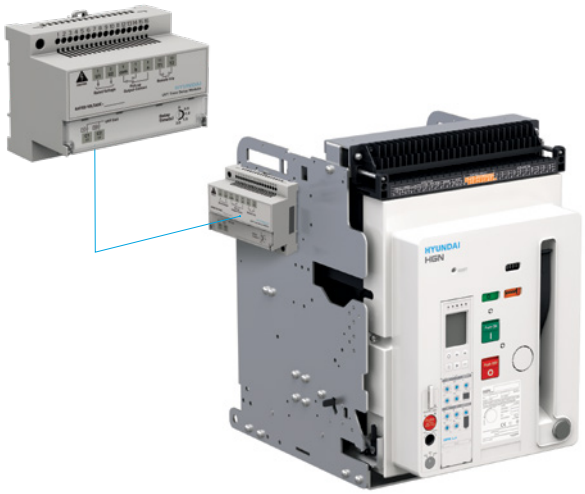
Name of Accessory	Order Name Placed upon Outgoing	Order Name for Separate Sales (HGNS-)	Detailed Specification
UVT Time Delay Controller	-	HGNS UT1	AC/DC 110 V
	-	HGNS UT2	AC/DC 220 V
	-	HGNS UT3	AC 380 V
	-	HGNS UT4	AC 440 V
	-	HGNS UT9	DC 125 V
UVT Time Delay Controller + UVT Coil	V1	HGNS V1	AC/DC 110 V
	V2	HGNS V2	AC/DC 220 V
	V3	HGNS V3	AC 380 V
	V4	HGNS V4	AC 440 V
	V9	HGNS V9	DC 125 V

Wiring and Operating Time



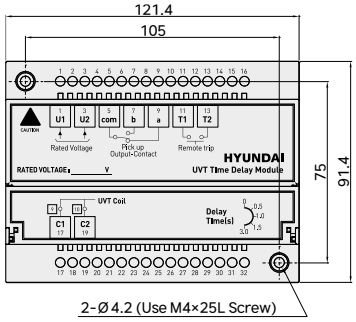
Can be selected from 0.5 – 1.0 – 1.5 – 3 (sec)

UVT Time Delay Controller



Dimension

Unit : mm

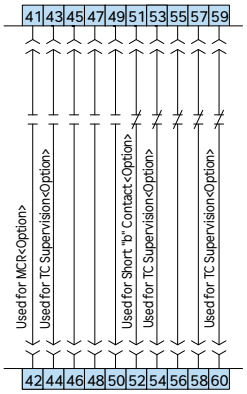


※ 5, 7, 9 switch status is the status when UVT is functioning.

AUX Switch

- It is an output contact to remotely monitor the On/Off state of the ACB.
- The default HGN DC Type provided is 4a5b without separate indication in the order. (MCR provided by default)
- AUX switch can be expanded up to 5a6b.
  - How to order HGS/HGN 6a6b : Add AA in the order form.
- When using trip coil monitoring contact, 3a3b can be used.
- When short "b" is added, it will be attached to 'b' contacts 51, 52 for outgoing and upon additional mounting, the short "b" sealed and released can be mounted additionally depending on the number of b contacts.

Connection Diagram



AUX Switch Ratings

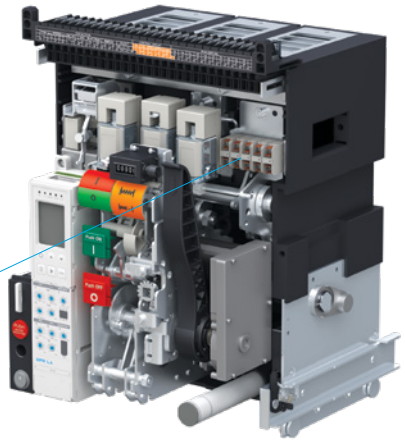
Applicable Load	Resistance	Lamp	Inductor	Motor
AC 125 V	5 A	0.7 A	4 A	1.3 A
AC 250 V	5 A	0.5 A	4 A	0.8 A
DC 125 V	≤ 2 A	-	≤ 1.0 A	-
DC 250 V	≤ 1 A	-	≤ 0.5 A	-

Wire Specification

Wire Specification	AWG	Insulation Level (V)
Specification	20	300



AUX Switch



Accessories

Counter

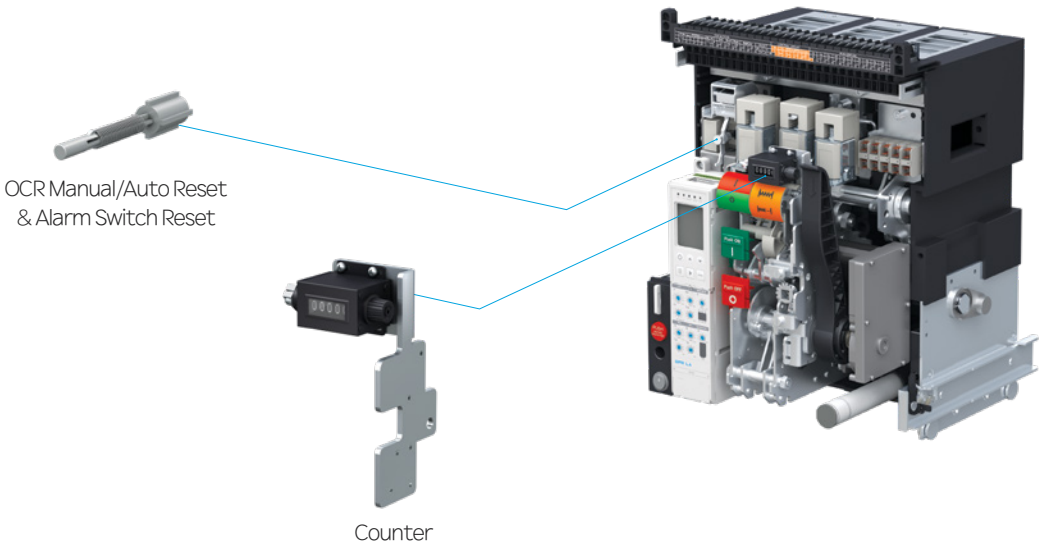
- It is a device that displays the number of operations, indicating On/Off operation of the circuit breaker in 1 cycle.
- It is a primary component so separate order is unnecessary.

OCR Manual/Auto Reset & Alarm Switch Reset

- It is a device that interlocks the closing of the circuit breaker until the electric line is restored after the user has completely removed the fault factor when a circuit breaker tripped by OCR due to overload or fault current in the electric line. (BR)
- BR : When the circuit breaker functions due to OCR, it interlocks to prevent closing of circuit breaker and is reset manually.
- BA : When the circuit breaker functions due to OCR, the circuit breaker's status is indicated as output contact 1a1b other than the interlock feature and this output contact is reset manually.
- BH : When the circuit breaker functions due to OCR, the circuit breaker's status is indicated as output contact high capacity switch other than the interlock feature and this output contact is reset manually.
- As a product mounted in the main unit, add BR/BA/BH in the order form when ordering the main unit.

Ordering Method

Name of Accessory	Order Name Placed upon Outgoing	Detailed Specification
OCR Manual Reset	BR	Interlock Manual Reset
OCR Manual Reset + Alarm S/W Reset	BA	Interlock Manual Reset + Alarm S/W Reset
OCR Manual Reset + High Capacity Alarm S/W Reset (1C)	BH	Interlock Manual Reset + Alarm S/W Reset (High Capacity DC 3 A)



Position Switch

- This switch, mounted at the side of the cradle, displays the position of ACB during draw-out.
- The position of Inserted/Isolated/Test/Connected can be displayed.
- Only one switch is applicable and it can only be mounted at the right side of the cradle.
- It can be purchased separately.

Position Switch



AUX Switch Ratings

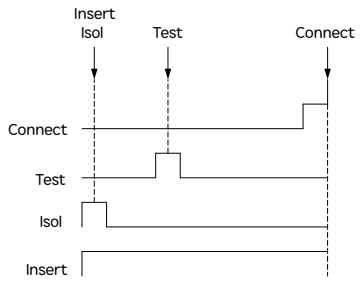
Applicable Load	Resistance	Lamp	Inductor	Motor
AC 125 V	10 A	1.5 A	6 A	2 A
DC 30 V	6 A	3 A	6 A	3 A
DC 125 V	0.6 A	0.1 A	0.6 A	0.1 A
DC 250 V	0.3 A	0.05 A	0.3 A	0.05 A

Ordering Method

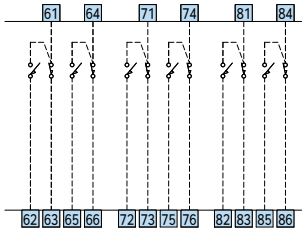
Order Name		Specification of Switch Output					Mounting Position
		Composition	Insert	Isol	Test	Connect	
AQ	HGNS AQ	Test 1C/Connect 1C	×	×	● (1C)	● (1C)	Right Side of Cradle
AR	HGNS AR	Connect 2C	×	×	×	● (2C)	
AS	HGNS AS	Test 2C	×	×	● (2C)	×	
AT	HGNS AT	Isol 1C/Insert 1C	● (1C)	● (1C)	×	×	
AU	HGNS AU	Insert 2C	● (2C)	×	×	×	
AV	HGNS AV	Isol 2C	×	● (2C)	×	×	
PQ	HGNS PQ	Isol 1C/Insert 1C/Test 1C/Connect 1C	● (1C)	● (1C)	● (1C)	● (1C)	
PS	HGNS PS	Isol 1C/Test 1C/Connect 2C	×	● (1C)	● (1C)	● (2C)	
P4	HGNS P4	Test 2C/Connect 2C	×	×	● (2C)	● (2C)	
PT	HGNS PT	Test 4C	×	×	● (4C)	×	
PR	HGNS PR	Insert 1C/Isol 1C/Test 3C/Connect 3C	● (1C)	● (1C)	● (3C)	● (3C)	
P8	HGNS P8	Insert 2C/Isol 2C/Test 2C/Connect 2C	● (2C)	● (2C)	● (2C)	● (2C)	
PU	HGNS PU	Test 4C/Connect 4C	×	×	● (4C)	● (4C)	

※ For position switches, because they are common parts with the U-Series product, the position switch may be marked "UANS".

Operating Sequence of Position Switch Operation Circuit



Connection Diagram



Test Position	Connection Position	Isolation Postion
61-62 a 61-63 b	71-72 a 71-73 b	81-82 a 81-83 b
64-65 a 64-66 b	74-75 a 74-76 b	INSERT POSITION
67-68 a 67-69 b	77-78 a 77-79 b	91-92 a 91-93 b

Dimension

Unit : mm



Accessories

Lifting Lug

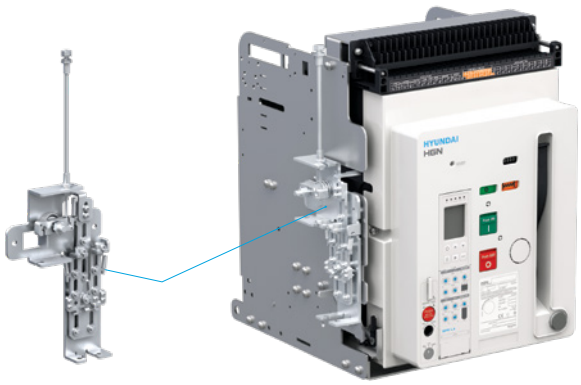
- It is a component that is used for transporting the circuit breaker.
- It can be hung to the handle at the side of the cradle, side of the arc shield and the main unit and the product can be moved by a crane or a person.
- When handling all the products, use the crane and in case of transporting independently using only the crane in case of products below 3,200 A, the products must be transported in accordance with the safety regulations.
- As a product sold separately, place an order for HGNS AL.

Lifting Lug



Mechanical Interlock

- MI is a device that mechanically interlocks the closing and tripping of each circuit breaker by connecting 2 or 3 circuit breakers.
- The MI is operated by a combination of the MI unit and the components of the interlock that is installed inside the main unit.
- The MI unit is a separate product and it is a device that is mounted additionally after the ACB has to be installed in the panel.
- When placing the order for the main unit, add B0 in the order from and only the component inside the interlock is mounted inside the main unit for release and as for the external mounting kit, place an additional order as a separate product.
- MI wire should be connected under 2 m between ACBs.
- In order to install, a square hole of 100×200 (mm) is required.



Mechanical Interlock

Ordering Method

Name of Accessory	Order Name for Separate Sales (HGNS-)	Accessory Spec.
Mechanical Interlock	HGNS DWB1	A & B & C & D & P & Q Frame Draw-Out (Fixed) Type 2 Way MI (External Mounting Kit Only)
	HGNS DWB2	A & B & C & D & P & Q Frame Draw-Out (Fixed) Type 3 Way MI (External Mounting Kit Only)
	HGNS FWB1	A & B & C & D & P & Q Frame Fixed Type 2 Way MI (External Mounting Kit Only)
	HGNS FWB2	A & B & C & D & P & Q Frame Fixed Type 3 Way MI (External Mounting Kit Only)
	HGNS B0DA	A, P Frame Interlock Part (B0)
	HGNS B0DB	B, Q, H, I Frame Interlock Part (B0)
	HGNS B0DC3	C Frame 3P Interlock Part (B0)
	HGNS B0DC4	C Frame 4P Interlock Part (B0)
	HGNS B0DD3	D Frame 3P Interlock Part (B0)
	HGNS B0DD4	D Frame 4P Interlock Part (B0)
	HGNS B0FA	A, P Frame Interlock Part (B0) + Fixed Type of Installation Bracket
	HGNS B0FB	B, Q, H, I Frame Interlock Part (B0) + Fixed Type of Installation Bracket
	HGNS B0FC3	C Frame 3P Interlock Part (B0) + Fixed Type of Installation Bracket
	HGNS B0FC4	C Frame 4P Interlock Part (B0) + Fixed Type of Installation Bracket
	HGNS B0FD3	D Frame 3P Interlock Part (B0) + Fixed Type of Installation Bracket
	HGNS B0FD4	D Frame 4P Interlock Part (B0) + Fixed Type of Installation Bracket
	HGNS DWB1A	A, P Frame Draw-Out Type 2 Way MI + Interlock Part (B0)
	HGNS DWB2A	A, P Frame Draw-Out Type 3 Way MI + Interlock Part (B0)
	HGNS DWB1B	B, Q, H, I Frame Draw-Out Type 2 Way MI + Interlock Part (B0)
	HGNS DWB2B	B, Q , H, I Frame Draw-Out Type 3 Way MI + Interlock Part (B0)
	HGNS DWB1C3	C Frame 3P Draw-Out 2 Way MI + Interlock Part (B0)
	HGNS DWB1C4	C Frame 4P Draw-Out 2 Way MI + Interlock Part (B0)

Name of Accessory	Order Name for Separate Sales (HGNS-)	Accessory Spec.
Mechanical Interlock	HGNS DWB2C3	C Frame 3P Draw-Out 3 Way MI + Interlock Part (B0)
	HGNS DWB2C4	C Frame 4P Draw-Out 3 Way MI + Interlock Part (B0)
	HGNS DWB1D3	D Frame 3P Draw-Out 2 Way MI + Interlock Part (B0)
	HGNS DWB1D4	D Frame 4P Draw-Out 2 Way MI + Interlock Part (B0)
	HGNS DWB2D3	D Frame 3P Draw-Out 3 Way MI + Interlock Part (B0)
	HGNS DWB2D4	D Frame 4P Draw-Out 3 Way MI + Interlock Part (B0)
	HGNS FWB1A	A, P Frame Fixed Type 2 Way MI + Interlock Part (B0)
	HGNS FWB2A	A, P Frame Fixed Type 3 Way MI + Interlock Part (B0)
	HGNS FWB1B	B, Q, H, I Frame Fixed Type 2 Way MI + Interlock Part (B0)
	HGNS FWB2B	B, Q, H, I Frame Fixed Type 3 Way MI + Interlock Part (B0)
	HGNS FWB1C3	C Frame 3P Fixed Type 2 Way MI + Interlock Part (B0)
	HGNS FWB1C4	C Frame 4P Fixed Type 2 Way MI + Interlock Part (B0)
	HGNS FWB2C3	C Frame 3P Fixed Type 3 Way MI + Interlock Part (B0)
	HGNS FWB2C4	C Frame 4P Fixed Type 3 Way MI + Interlock Part (B0)
	HGNS FWB1D3	D Frame 3P Fixed Type 2 Way MI + Interlock Part (B0)
	HGNS FWB1D4	D Frame 4P Fixed Type 2 Way MI + Interlock Part (B0)
	HGNS FWB2D3	D Frame 3P Fixed Type 3 Way MI + Interlock Part (B0)
	HGNS FWB2D4	D Frame 4P Fixed Type 3 Way MI + Interlock Part (B0)

Accessories

Key Lock

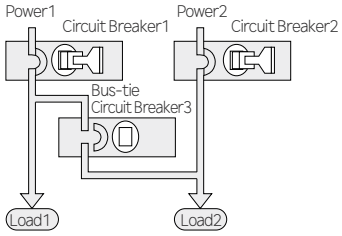
- It is a device that maintains the interlock condition to prevent electric/manual closing when the circuit breaker is open.
- When the key has been used to unlock, the electric/manual On/Off operation is possible. When the key has been removed from the circuit breaker, it becomes interlocked. When the key has been used to lock, the Off button has to be pressed to turn the key to the lock position.
- As a product mounted on the main unit, add AB in the order form when placing an order for the main unit.



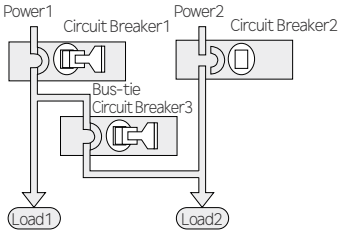
Key Lock & Key Interlock

- 3 circuit breakers mounted with the key lock device for preventing On can be used to make up the system. Only 2 circuit breakers can be allowed to close using 2 keys and the remaining 1 unit can be interlocked to provide stable load.

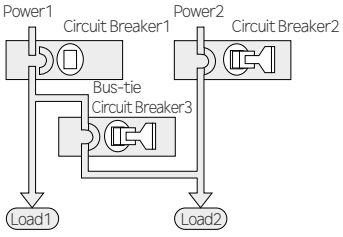
Circuit Breaker 3 cannot be closed



Circuit Breaker 2 cannot be closed



Circuit Breaker 1 cannot be closed



※ Example : Parallel feed at 2 power is prevented in case bus-tie circuit breaker is used

On/Off Button Lock

- It is a safety device that prevents manual On/Off operation caused by the user's mistaken. The buttons are covered with a plastic cover to prevent arbitrary operation of manual close/trip button and padlock is used.
- As a product mounted on the main unit, add AM in the order form when placing an order for the main unit.
- Padlock is not provided by our company. (Ø5 - Ø8)

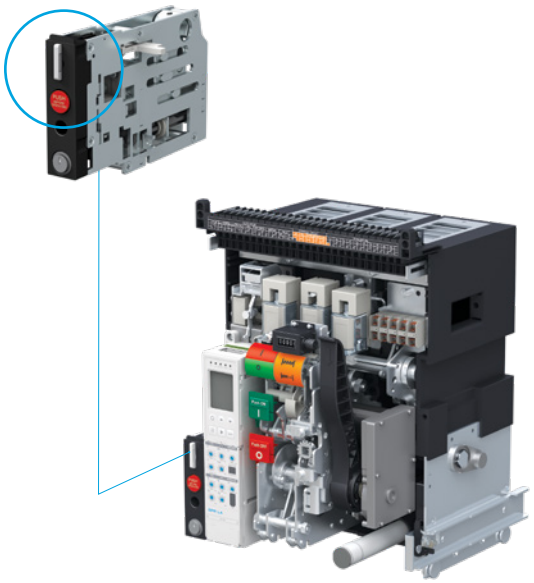
DR Device Unit & Position Pad Lock

- As for the draw-out type of circuit breaker, DR device is mounted on the main unit.
- DR device is comprised of a groove for inserting draw-out handle, draw-out push button, position indicator and a padlock device.
- As for the padlock device, it is provided as a standard safety device that prevents arbitrary draw-out by using a padlock in the connection/test/isolated position but the padlock is not provided by our company. (Ø5 - Ø8)
- It can be purchased separately.

On/Off Button Lock



DR Device Unit &
Position Pad Lock



Accessories

Auto Connection Type & Manual Control Plug Type

- As for the method of connecting the control power, auto connection type in which the control power is automatically connected/isolated when the main unit is drawn out and manual connection type in which the user directly inserts the connector with the control power connected to the main unit are available.
- Fixed type is provided with manual control plug type as standard so that type of connection terminal of the main power can be designated without separate indication.
- When completely removing the manual connection type of product from the cradle, the main unit has to be separated after removing the connector.
- As a product mounted on the cradle, select A for automatic connection type and J for manual connection type.

Short “b” Contact

- When the circuit breaker is moved from the Connected position to the Test position, the auxiliary contact AUX “b” is disconnected to maintain the external control circuit in normal, ultimately maintaining an accurate sequence operation.
  As for the number of shorting “b” contact, it corresponds to the number of AUX “b” (5b).
- Upon outgoing, only 1b (51, 52) is mounted and as for the remaining 4b, it is sealed together with the manual pack so it can be mounted as much as needed for use.
- As a product mounted on the main unit, add AK in the order form when placing an order for the main unit.

Indication of Contact Status
(When AUX “b” and Short “b” are Connected)

ACB Operating / ACB Position	ACB Close (AUX “b” Off)	ACB Open (AUX “b” On)
Connected Position (Short “b” Off)	OFF AUX "b" Short "b"	ON AUX "b" Short "b"
Test Position (Short “b” On)	OFF AUX "b" Short "b"	ON AUX "b" Short "b"

Manual Control Plug Type

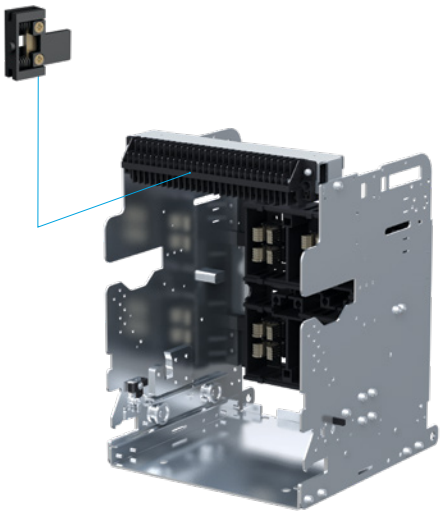


Auto Control Plug Type





Short “b” Contact



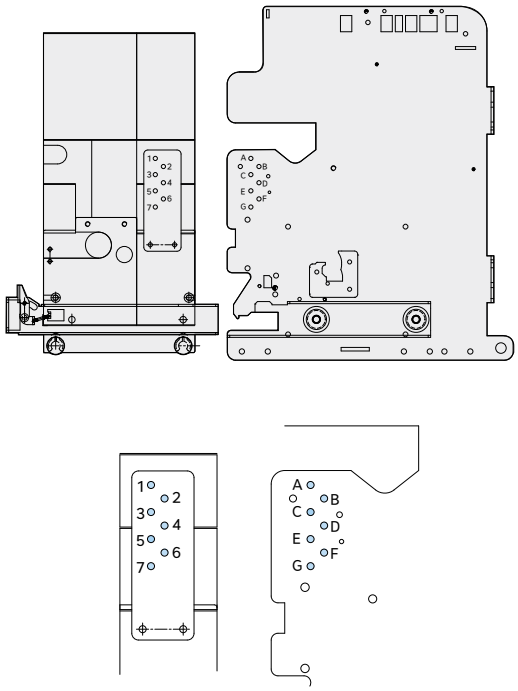
Miss-Insertion Preventor

- Although the external size is equivalent, in case the rating does not conform when the main unit of the circuit breaker (ACB) is inserted in the cradle in the composition of another circuit breaker, this device prevents it from being inserted.
- The configuration is based on the product frame.
- Each component is mounted on the main unit and cradle respectively. Add AW in the main unit and cradle's order form respectively when placing the order.

Ordering Method

Product Frame	Cradle	ACB
800 A	ABDE	367
1,000 A	ABCG	456
1,250A	ABCF	457
1,600 A	ABCE	467
2,000 A	ABCD	567
2,500 A	BCDE	167
3,200 A	BCDF	157
4,000 A	BCDG	156

Component Attachment Position by Product Frame





Accessories

Phase Insulation Barrier

- Phase insulation barrier is a part installed between phases that prevents short circuit between phases in advance by closing the arc which may occur between phases.
- 2 ea are mounted for 3P Type and 3 ea are mounted for 4P Type and as they are sold individually, select the required quantity when placing the order.
- As a product sold separately, place an order in HGNS P BAR.

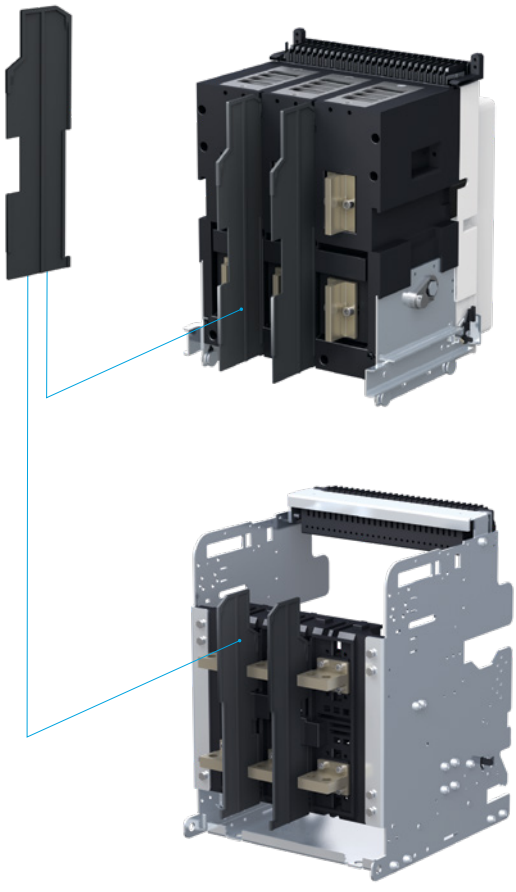
Safety Shutter & Shutter Lock

- It is a safety device that prevents the contact between the conductor part and the front part when the main unit of the circuit breaker mounted on the cradle is drawn out.
- The structure automatically opens and closes when the circuit breaker is drawn out and it comes with a shutter lock device that prevents it from opening when the shutter is closed in the event the main unit is drawn out. The padlock is not provided by our company. (Ø5 - Ø8)
- As for the shutter, the power side and load side operates separately, each equipped with an operation device and they can be locked individually.
- As a product mounted on the cradle, add AE in the order form when placing the order for the cradle.

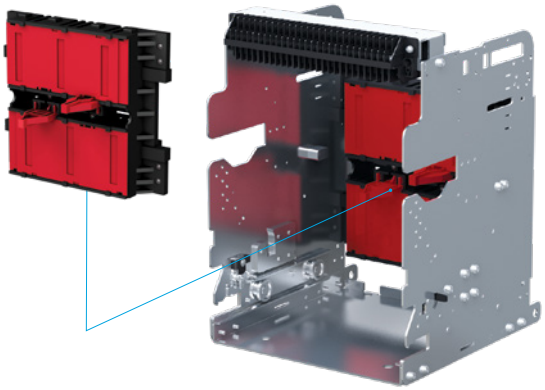
Ordering Method

Name of Accessory	Order Name Placed upon Outgoing	Order Name for Separate Sales (HGNS-)	Detailed Specification
Safety Shutter	AE	HGNS AEA3	A, P Frame, 630 - 2,000 A, 3P
		HGNS AEA4	A, P Frame, 630 - 2,000 A, 4P
		HGNS AEB3	B, Q, H, I Frame, 2,000 - 4,000 A, 3P
		HGNS AEB4	B, Q, H, I Frame, 2,000 - 4,000 A, 4P
		HGNS AEC3	C Frame, 3,200 - 5,000 A, 3P
		HGNS AEC4	C Frame, 3,200 - 5,000 A, 4P
		HGNS AED3	D Frame, 4,000 - 6,300 A, 3P
		HGNS AED4	D Frame, 4,000 - 6,300 A, 4P

Phase Insulation Barrier



Safety Shutter & Shutter Lock



Fixing Block

- It is a safety device for reinforcement in order to prevent abnormal operation caused by vibration.
- It is capable of simultaneously fixing the main unit and the cradle and reinforcing the cradles and panel supporters.
- Each component is mounted on the main unit and the cradle. Add AF in the respective order form of the main unit and cradle when placing the order.

Control Terminal Protection Cover

- It is a safety cover that is placed on the connection terminal to prevent contact with external foreign substances or physical contact after the control power has been connected to the automatic connection type of terminal.
- It cannot be applied to fixed type ACB and draw-out type ACB with manual connection terminal.
- As a product mounted on the cradle, add BC when placing an order for the draw-out type with automatic connection type.



Accessories

Arc Shield

- Arc which occurs while breaking the circuit breaker is extinguished through the arc chamber within the main unit or as the residual arc is discharged to the external upper part, the basic distance required for insulation has to be maintained.
- When mounting the arc shield, the insulation distance can be minimized to “0” due to the residual arc discharged to the upper part.
- When mounting the arc shield, it can be piled in 4 stories.
- Arc shields are provided by default for the HGN DC model.



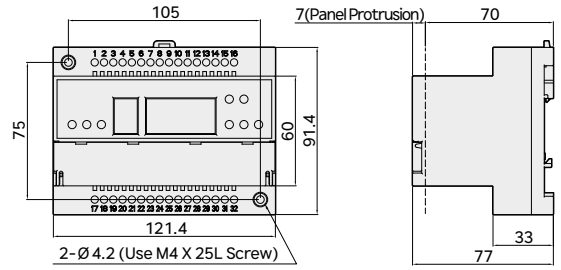
RCTU

- This product is a module that closes and prevents the ACB remotely by using communication.
- This product uses Modbus/RS-485 communication.
- Close/Open control of ACB assures its reliability through SBO (Select Before Operation) function.
- It can be used together with the temperature monitoring device module.

Contact Specification

For ACB Control	Applicable Range
Contact Ratings	10 A 240 VAC, 30 VDC
Max Switching Power	10 A 240 VAC, 30 VDC

Dimension



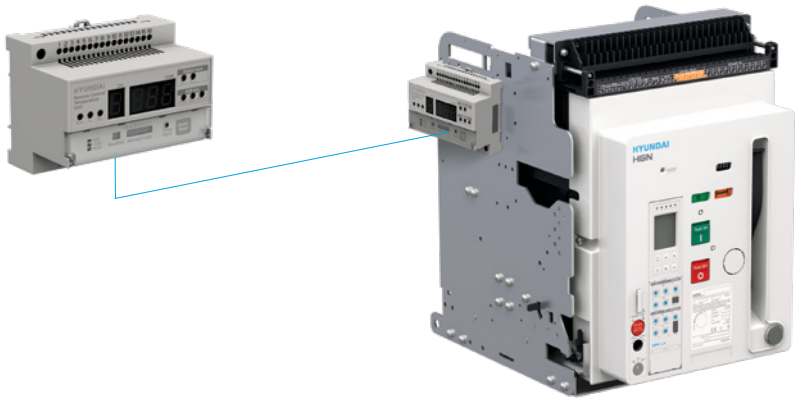
Temperature Monitoring Device Module

- This product detects the bus bar temperature of the ACB through the temperature sensor installed on the temporary location of ACB and displays it.
- Up to 3 temperature sensor can be installed and it can trigger the designated temperature alarm through a separate temperature contact.
- It supports Modbus/RS-485 communication and the each channel's temperature value and the maximum temperature value out of the channel are transmitted through communication.
- The alarm for operating temperature value can be set through communication.
- The temperature value of each channel can be checked through the Segment LED mounted at the front and once it reaches the temperature alarm set by the user, the temperature value of the relevant channel is displayed.
- As a product sold separately, it can be installed in the ACB and the internal panel. It can also be used together with the remote breaking module.
- This product is an optional product of remote breaking module and it is only sold as a combination type with the remote breaking module.

Contact Specification

Temperature Alarm	Applicable Range
Contact Ratings	10 A 120 VAC/5 A 240 VAC, 5 A 30 VDC
Max Switching Power	1,200 VA, 150 W

Remote Control Trip Unit (RCU) /
Temperature Monitoring Device Module



Accessories

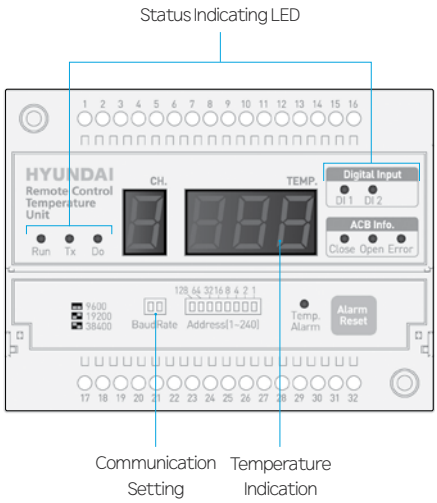
Temperature Sensor Unit

• Detailed specification of Remote Control Temperature Unit (RCTU)
- RCTU is equipped with a contact that enables remote insertion/disconnection of ACB using communication. Through the 4 temperature channels, the temperature of ACB is measured.
- RCTU supports RS-485/Modbus-RTU communication.
- Insertion/disconnection control of ACB assures its reliability through SBO (Select Before Operation) function.
- If the temperature rises over the value set by the user through communication, it can be checked through the alarm contacting point (Additional connection required).
- The temperature of ACB can be checked through the Segment LED at the front.
- RCTU can be installed in the ACB's cradle or panel.
- RCU module is equipped with the same function other than the temperature monitoring function in the RCTU module.

Status Indicating LED		Details
RCTU Status	Run LED	RCTU Operation Status LED
	Com LED	Communication LED
	Temp. Alarm LED	Temperature Alarm LED
	Alarm DO LED	Temperature Alarm Output Contact LED
Universal Digital Input	DI1	Dry Contact (5 V)
	DI2	Dry Contact (5 V)
Temperature Alarm	CB Close	ACB Close Status LED
	CB open	ACB Open Status LED
	CB Error	ACB Close/Open Terminal Non Contacted and Control Error Status

Contact Specification

Item		Applicable Range
ACB Control	Contact Ratings	10 A 240V AC, 30 VDC
	Max Switching Power	2,400 VA, 300 W
Temperature Alarm	Contact Ratings	10 A 120 VAC/5 A 240 VAC 5 A 30 VDC
	Max Switching Power	1,200 VA, 150 W



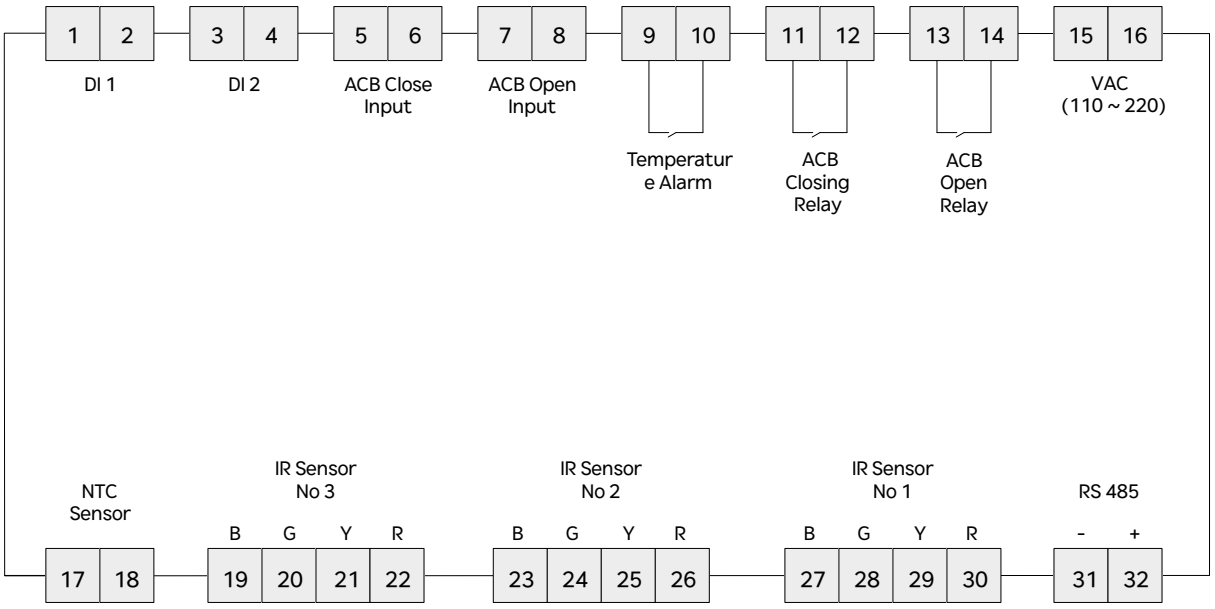
Terminal Block

No.	Definition	No.	Definition
1	DI1 +	17	NTC IR Sensor
2	DI1 -	18	NTC IR Sensor
3	DI2 +	19	IR Sensor 3 AOR (Black)
4	DI2 -	20	IR Sensor 3 GND (Green)
5	DI Close +	21	IR Sensor 3 AOT (Yellow)
6	DI Close -	22	IR Sensor 3 Power (Red)
7	DI Open +	23	IR Sensor 2 AOR (Black)
8	DI Open -	25	IR Sensor 2 GND (Green)
9	DO Temp. Alarm	25	IR Sensor 2 AOT (Yellow)
10	DO Temp. Alarm	26	IR Sensor 2 Power (Red)
11	DO Close	27	IR Sensor 1 AOR (Black)
12	DO Close	28	IR Sensor 1 GND (Green)
13	DO Open	29	IR Sensor 1 AOT (Yellow)
14	DO Open	30	IR Sensor 1 Power (Red)
15	AC Power	31	RS485 (-)
16	AC Power	32	RS485 (+)

⚠ Caution

1. IR sensor differs in value by the reflectivity of metal surface. Measuring point should be varnished or painted with matt black paint depending on material.
2. Measuring point and IR distance differ measuring spot size by D : S ratio. This sensor has 8 : 1 scale.
3. IR sensor should be indicated at the point where there is no gloss surface.
4. IR sensor distance : The ratio of measurement locations is 8 : 1. The diameter measured by the IR sensor is 1 cm when it is 8 cm away from the measurement position.

Block Diagram



Accessories

Temperature Sensor

- The temperature sensor is mounted on the designated position as standard but another location can be designated by the user.
- It is sold separately.

Ordering Method

Name of Accessory	Order Name for Separate Sales (HGNS-)	Detailed Specification
Temp Sensor	HGNS TSN	Temperature Sensor
Remote Control Trip Module	HGNS RCU	Remote Control Trip Module
	HGNS RCTU	Remote Control Trip Module + Temperature Monitoring Device Module
	HGNS RCTUN	Remote Control Trip Module + Temperature Monitoring Device Module + Temperature Sensor

Physical Medium

- Shielded Twister Pair cable used

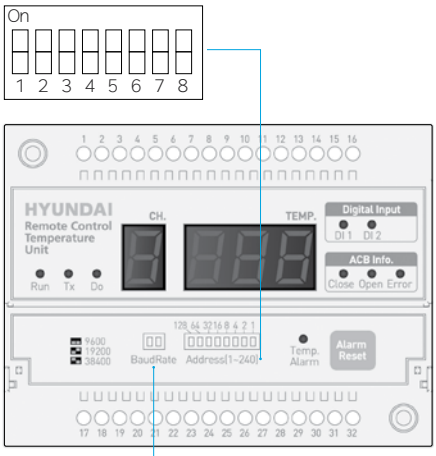
⚠ Caution

1. The use of Shielded Twisted Pair cable is recommended.
2. The maximum length for connection can be 1 km but the communication length may become shorter depending on the installation environment and the number of device connected to the communication line.
3. The communication length may differ depending on the communication speed.

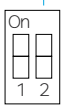
Communication Specification

- RS - 485 (Modbus-RTU)
- Baud rate : 9,600, 19,200, 38,400 bps (default : 9,600)
- Data Bits : 8 bits
- Parity : None
- Stop Bits : 1 bits

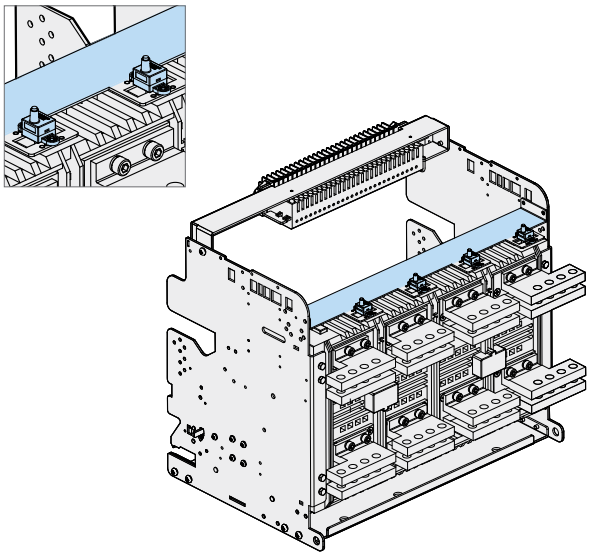
Address Setting : 1 - 240



Baud Rate Setting

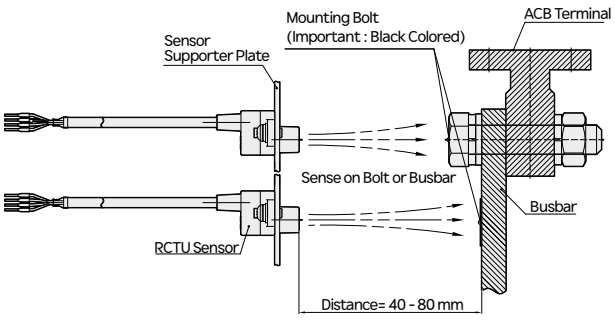


Set		Description
Off	Off	9,600
Off	On	19,200
On	Off	38,400

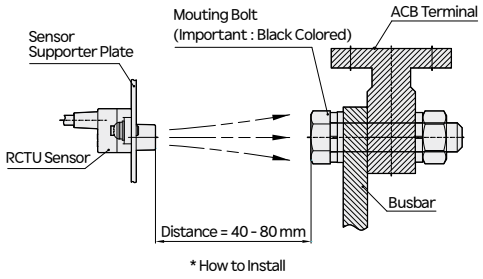


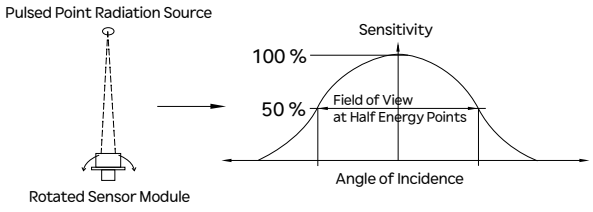
IR (Infrared Sensor) Specification (Option)

IR Sensor



Example of IR Sensor Application





Parameter	Limits			Unit	Conditions
	Min	Typ	Max		
Cable Length		3		m	
Optical Axis	- 3.5	0	+ 3.5	Degree	Sensor View Direction
Field of View		6	11	Degree	50 % Energy Points
D:S Ratio		8:1			Distance (mm) to Spot Size

IR Sensor Measurement

- IR temperature sensor has to be installed with sufficient insulation distance from the measuring point.
- The recommended distance between the measuring point and the temperature sensor is 50 - 80 mm.
- The measuring point must be a surface without reflection due to the characteristics of the IR sensor and black matte painting is recommended at the measuring point.

⚠ Caution

1. As for the IR sensor, the measurement value differs depending on the reflection rate of the metal surface. Measuring point with surfaced painted with matte black or varnish other than metallic varnishing must be measured.
2. As for the surface to be measured and the IR distance, the size of measurement area differs depending on the D : S Ratio. This sensor has a ratio of 8 : 1.

Accessories

Temperature Sensor

Connection Diagram

ACB Information Pin

a Contact	b Contact	Test	Connection
45, 47, 49	55, 57	61	71
46, 48, 50	56, 58	62	72

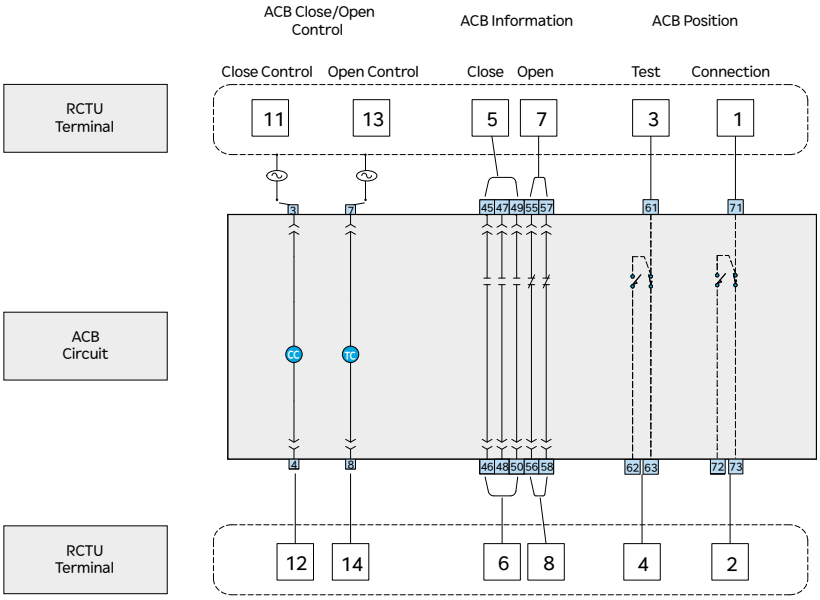
ACB Open/Close Control Pin

CC	TC
3	7
4	8

HGN



Example : When DI input is used as ACB position



INTC Thermistor Specification (Option)

- The electrical resistance value changes depending on the ambient temperature and by measuring the change in value, the temperature is displayed.
- The length of the cable is 1,000 mm and it measures the value of ambient temperature where in the RCTU has been installed.
- The range that can be measured is -50 - 250 ℃.

Parameter	Content	Conditions
Length of Cable	1,000 mm	
Resistance Value	10 kΩ ± 1 %	Ambient Temperature of 25 ℃

UET (Unified ETU Tester)

- The Unified ETU Tester (UET) performs relay test operations of OCR (Long time/short time delay/ instantaneous/ground fault).
- It is possible to set the current size and phases using button for the OCR test.
- The relay test factors can be set through the front LCD and the operating time can be checked.
- It is equipped with a battery so it can be used without a separate external power.
- As for the cable connector of the UET, connect when the control power of OCR is off when inserting the OCR.
- UET is not used as a measuring equipment, but is simply a device that displays values from the operation results of ACB OCR.
- The equipment output is digitally controlled by CPU and does not require calibration.

※For more information, please refer to the manual on our website.



Ratings

Item		Details
Operating Voltage (ADAPTOR provided)	Input	AC 100 - 240 V, 50/60 Hz, 0.5 A
	Output	DC 24 V, 1 A
Battery		Lithium polymer battery 3.7 V, 1,800 mAh
Operating Temperature		0 - 50 ℃
Storage Temperature		0 - 60 ℃
Trip Time Measuring Range	Resolution	1 ms
	Range	0 - 999.999 sec
Fault Simulation Signal		Digital Signal
Power Consumption		120 W

※Please be sure to use the adaptor we provide. We are not responsible for any problems caused by using an adaptor other than the adaptor we provide with the product.

Ordering Method

Name of Accessory	Order Name for Separate Sales	Accessory Spec.
OCR Checker	HGNS OC	PORTABLE OCR CHECKER

Accessories

CTD-Condenser Trip Device

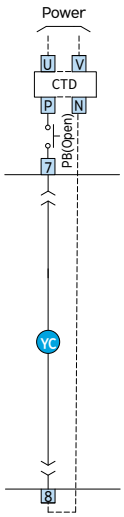
- In case of using AC/DC power for trip control of the circuit breaker, it is a device that can electrically trip the circuit breaker once by using the power charged in the CTD when the supply of control voltage has been stopped due to black-out and others.
- It is a product sold separately.



Ordering Method

Name of Accessory	Order Name for Separate Sales (HGNS-)	Detailed Specification
CTD	HGNS CTD1	AC 110 V
	HGNS CTD2	AC 220 V

Connection Diagram

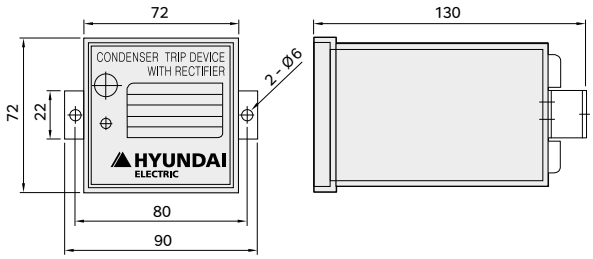


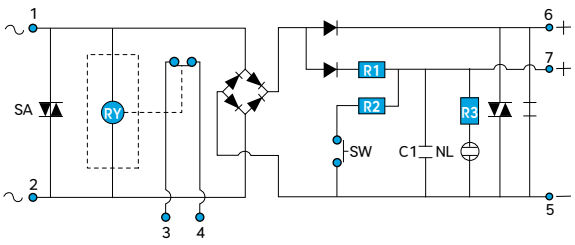
Ratings

Item	Details	
Rated Input Voltage	AC 110 V	AC 220 V
Normal Charging Voltage	DC 145 V	DC 290 V
Normal Current Capacity	DC 2 A	
Rated Frequency	50/60 Hz	
Delay Circuit Time	Within 1.5 sec	
Applicable Standard	IEC 60694/KSC 4611	

Operating Sequence of CTD External Operation Circuit

Unit : mm





- AC Input Power NO. 1, 2
- CTD Trip Power NO. 7 (+), 5 (-)
- DC Normal Power NO. 6 (+), 5 (-)
- Delay Contact NO. 3, 4

Test Jumper

- It is a device that enables the main unit to be operated independently by separating the automatic connection type of main unit from the cradle.
- As a product sold separately, place an order for HGNS AJ when ordering.



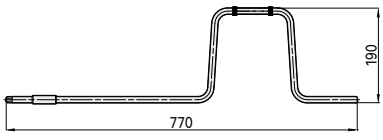
Draw-In/Out Handle

- The draw-in/out handle for drawing the draw-in type of circuit breaker is provided basically.
- In case the standard handle is uncomfortable when drawing the circuit breaker in and out, Long Type Handle and Universal Joint Handle which are sold separately can be used.
- As a product sold separately, place an order of HGNS LHANDLE or HGNS UHANDLE.

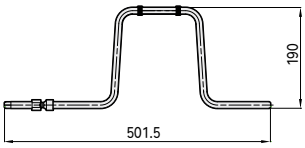


Type

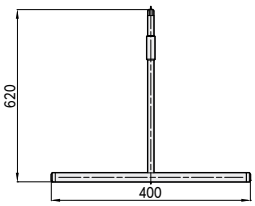
Long Type Handle



Universial Joint Handle



T-Type Handle



Ordering Method

Name of Accessory	Order Name Placed upon Outgoing	Order Name for Separate Sales (HGNS-)	Detailed Specification
Draw-In/Out Handle	-	HGNS HANDLE	Standard Handle (Provided as Standard)
		HGNS LHANDLE	Long Type of Draw-In/Out Handle (For ACB)
		HGNS UHANDLE	Universal Joint Type of Draw-In/Out Handle
		HGNS THANDLE	T-Type, Long Type Handle

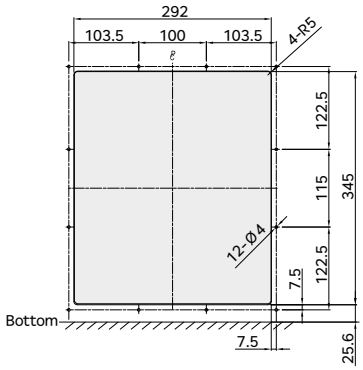
Accessories

Door Flange

- It is an auxiliary device mounted on the cutting part of the panel door at the front protrusion mounting type of ACB.
- The door flange of our company is IP30.
- As for the cutting dimension of the panel, refer to the external drawing.
- As for the panel cutting dimension, it is equivalent for both the draw-out and fixed type and it is the same even if the dust cover is installed.
- As a product sold separately, place an order for HGNS AG.

Panel Door Cutting Dimension

Unit : mm

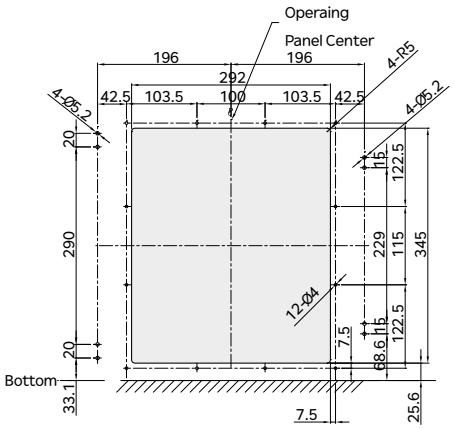


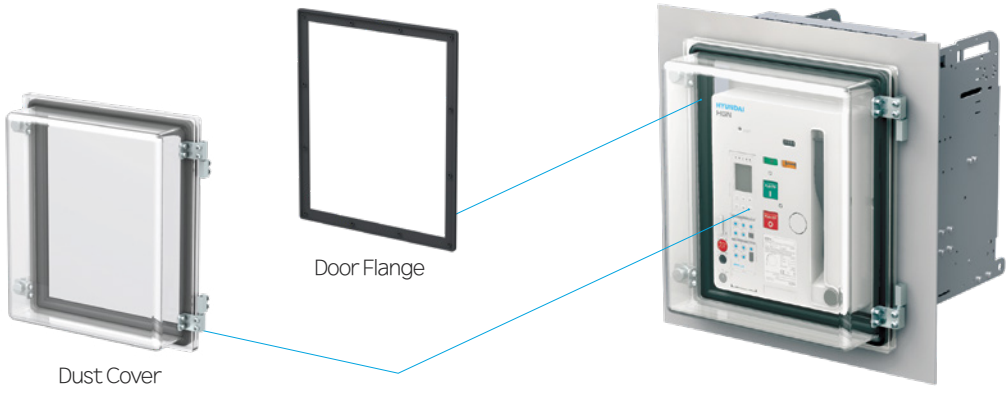
Dust Cover

- It is a device installed to protect against dust and moisture mounted on the cutting part of the panel door at the front protrusion mounting type of ACB.
- The dust cover of our company is IP52.
- The cover can be locked in the test and connected position.
- The cutting dimension of the panel must be referred to.
- As a product sold separately, place an order for HGNS DC.

Panel Door Cutting Dimension

Unit : mm





Technical Data

Standard Operating Conditions

Load Rate (I/In)

- ACB cannot be used in outdoor panel in principle. In case of inevitable installation, the ambient temperature must be below 40 °C and the load rate must be below 50 % of the rated current. In case it is not complied with, it may cause heating and burning in the ACB.
- As for the total harmonic distortion (THD) rate, it must be managed below 5 % in the distribution system phase. In case it is not complied with, it may cause heating in the product.

Load Rate	Hours of Use	Effect	Installation
I/In ≤ 80 %	24/24 Hours	-	Normal Condition (Recommended)
80 < I/In ≤ 90 %	24/24 Hours	-	Periodic Inspection
I/In = 100 %	24/24 Hours	Change in Plastic Insulation Material Color	Additional Exhaust

Altitude

- Use below 2,000 m.

Derating Table for use above 2,000 m

Altitude	≤ 2,000 m	3,000 m	3,000 m	5,000 m
Rated Operational Voltage (V)	1,500	1,350	1,200	1,050
Rated Current (%)	100	98	96	94

Protection Degree Provided by Enclosures (IP Code)

- Standard : IP30 (When the ACB front cover part of the switchboard door is cut out)
- With Dust Cover : IP52

Classification of Protection Degree Provided by Enclosures (IP code) IEC 60529

Indication : I P □ □

- Second Characteristic Numeral : Level of preventing liquid from permeating into the external box (0 - 8)
- First Characteristic Numeral : Level of preventing particulate materials from permeating into the external box (0 - 6)

※ In the event indicating the characteristic number is not required, mark the said area with 'X'.

Classification Table Following the Degree

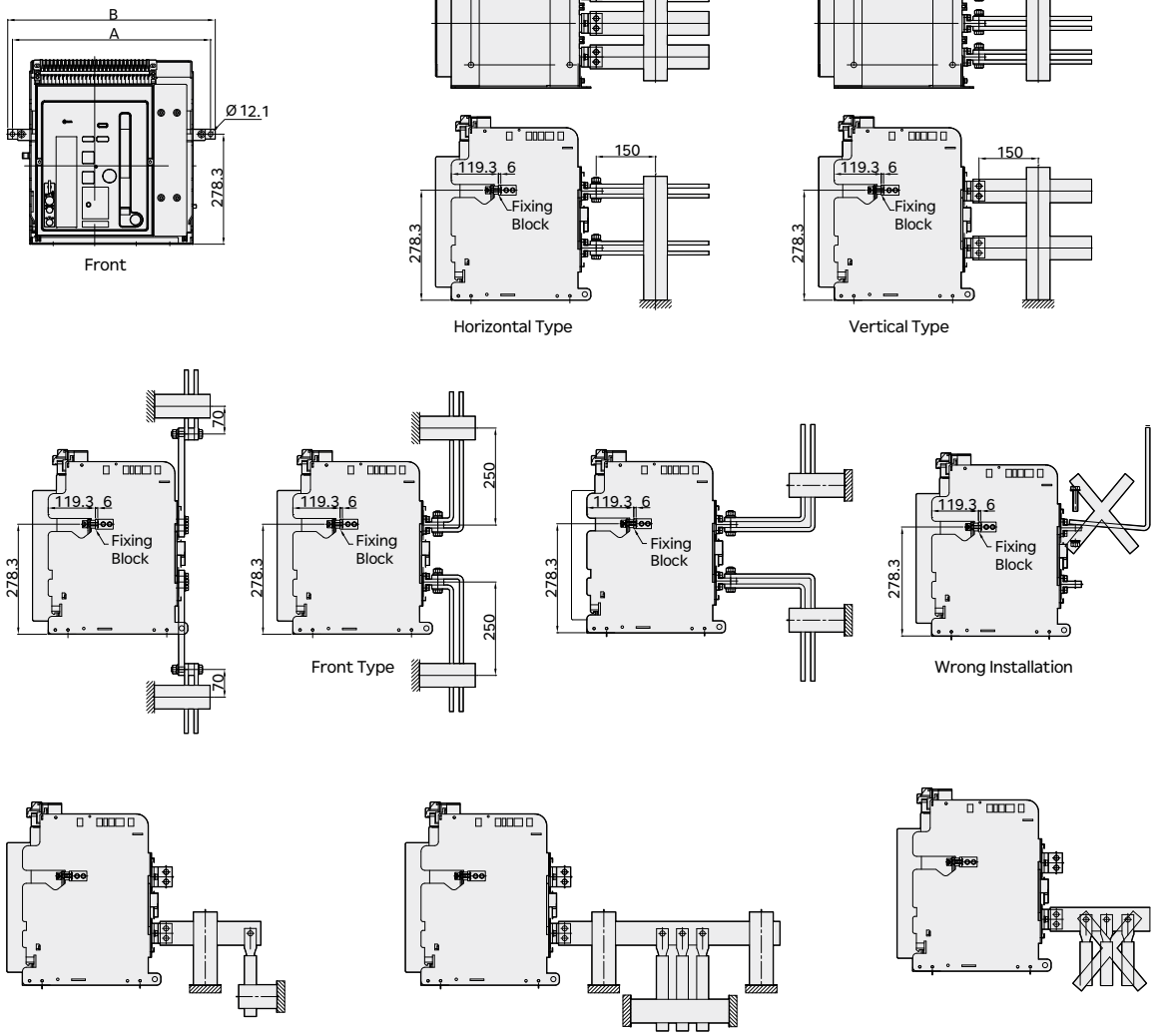
Numeral	Protection Degree	
	First Characteristic Numeral (Particulate)	Second Characteristic Numeral (Liquid)
0	No protection (Open)	No protection (Open)
1	Protected against external objects with a diameter of 50 mm and above	Protected against vertical fall
2	Protected against external objects with a diameter of 12.5 mm and above	Protected against falling with a gradient of 15 °
3	Protected against external objects with a diameter of 2.5 mm and above	Protected against falling with a gradient of 60 °
4	Protected against external objects with a diameter of 1.0 mm and above	Protected against spraying water from up, down, left, right (Splash proof)
5	Protected against deposition risk of dust and others	Protected against spraying water from up, down, left, right (Hose proof)
6	Complete protection without deposition of dust and others	Protected against heavy rainfall and tsunami (Deckwater proof)
7	-	Protected against submersion under certain pressure and certain time (Immersible)
8	-	Protected against complete submersion (Submersible)

Technical Data

Product Installation Condition

- In order to prevent burning of products caused by electrical fault and to enhance structural safety of the panel when installing the product in areas with concerns of vibration, mount the fixing block and install the terminal supporting position like the figure.
- In particular, when executing short-time/short-circuit panel testing, it must be installed for structural safety.
- For the support dimension of the product, refer to the drawing.
- In the event of switchboard (Panel) development test (Short-time, short-circuit), the bus bar inside the panel must be supported firmly according to the following conditions.

Unit : mm



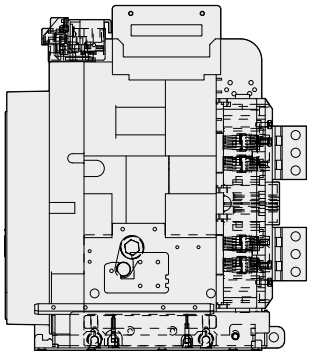
Model Name	A	B
HGN 16-40I 3P	487	511
HGN 08-40H 4P, HGN 16-40I 4P	602	626

Convenient Connection Method

As for HG Series air circuit breaker, 4 types of mounting (Connected, test, isolated, removed) are possible and offer easy maintenance.

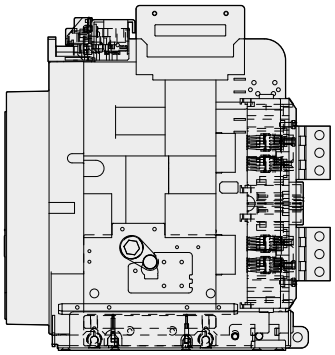
Connected Position

As a commonly used status, the main circuit and control circuit are both connected.



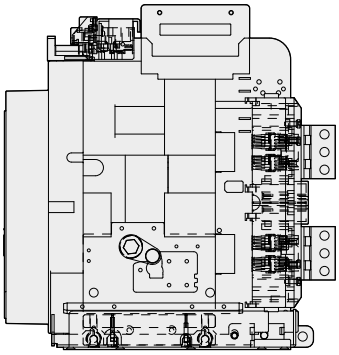
Test Position

As a status in which the main circuit is isolated and the control circuit is connected, the circuit breaker can be turned On/Off with the switchgear door closed.



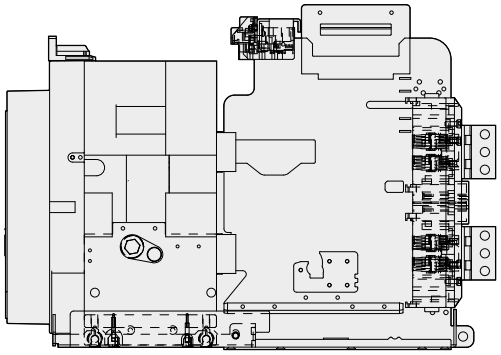
Isolated Position

With the main circuit and control circuit both isolated, the air circuit cannot be turned On/Off.



Removed Position

The air circuit has completely been removed from the cradle.



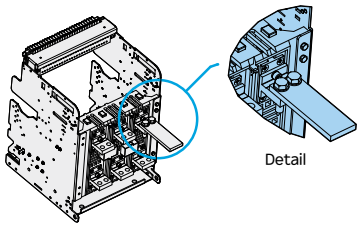
Technical Data

Internal Resistance and Power Consumption

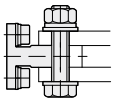
Model Name	Rated Current (A)	Fixed Type		Draw-Out Type	
		Internal Resistance (μΩ)	Power Consumption (W)	Internal Resistance (μΩ)	Power Consumption (W)
H Frame HGN20 ~ 32	2,000	10	120	20	240
	2,500	10	188	20	375
	3,200	10	307	20	614
H Frame HGN40	2,000	10	120	20	240
	2,500	10	188	20	375
	3,200	10	307	20	614
	4,000	8	384	11	528

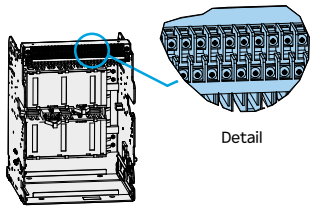
※ The above power consumption is the total power consumption of the circuit breaker based on each rated current.
The internal resistance is the resistance value per pole.
※ The internal resistance mentioned above is the factory default value. It may slightly increase depending on the operating environment (load conditions, short circuits, load switching, and other ambient conditions) and the duration of use.
Regular inspections and maintenance are recommended.
※ Contact resistance may increase after 1-2 years. Wiping the contact surface with an alcohol-dampened towel can reduce contact resistance.
※ Inquire with us separately for DC ACB (I frame).

Tightening Torque



Main Circuit
Tightening Torque
M10 : 270 ± 32 kgf-cm
M12 : 480 ± 57 kgf-cm





Control Circuit
Tightening Torque
M3.5 : 8.8 ± 1.1 kgf-cm

Insulation Voltage

When designing the switchboard, maintain the insulation distance in accordance with the table below.

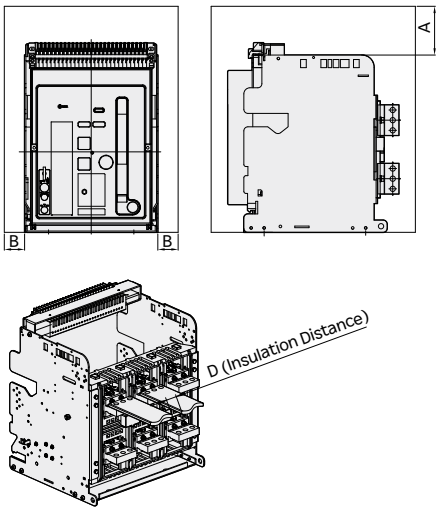
Insulation Distance from ARC

Unit : mm

Item	A	B
Fixed Type	150	50
Draw-Out Type with Arc Shield	0	50

Minimum Insulation Distance at Charging Side

Insulating Voltage	Minimum Insulation Distance (mm)
Below DC 1,500 V	18



Calibration of Rating Current

Calibration of Rating Current Following Ambient Temperature and Motion Specification

Model Name	Rating Current	ACB Terminal	Applicable Motion Standard					
				Vertical Type				
				40 ℃	45 ℃	50 ℃	55 ℃	60 ℃
H Frame HGN08 - 25 I Frame HGN16 - 25	800 A	20 t×75×1 ea	5 t×50×2 ea	800 A	800 A	800 A	800 A	800 A
	1,000 A		5 t×60×2 ea	1,000 A	1,000 A	1,000 A	1,000 A	1,000 A
	1,250 A		5 t×80×2 ea	1,250 A	1,250 A	1,250 A	1,250 A	1,250 A
	1,600 A		5 t×100×2 ea	1,600 A	1,600 A	1,600 A	1,600 A	1,600 A
	2,000 A		5 t×100×3 ea	2,000 A	2,000 A	2,000 A	2,000 A	2,000 A
	2,500 A		5 t×100×4 ea	2,500 A	2,500 A	2,500 A	2,500 A	2,500 A
H Frame HGN32 I Frame HGN32	3,200 A	20 t×90×1 ea	10 t×100×3 ea	3,200 A	3,200 A	3,120 A	3,050 A	2,950 A
H Frame HGN40 I Frame HGN40	4,000 A Vertical	15 t×125×2 ea	10 t×100×4 ea	4,000 A	4,000 A	3,950 A	3,800 A	3,680 A

※ The ambient temperature is based on the external temperature of the circuit breaker and the motion that can be applicable to the circuit breaker is applied.
The motion is based on copper material with no separate painting.
The maximum allowable temperature of the motion is below 100 ℃.
※ The above table represents the correction values for individual applications.
※ When installing inside a panel, please apply the Derating Table.

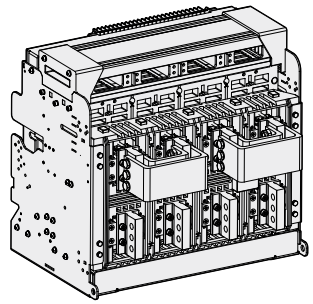
Calibration of Rating Current Following Altitude

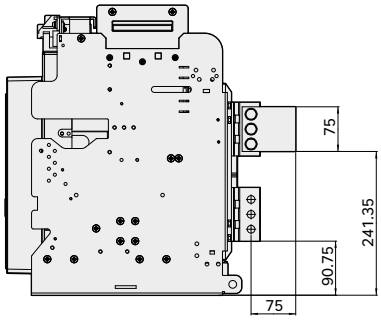
ACB is designed to be used at altitudes below 2,000 m.
When used at above 2,000 m, change the ratings depending on the surrounding environment condition for use.

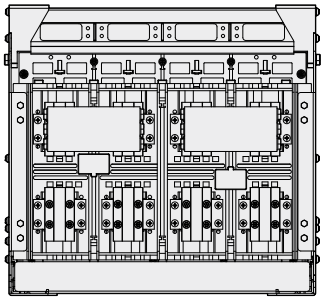
Dimensions

HGN DC Draw-Out Type 1,600 (800) - 3,200 A (Vertical Type Lower Supply)

Unit : mm

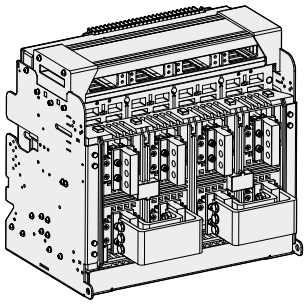


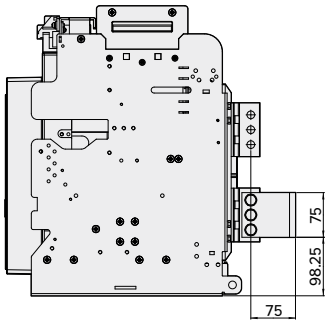


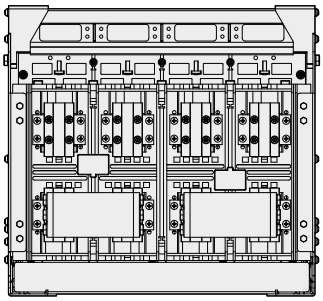


HGN DC Draw-Out Type 1,600 (800) - 3,200 A (Vertical Type Upper Supply)

Unit : mm

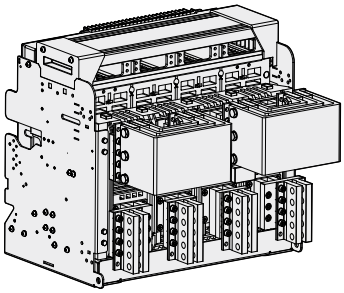


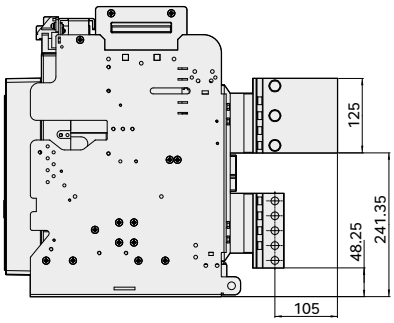


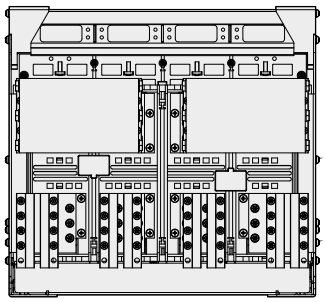


HGN DC Draw-Out Type 4,000 A (Vertical Type Lower Supply)

Unit : mm

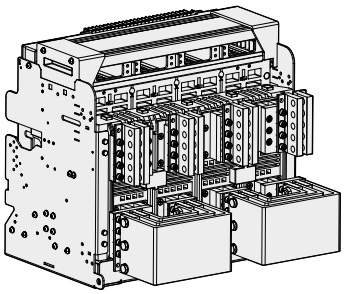


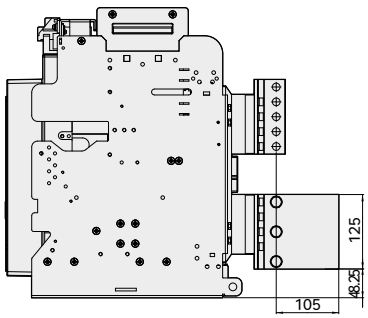


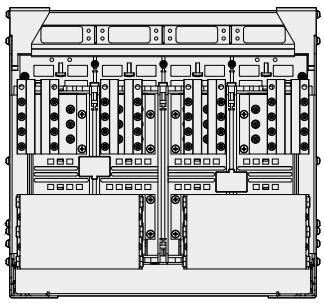


HGN DC Draw-Out Type 4,000 A (Vertical Type Upper Supply)

Unit : mm





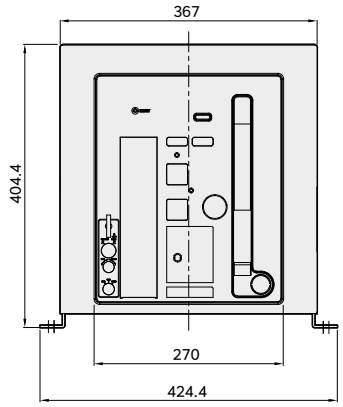


Dimensions

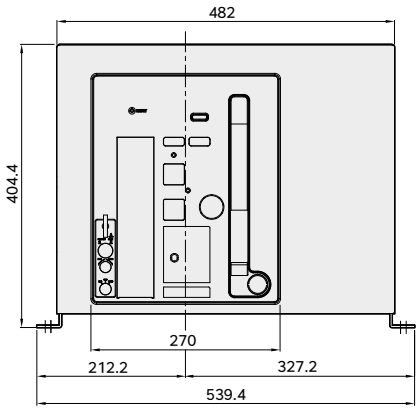
HGN DC Fixed Type 1,600 (800) - 3,200 A

Unit : mm

Front

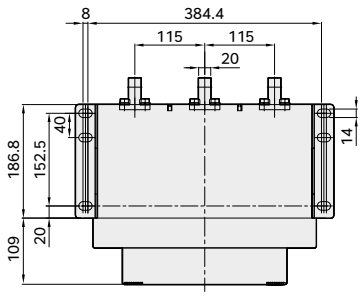


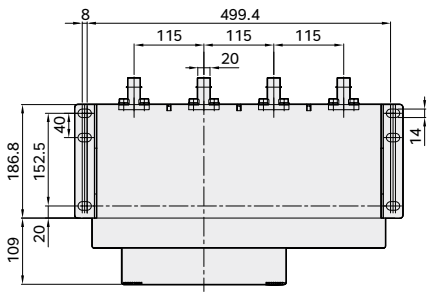
1,000 V



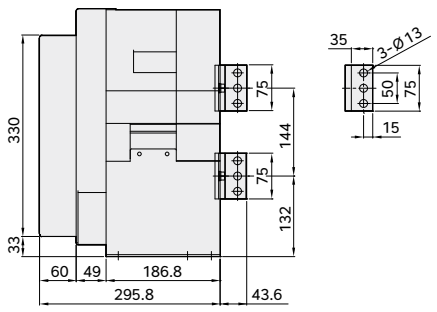
1,500 V

Vertical Type

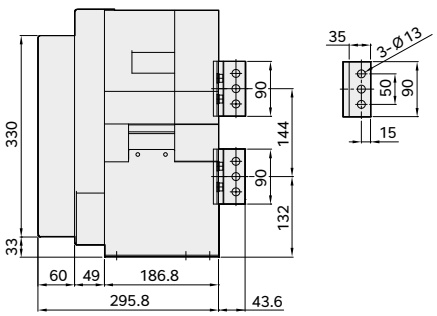




[800 - 2,500 A]



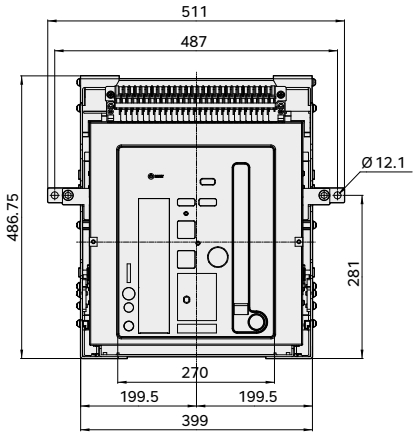
[3,200 A]



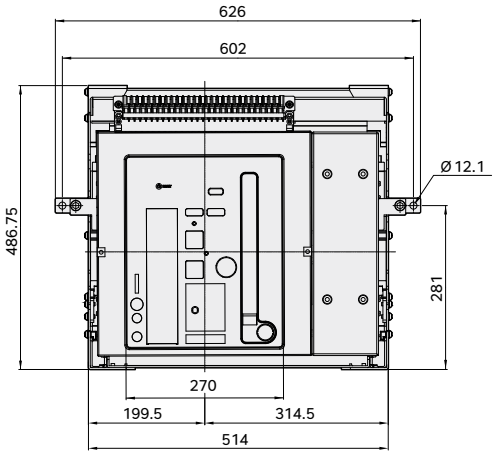
HGN DC Draw-Out Type 1,600 (800) - 3,200 A

Unit : mm

Front

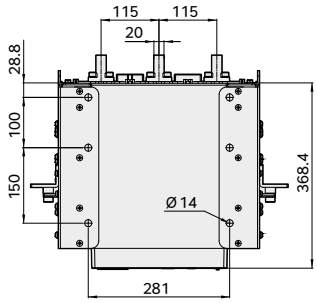


1,000 V

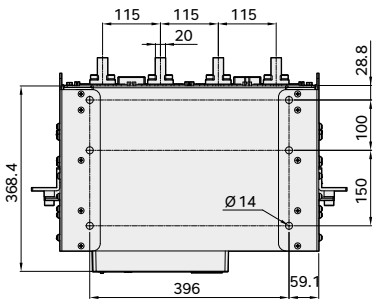


1,500 V

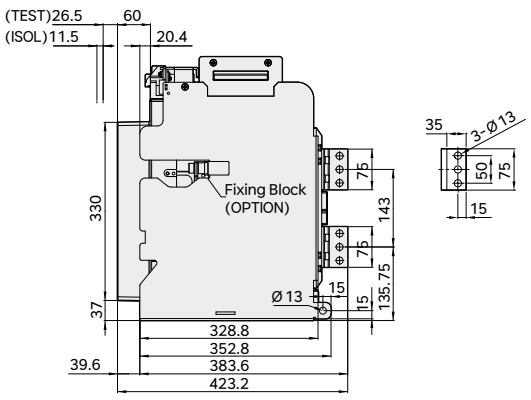
Vertical Type

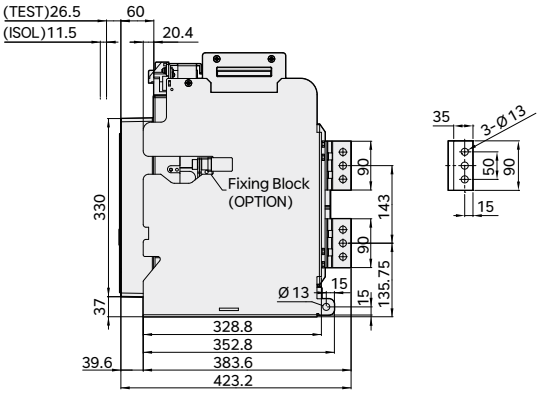


[800 - 2,500 A]



[3,200 A]



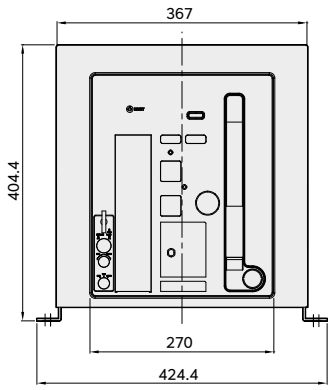


Dimensions

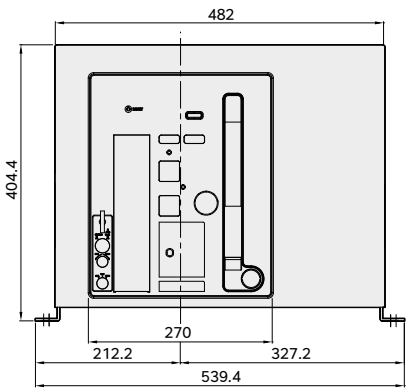
HGN DC Fixed Type 4,000 A

Unit : mm

Front

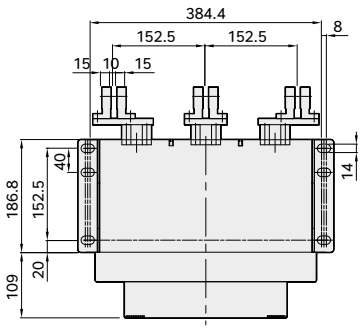


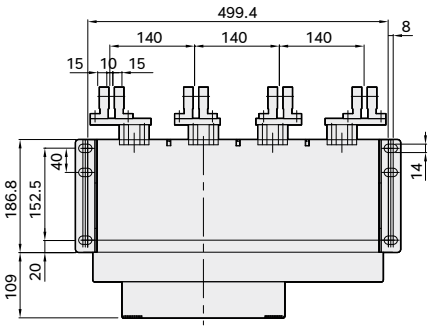
1,000 V



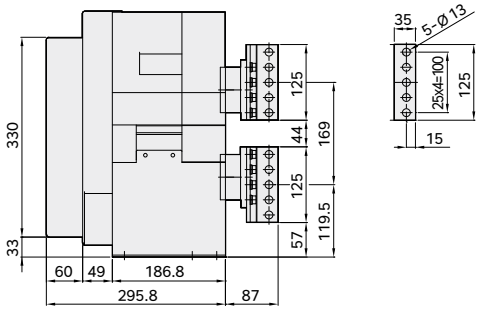
1,500 V

Vertical Type





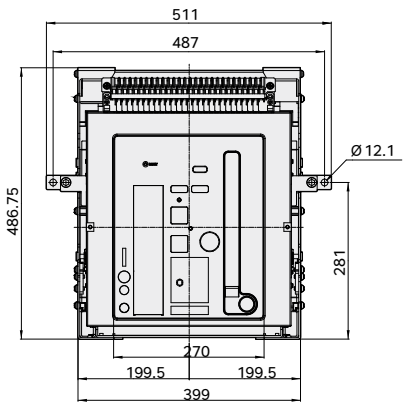
[4,000 A]



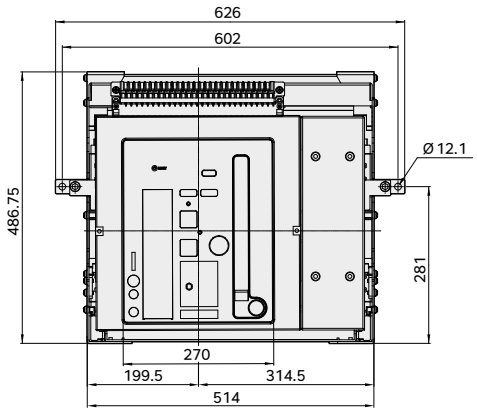
HGN DC Draw-Out Type 4,000 A

Unit : mm

Front

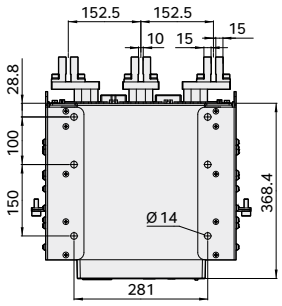


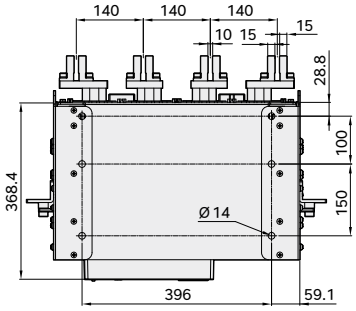
1,000 V



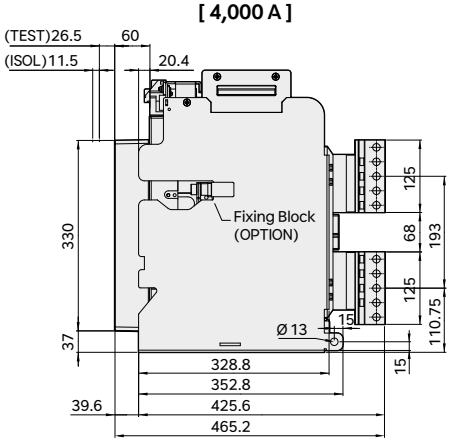
1,500 V

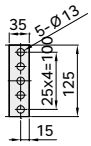
Vertical Type





[4,000 A]



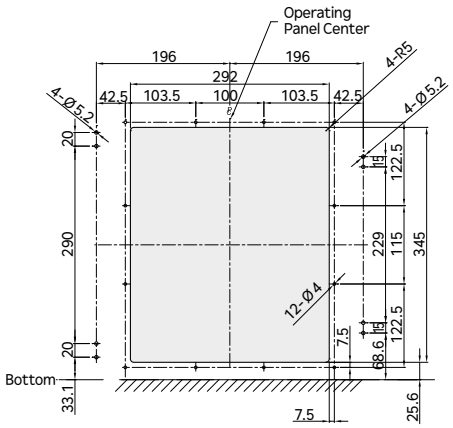


Dimensions

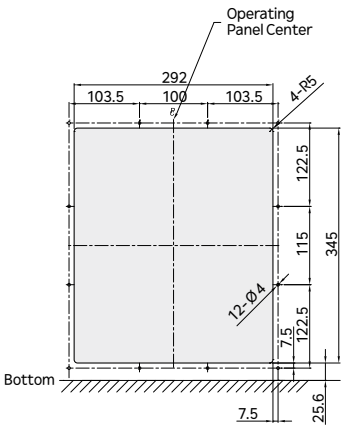
Panel Cutting Dimension of HGS/HGN Fixed Type

Unit : mm

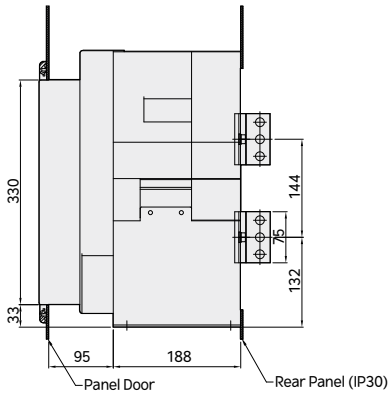
Panel Door Cut-Out (For Dust Cover)



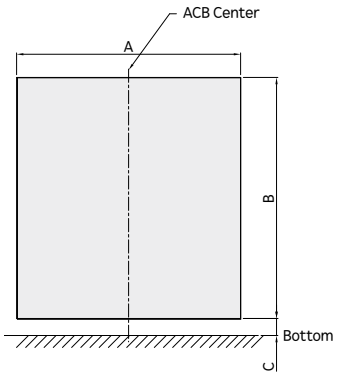
Panel Door Cut-Out (For Door Flange)



Side View



Rear Panel Cutting Size (For IP30)



Detailed Dimensions of Real Panel Cutting

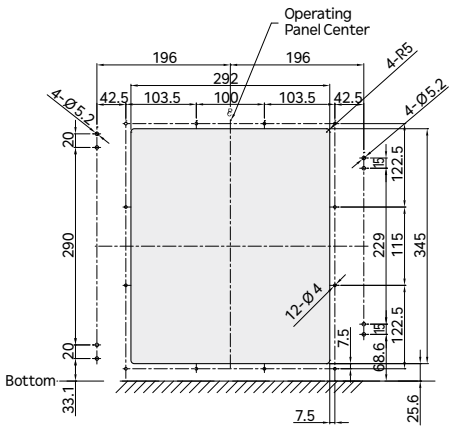
Model Name	A	B	C
HGN 16I3 - 40I3	354	355	25
HGN 08H4 - 40H4, HGN 16I4 - 40I4	469	355	25

※ The drawing dimension of this page may be subject to change without prior notice.

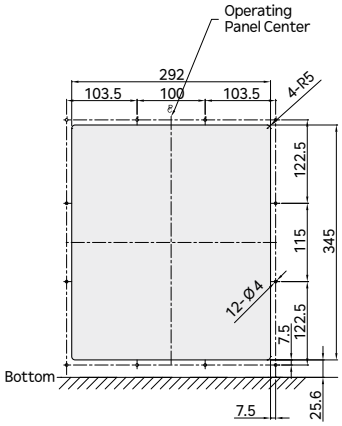
Panel Cutting Dimension of HGS/HGN Draw-Out Type

Unit : mm

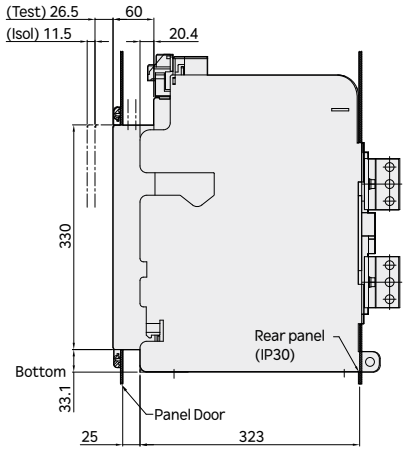
Panel Door Cut-Out (For Dust Cover)



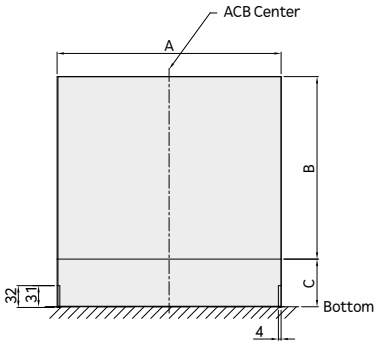
Panel Door Cut-Out (For Door Flange)



Side View



Rear Panel Cutting Size (For IP30)



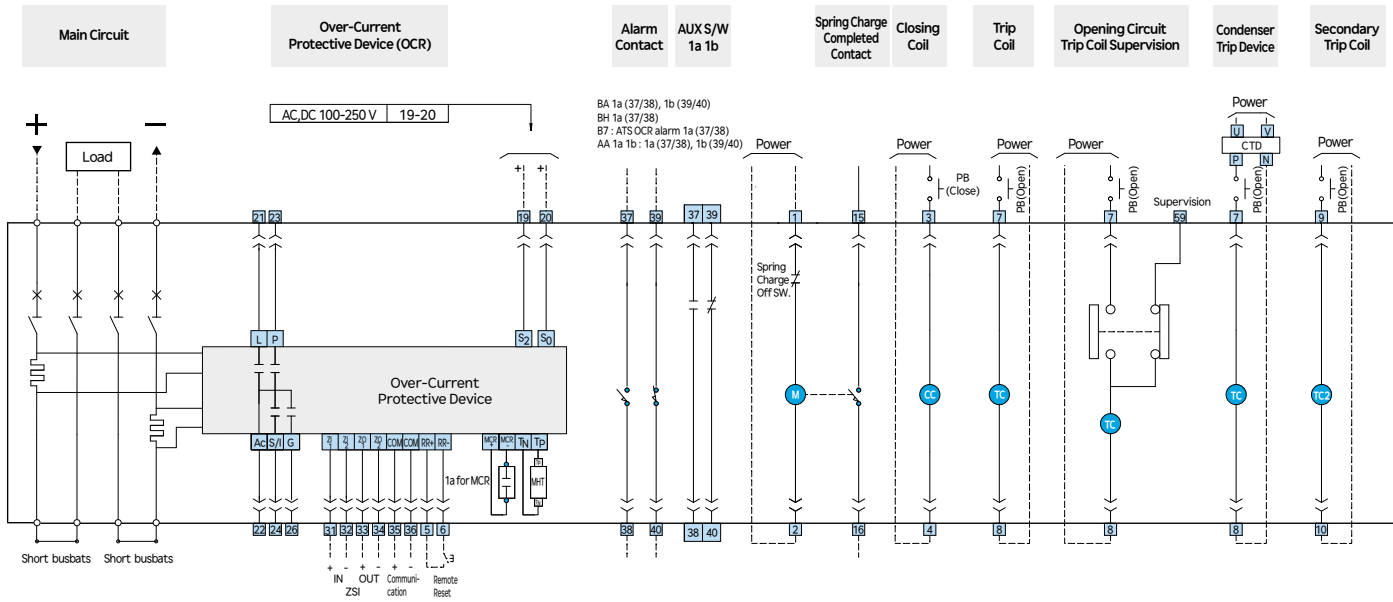
Detailed Dimensions of Real Panel Cutting

Model Name	A	B	C
HGN 16I3 - 40I3	400	298	55
HGN 08H4 - 40H4, HGN 16I4 - 40I4	515	298	55

※ The drawing dimension of this page may be subject to change without prior notice.

Circuit Diagram

Air Circuit Breaker (HGN DC Type)



Symbol Description

L	LTD Terminal
PT	Pre-Trip Alarm
G	Ground Fault Contact
S / I	STD/INST Contact
Ac	Common Contact
ZI	Zone Selective Interlock Input
ZO	Zone Selective Interlock Output
MCR +,-	MCR Input Terminal
Tp / Tn	MHT Output Terminal
T	Charging Motor
CC	Close Coil
TC	Trip Coil
UVT	Under-Current Voltage Trip Coil
MHT	Magnetic Hold Trigger
SO /S2	OCR Power (AC, DC 100-250 V)
S1 / S3	OCR Power (DC 24 V)

Terminal Description

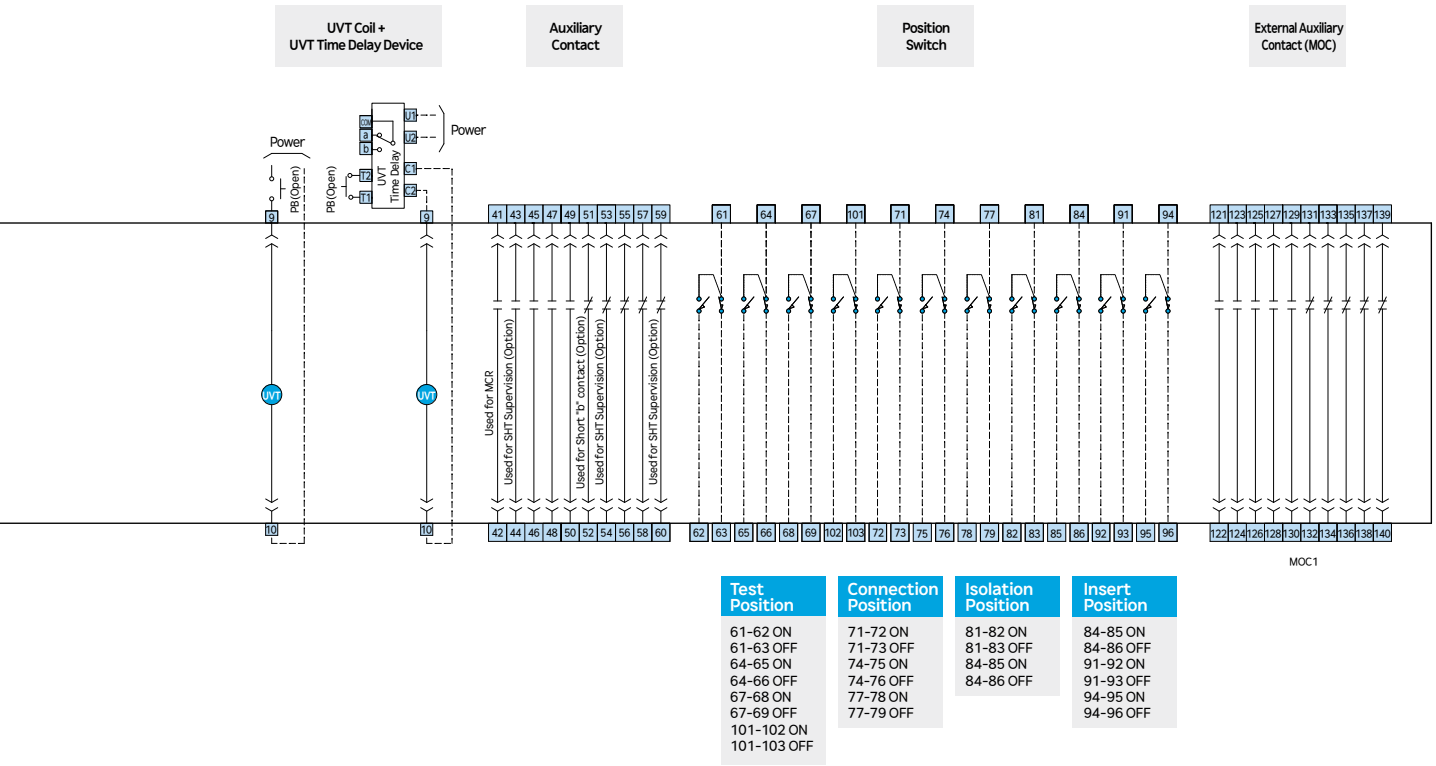
1 - 2	Charging Motor Power
3 - 4	Closing Coil Power
5 - 6	Remote Reset
7 - 8	Trip Coil Power
9 - 10	UVT Coil Power
15 - 16	Spring Charge Switch
19 - 20	OCR Power (AC, DC 100-250 V)
27 - 28	OCR Power (DC 24 V)
22 - 21	LTD Contact
22 - 23	Pre-Trip Alarm/Temperature Alarm Contact
22 - 24	STD/UBST Contact
22 - 26	GFT Contact
31 - 34	ZSI (Zone Selective Interlock)
35 - 36	Communication (RS485)
41 - 60	Auciliary Contact
61 - 93	Position Switch

- RR : Remote Reset
- VM : Voltage Module
- VR - VN : Voltage phase input
- R - N : Current input
- Rn - Nn : Current input

—— Manufacture's Wiring

---- User's Wiring

Disconnecting Device (Draw-Out Type)



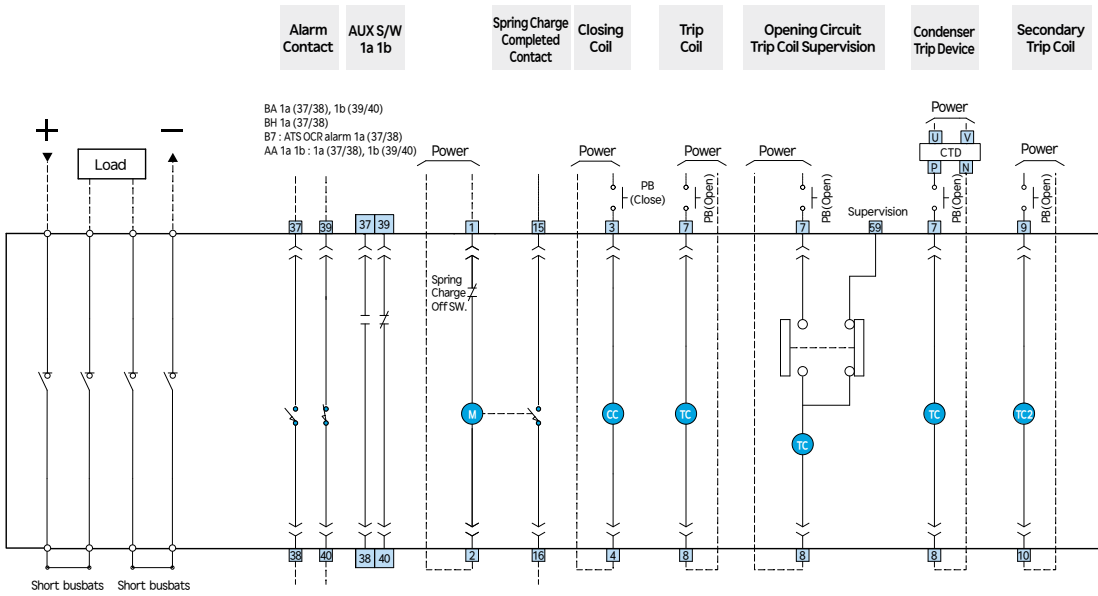
Test Position	Connection Position	Isolation Position	Insert Position
61-62 ON 61-63 OFF 64-65 ON 64-66 OFF 67-68 ON 67-69 OFF 101-102 ON 101-103 OFF	71-72 ON 71-73 OFF 74-75 ON 74-76 OFF 77-78 ON 77-79 OFF	81-82 ON 81-83 OFF 84-85 ON 84-86 OFF	84-85 ON 84-86 OFF 91-92 ON 91-93 OFF 94-95 ON 94-96 OFF

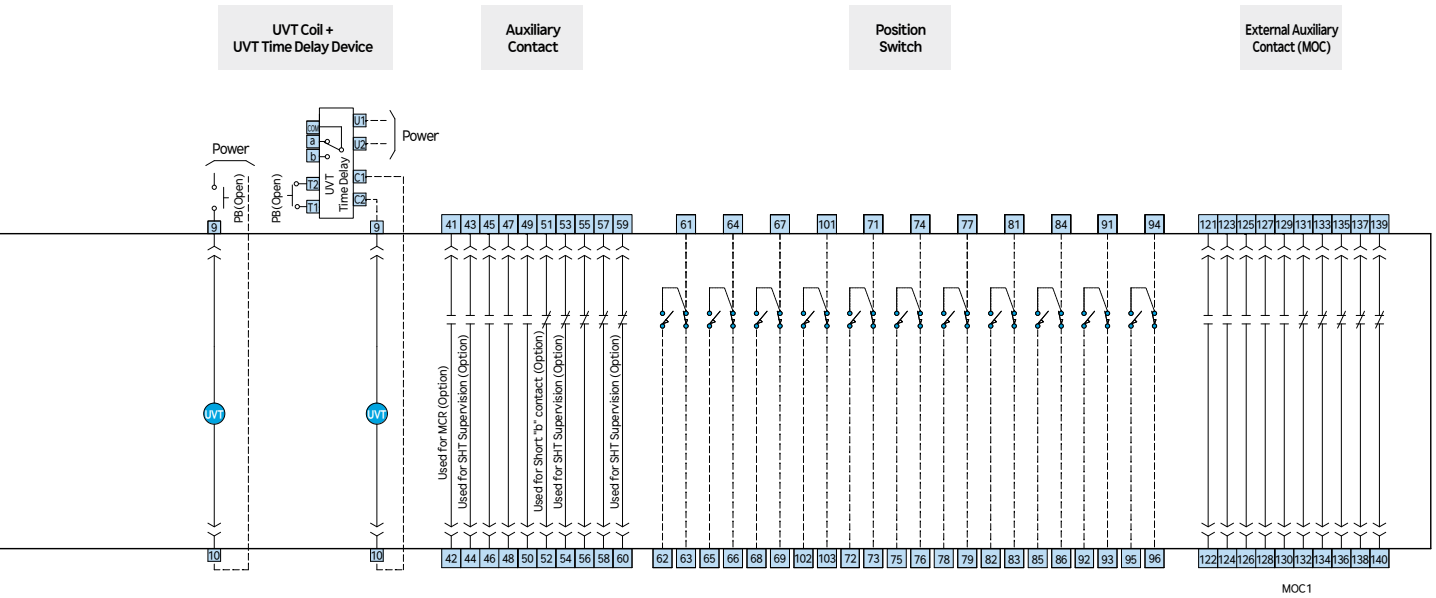
Control Jack Lay-out (Auto Connection Type)

OCR													Operating					Auxiliary switch									
POW	OCR Contact					DC POW	RR	ZSI		COM	Temp		M	CC	TC	UVT	CHA	MCR	2a	3a	4a	5a	1b	2b	3b	4b	5b
	COM	L	S/I	P/T	G																						
19(+)	21		23		27(+)		5	31	33	35	37	39			1	3	7	9	15	41	43	45	47	49	51	53	55	57	59
20(-)	22		24		26		28(-)	6	32	34	36	38	40			2	4	8	10	16	42	44	46	48	50	52	54	56	58	60
OCR Protection relay													Operating circuit					Auxiliary switch									

Circuit Diagram

Switch-Disconnector (HGN DC Type)





Test Position	Connection Position	Isolation Position	Insert Position
61-62 ON	71-72 ON	81-82 ON	84-85 ON
61-63 OFF	71-73 OFF	81-83 OFF	84-86 OFF
64-65 ON	74-75 ON	84-85 ON	91-92 ON
64-66 OFF	74-76 OFF	84-86 OFF	91-93 OFF
67-68 ON	77-78 ON		94-95 ON
67-69 OFF	77-79 OFF		94-96 OFF
101-102 ON			
101-103 OFF			

Order Code

HGN

HGN	12	A	3	J	M2
Model Name	Rated Frame	Frame & Product Classification	No. of Poles	Mounting Method	Charging Motor Power

Model Name	
HGN	Air Circuit Breakers

Rated Frame	
Air Circuit Breakers (DC)	
16	1,600 A
25	2,500 A
32	3,200 A
40	4,000 A
Switch-Disconnector (DC)	
08	800 A
10	1,000 A
12	1,250 A
16	1,600 A
20	2,000 A
25	2,500 A
32	3,200 A
40	4,000 A

Frame & Product Classification	
I	Air Circuit Breakers
H	Switch-Disconnector

No. of Poles	
3	3 Pole
4	4 Pole

Mounting Method	
Independent Release of Draw-Out Type (Control Power) of Main Unit	
A	Auto Connection
Fixed Type (Terminal Form)	
V	Vertical Type
Draw-Out Type (Combined with Cradle)	
C	Auto Connection + Vertical Type

Charging Motor Power	
M0	Manual Type
M1	AC/DC 110 V
M2	AC/DC 220 V
M7	DC 24 V
M8	DC 48 V
M9	DC 125 V

※ 1) Only 4P is available for H Frame Switch-Disconnector.

C2 – Closing Coil

C0	N/A
C1	AC/DC 110 V
C2	AC/DC 220 V
C3	AC 380 V
C4	AC 440 V
C7	DC 24 V
C8	DC 48 V
C9	DC 125 V

S2 – Trip Coil

S0	N/A
S1	AC/DC 110 V
S2	AC/DC 220 V
S3	AC 380 V
S4	AC 440 V
S7	DC 24 V
S8	DC 48 V
S9	DC 125 V

Trip Supervision Coil

T1	AC/DC 110 V
T2	AC/DC 220 V
T3	AC 380 V
T4	AC 440 V
T7	DC 24 V
T8	DC 48 V
T9	DC 125 V

61 – Over-Current Trip

General Feeder

00	N/A
D1	AC/DC 110/220 V

※ D1 can only be applied to DC circuit breakers.

K – Rating Current

0	OCR N/A
H	800 A
J	1,000 A
K	1,250 A
L	1,600 A
M	2,000 A
N	2,500 A
P	3,200 A
Q	4,000 A

AB – Parts for Main Unit

INST (UVT Coil)

U1	AC/DC 110 V
U2	AC/DC 220 V
U3	AC 380 V
U4	AC 440 V
U7	DC 24 V
U8	DC 48 V
U9	DC 125 V

Secondary Trip Coil

S1	AC/DC 110 V
S2	AC/DC 220 V
S3	AC 380 V
S4	AC 440 V
S7	DC 24 V
S8	DC 48 V
S9	DC 125 V

AM	On/Off Button Lock Device
AA	Auxiliary Switch (Additional 1a 1b)
AB	Key Lock Device (Prevent ON)
BR	OCR Manual Reset Function
BA	OCR Manual Reset & Alarm S/W 1a1b
BH	CR Manual Reset + High Capacity Alarm S/W 1a
B0	Mechanical Interlock Device
B6	Spring Charge Switch
AG	Door Flange
BT	Closing Preparation Contact (B6 and CT cannot be Combined)

AB – Parts Mounted on Cradle (Only Applicable when Placing an Order for Combination Type)

Time Delay Type (UVT Coil + UVT Controller)

V1	AC/DC 110 V
V2	AC/DC 220 V
V3	AC 380 V
V4	AC 440 V
V9	DC 125 V

Position S/W

AQ	Test 1C, Connected 1C
AR	Connected 2C
AS	Test 2C
AT	Isolated 1C, Inserted 1C
AU	Inserted 2C
AV	Isolated 2C
P4	Test 2C, Connected 2C
PT	Test 4C
PS	Isolated 1C, Test 1C, Connected 2C
PQ	Inserted 1C, Isolated 1C, Test 1C, Connected 1C
P8	Inserted 2C, Isolated 2C, Test 2C, Connected 2C
PR	Inserted 1C, Isolated 1C, Test 3C, Connected 3C
PU	Test 4C, Connected 4C

BC	Auto Connection Control Terminal's Safety Cover
MC	External Auxiliary Contact (MOC) 10a10b
AK	Short "b" Contact
AE	Safety Shutter
AX	Arc Shield
AF	Fixing Block
AW	Miss Insertion Preventer
TM	Temperature Sensor Bracket

- U☐ is only applicable when instantaneous type and only the UVT coil is installed. When using time delay type, place a separate order for the time delay type of controller (HGNS -) and install externally.
- V☐ is only applicable when time delay type is applied and the UVT coil is installed in the main unit while the time delay type of controller is installed at the side of the cradle. In case of dimension issues, place an order for HGNS V☐ for separate installation.
- B0 : The mechanical interlock device has to be ordered separately and when B0 is marked, only the components interlocked inside the main unit are assembled for release. (Refer to additional components).
- When S☐ Secondary trip is applied, UVT coil cannot be applied at the same time.
- HGN DC Type, can be used as 4a5b by using MCR function, and 3a3b can be used when using T☐ trip coil monitoring contact.
- In the order code for the main unit only (A/J/Fixed type), combination with cradle accessories is not possible and the cradle accessory arrangement is only possible when purchased with cradle arrangement. For the independent release of the main unit, place a separate order (HGNS -) (refer to additional components).
- Position switch cannot be overlapped for use and can only be attached only at the right side of cradle.
  In case a combination other than the combination indicate is required, make an inquiry separately.
- Mechanical interlock device (B0), external auxiliary contact (MC) and fixing block (AF) cannot be used together.
- BC : The control terminal's safety cover is only available for the auto connection.
- AK : Short "b" is only available for auto connection and is attached to 51, 52. It can be attached as much as the number of "B" contact so upon additional mounting, place a separate order (HGNS -).
- AG : Originally, it is attached separately but upon release, the front cover is inserted when shipped.
- MR : AE, AG, AF, AM are attached as a standard option, so the relevant accessories are not marked separately.
- AA and BA (BH) cannot be attached at the same time.
- UVT and Secondary Trip Coil cannot be attached at the same time.

Order Code

HGN (When Placing an Order for the Cradle)

DHN	12	A	3	A	H	AE
Model Name	Rated Frame	Frame Category & Neutral Pole	No. of Poles	Control Terminal Connection	Terminal Bus Bar Arrangement	Accessories
HGN Air Circuit Breakers Cradle	32 DHN06 - 32I (X) 40 DHN40I (X)	I Common use for H&I Frame	3 3 Pole 4 4 Pole	A Auto Connection	V Vertical Type	AE Safety Shutter

※ Marking upon Shipping of Cradle
·DHN06/08/10/12/16/20/25I (X) > DHN06 - 32I (X)
·DHN40I (X) > DHN40I (X)

※ AE :
Safety shutter lock is supplied with safety shutter.

※ 1) Only 4P is available for H Frame Switch-Disconnector.

Accessories

Name of Accessory	Order Form		Detailed Specification	Release Form	Page
	Mounted	Sold Separately			
Auxiliary Switch [1]	MC	HGNS MOC	External Auxiliary Contact 10a10b (Mechanical Opearting Cell Switch)	Mounted upon Release and Sold Separately	-
Automatic Connection Control Terminal Protection Cover	BC	HGNS BC	Automatic Connection Control Terminal Cover (Cannot be Mounted on Fixed Type, Manual Connection Type)	Mounted upon Release and Sold Separately	-
Condenser Trip Device	-	HGNS CTD1	AC 110 V	Sold Separately	-
	-	HGNS CTD2	AC 220 V		
UVT Time Delay Module [2]	-	HGNS UT1	AC/DC 110 V & DC 125 V	Sold Separately	-
	-	HGNS UT2	AC/DC 220 V		
	-	HGNS UT3	AC 380 V		
	-	HGNS UT4	AC 440 V		
	-	HGNS UT9	AC/DC 110 V & DC 125 V		
UVT Time Delay Module + UVT Coil	V1	HGNS V1	AC/DC 110 V	Mounted upon Release and Sold Separately	-
	V2	HGNS V2	AC/DC 220 V		
	V3	HGNS V3	AC 380 V		
	V4	HGNS V4	AC 440 V		
	V9	HGNS V9	DC 125 V		
OCR Acce	-	HGNS OC	OCR Portable Checker	Sold Separately	-
	-	HGNS HROC	Higher OCR Checker		
	-	HGNS VM	Voltage Module		
Key Lock	AB	HGNS AB	Key Lock	Mounted upon Release and Sold Separately	-
Safety Shutter	AE	HGNS AEA3	06A3 - 20A3, 06P3 - 20P3	Mounted upon Release and Sold Separately	-
		HGNS AEA4	06A4 - 20A4, 06P4 - 20P4		
		HGNS AEB3	20B3 - 40B3, 20Q3 - 40Q3		
		HGNS AEB4	20B4 - 40B4, 20Q4 - 40Q4		
		HGNS AEC3	32C3 - 50C3		
		HGNS AEC4	32C4 - 50C4		
		HGNS AED3	40D3 - 63D3		
		HGNS AED4	40D4 - 63D4		
Fixing Block	AF	HGNS AF	For Main Unit + Cradle	Mounted upon Release and Sold Separately	-
		HGNS AFB	For Main Unit + Cradle	Sold Separately	-
		HGNS AFC	For Cradle		
Position Switch [3]	AQ	HGNS AQ	Test 1C, Connected 1C	Mounted upon Release and Sold Separately	-
	AR	HGNS AR	Connected 2C		
	AS	HGNS AS	Test 2C		
	AT	HGNS AT	Isolated 1C, Inserted 1C		
	AU	HGNS AU	Inserted 2C		
	AV	HGNS AV	Isolated 2C		
	P4	HGNS P4	Test 2C, Connected 2C		
	PS	HGNS PS	Isolated 1C, Test 1C, Connected 2C		
	PT	HGNS PT	Test 4C		
	PQ	HGNS PQ	Inserted 1C, Isolated 1C, Test 1C, Connected 1C		
	P8	HGNS P8	Inserted 2C, Isolated 2C, Test 2C, Connected 2C		
	PR	HGNS PR	Inserted 1C, Isolated 1C, Test 3C, Connected 3C		
	PU	HGNS PU	Test 4C, Connected 4C		

※ 1) Contact us when placing an order for set order required for installation.
2) UVT time delay type of controller is mounted outside the main unit or the cradle.
3) Position switch cannot be used together and can only be mounted at the right side of the cradle.
For position switches, because they are common parts with the U-Series product, the position switch may be marked "UANS".

Order Code

Accessories

Name of Accessory	Order Form		Detailed Specification	Release Form	Page
	Mounted	Sold Separately			
Miss Insertion Preventor	AW	HGNS AW	Add Order Form in the Main Unit + Cradle, Main Unit/Cradle Respectively	Mounted upon Release and Sold Separately	-
Arc Shield [4]	-	HGNS AXI3	16I3 - 40I3	Mounted upon Release and Sold Separately	-
		HGNS AXI4	16I4 - 40I4, 08H4 - 40H4		
Mech Interlock [5]	-	HGNS DWB1	Drawout/Wire Type 2way MI	Sold Separately	-
	-	HGNS DWB2	Drawout/Wire Type 3way MI		
	-	HGNS FWB1	Fixed/Wire Type 2way MI		
	-	HGNS FWB2	Fixed/Wire Type 3way MI		
	-	HGNS DWB1B	B, Q, H, I Frame, Draw-Out Type, Wire Type 2 Way MI + Interlock Parts (B0)		
	-	HGNS DWB2B	B, Q, H, I Frame, Draw-Out Type, Wire Type 3 Way MI + Interlock Parts (B0)		
	-	HGNS FWB1B	B, Q, H, I Frame, Fixed Type, Wire Type 2way MI + Interlock Parts (B0)		
	-	HGNS FWB2B	B, Q, H, I Frame, Fixed Type, Wire Type 3way MI + Interlock Parts (B0)		
	BO	HGNS B0DB	B, Q, H, I Frame Interlock Parts (B0)	Mounted upon Release and Sold Separately	-
Temperature Sensor Bracket	-	HGNS TMSUPB3	B, Q, H, I Frame 3P, Temperature Sensor Bracket	Mounted upon Release and Sold Separately	-
	-	HGNS TMSUPB4	B, Q, H, I Frame 4P, Temperature Sensor Bracket		
Door Flange	AG	HGNS AG	IP30	Mounted upon Release and Sold Separately	-
Dust Cover	-	HGNS DC	IP52	Sold Separately	-
Test Jumper	-	HGNS AJ	For Automatic Connection Type of Test	Sold Separately	-
Short "B" Contact	AK	HGNS AK	Short Circuit "B" Contact	Mounted upon Release and Sold Separately	-
Lifting LUG	-	HGNS AL	2 ea = 1 set, C Frame 5,000 A is Provided as Standard	Sold Separately	-
On/Off Button Lock	AM	HGNS AM	On/Off Button Cover Lock	Sold Separately	-
Insulation Barrier	-	HGNS PBAR	Insulation Barrier	Sold Separately	-
Draw In-Out Handle	-	HGNS LHANDLE	Long Head Type Draw In-Out Handle	Sold Separately	-
	-	HGNS UHANDLE	Universal Join Head Draw In-Out Handle		
	-	HGNS THANDLE	T Grip Long Type Draw In-Out Handle		
	-	HGNS HANDLE	Standard Draw In-Out Handle		

※ 4) Arc shields are provided by default for the HGN DC model.
5) Mechanical interlock is a wire type. The MI unit is mounted outside the main unit or on the cradle and the interlock part is mounted inside the main unit.

Handling and Maintenance Inspection

Standard Operation Conditions

| Ambient Temperature | -5 - +40 °C
(Daily average temperature of below 35 °C)
| Altitude | Below 2,000 m above sea level
| Relative Humidity | Below 85 % (Below daily average of 85 %)

The surrounding environment may affect the insulation performance and durability of the circuit break so the operation conditions must be checked clearly before application.



In case of operating in high ambient temperature exceeding 40 °C, use according to the derating calibrated depending on the ambient temperature indicated in the catalogue.



In case of operating in places with high humidity and frequent rain, the insulation or electrical performance may drop.



In case of storing and using in places with a lot of dust and humidity, the use of a dust cover or dehumidifier is recommended. Excessive impact and vibration may cause damage in the operation mechanism.



When storing circuit breaker without electric circuit : -25 °C - 85 °C
When storing circuit breaker with electric circuit : -15 °C - 70 °C
After storing in low temperature, store for more than 8 hours in room temperature then store in an open location for the type with charging motor.



In case there is a lot of corrosive gas, it must be stored in an enclosed protective structure to minimize damage.



In case of highland with high altitude, the insulation performance drops so please apply the highest rating product after checking the insulation performance calibration coefficient.

Special Operating Conditions

The HG-Series air circuit breaker has been designed/manufactured to operate in a standard operating environment designated by the IEC standard and in case of operating the circuit breaker in the following special environment, contact our company.

- In case the altitude above sea level and the ambient temperature exceeds the standard operating environment
- In case of a region with a lot of sea wind or salinity
- In case of a region with snow and freezing
- In case of a region of frequent shock and vibration
- In case the relative humidity exceeds the standard operating environment
- In case of a region with a lot of humidity and frequent rainfall
- In case of a region with a lot of dust, vapor, corrosive and combustible gas, wet steam
- In other special environment that exceeds the standard environment

※ Special operating environment and condition may cause damage in the product (Increase in resistance at main circuit, rust, change in coating).

Handling and Maintenance Inspection

Maintenance Inspection

Regular Inspection

Inspection Form	Inspection Item			Inspection Period
	Main Review Item	Details and Countermeasures		
General Regular Inspection	Foreign Substance	Visual examination must be executed and there must be no foreign substance inside. Blow away the foreign substance using compressed air.		Every 6 months
	Gap	Execute visual examination and there must be no cracks or gaps.		Replace, repair
	Operation	Circuit breaker must be closed manually to inspect the operation area during normal status.		Once a year
	Greasing	Apply an adequate amount of grease on each pin, axle and bearing. Excessive greasing may cause dust and stain from building up. Electrical grease : HITALWBE 28G Lubiricant grease : Kamro Chemical/CHEMAX HHI 5000#1.		Every 6 months
	Spring Type	The status of the spring must be checked to inspect the damaged area.		Replace, repair
	Screw/Bolt Type	Tighten loose areas using screw or bolt.		Every 6 months
	Stain on the Fastening Bolt Area	There must be no dust or stain at the fastening area. Wipe the area well using clean cloth dipped in alcohol.		All the time
Mechanical Durability Warranty Lifespan Inspection (Based on IEC60947 standard)	Below 2,500 AF	Total operation frequency : 3,000 times	Non-current 2,500 time, current 500 time	Paid inspection is required (Precision inspection) Inspection period : At least 6 months after exceeding the opening/closing frequency Precision inspection once a year is recommended Contact our company to determine is replacement is required
		Open/closing frequency : 500 times	Open/close within the range of rated current	
		Open/closing frequency : 25 times	Open/close in over-load (Around 6 times the rated current) area	
		Immediately upon breaking	Prevent short-circuit/earth fault	
	Above 3,200 AF	Open/close frequency : 2,000 times	Non-current 1,500 time, current 500 time	
		Open/close cycle : 500 times	Open/close within the range of rated current	
		Open/close cycle : 25 times	Open/close in over-load (Around 6 times the rated current) area	
		Immediately upon breaking	Prevent short-circuit/earth fault	
Arc Chamber (Grid)	Contact with Foreign Substance	In case foreign substance has melted onto the grid's side plate, replace the ionized pieces inside the arc chamber and others.		Replace, contact our company
Arc Contact	Contact Tip Surface	Remove foreign substance, dust and oil.		Visual inspection, no problem Replace, contact our company
		Melting stains on the contact tip surface is caused by open/closing arc.		
		In case the extent of melting caused by arc reduces to 1/3, replace the entire operation and fixing moving contact unit.		
		However, in case there is foreign substance on the contact, precision diagnosis is required to determine if replacement is necessary through our company's paid service.		
Main Contact	Damage of Main Contact	In case the main contact has been worn off and is rough, the surface must be managed neatly.		At all times
		However, in case there is foreign substance on the contact, precision diagnosis is required to determine if replacement is necessary through our company's paid service.		

⚠ Caution

1. In the event the contact or part is replaced, turn the circuit breaker off and replace after drawing out the disconnected position to isolated.
2. Execute the inspection work after checking whether the remaining heat in the conduction part has been dissipated. It may cause burns.
3. When rubbing the contact tip, ensure that the sludge does not enter the circuit breaker's mechanical device. After rubbing, wipe the contact tip neatly.
4. Please refer to the instruction manual for detailed information when disassembling and assembling.

Essential for Today
Potential for Tomorrow

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